Abra Travels: Master Build Plan

One document for the whole project, from the basics to the last endpoint.
Written 13 October 2026. Read Part 0 first.

Part 0. Start here (for anyone new, including another session)

Abra Travels is a bus booking platform: travellers book seats, bus owners (operators) sell them, Abra (admin) runs the marketplace, drivers run the trips. This document explains **what exists, how everything connects, the order to build it in, and the exact screens, buttons and endpoints to build**, so that nobody has to guess.

0.1 Facts you need in the first five minutes

Item	Value
Repository	`D:\New_project\abra-travels` (monorepo)
Functional source of truth	`Abra Travels.pdf` (122 sections) in the repo root; the older design notes are in `documentation/`
The demos (the specification)	`prototypes/index.html` opens five working HTML demos: admin, client (operator), customer, driver, website. They share one data store and sync live. Everything in this document was read from them.
Backend	NestJS + TypeScript in `backend/`. 26 controllers and 113 routes exist today (Appendix I).
Database	MongoDB Atlas, database `abra_travels`. The connection string is in `backend/.env.development` (git-ignored). **Never put it in a document, a chat or git.** Connection tested: working.
Redis	Needed for seat holds, queues and live updates. Runs in Docker from `infrastructure/docker-compose.development.yml` (container `abra-redis`); start Docker Desktop first.
Flutter apps today	`apps/customer_app` (built on mock data, some screens missing), `apps/admin_app` and `apps/client_portal` (partly built: sign-in, KYC, banners, catalogue, settings). Shared code is in `packages/shared_*`.
Apps not started	Driver app, public website (Next.js), partner app
UI rule (mandatory)	**The real screens must look and behave exactly like the demo**: same layout, same labels and wording, same order of buttons, same colours and gradient, same dialogs and fields, same empty, error and loading states, same table and list views, same horizontal and vertical layouts. The demo is the design, not a sketch.
Design rule	Navy and blue palette with the brand gradient, from `documentation/design-system.md`. UI and functionality ship together. Every new screen gets a visual check.
Current phase	**Phase 0 built and checked, waiting for your yes; Phase 1 in progress on a separate branch** (updated 13 October 2026). Phase 0: access matrix, messages log, integration settings, assistant, live updates, feature flags with maintenance and the demo seed are built and tested; the 27-step checklist passes. Phase 1: steps 0 to 4 are done on branch `phase1-catalogue-kyc` (not merged). Full detail and what is not proven: Part 9.

0.2 How the demos map to the real build

In the demo	In the real application
`AbraStore`: one shared data store in the browser	MongoDB collections behind one NestJS API (Appendix A lists every collection and field)
`route()`: turns an event into notifications	An event service in the backend that creates notifications, push, WhatsApp and email (Appendix B)
`msgSend` / message log	A message queue worker plus the `messages` collection and the WhatsApp and email providers
Saving to the browser and syncing between tabs	The API plus Socket.IO live updates
`PERM`: role access matrix (None, View, Edit, Edit+Delete)	RBAC on the server, editable by Super Admin only (Appendix C)
Proof attachments stored in the browser	The files service (Phase 0), with the same limits and the same viewer
Scripted assistant	One backend endpoint that calls Groq and answers only from the caller's own data
`_ver` reseed	A seed script that loads the same sample data (Phase 0)
Demo buttons such as "simulate a payment"	Do not exist in production; they are replaced by Razorpay and the driver app

0.3 How to read this document

- **Part 1** the product and the flow. **Part 2** architecture and decisions. **Part 3** the file rule (three files per feature). **Part 4** the phases. **Part 5** the rules that apply to every feature. **Part 6** testing. **Part 7** open items and who owns them. **Part 8** how we work. **Part 9** the progress log: what is built, how it was checked, what is not done and the known problems.
- **Appendices D to H** list every page, tab, filter, table column, button, dialog field and the endpoint behind each button, exactly as the demos show them. **Appendix J** lists the three files of every feature.
- If a number or label here and the demo ever disagree, the demo wins and this document is fixed.

0.4 What happens next

The current list is in Part 9.5. In short: you run the Phase 0 checklist (Part 4), you say "Phase 0 is good", the two branches are merged and Phase 1 continues. No sample data is loaded into the real database: the first data is created by testing through the apps.

Part 1. The product and the flow

1.1 Who uses what

Who	App	What they do
Traveller (customer)	Customer app (Flutter: Android, iOS, web)	Search, pick seats, pay, get the ticket, track the bus, cancel and refund, raise SOS, get support
Abra admin, support, accounts, Super Admin	Admin app (Flutter web)	Run the marketplace: onboard operators, set commission, watch trips and money, handle safety and support
Bus owner and their staff (the operator)	Client portal (Flutter web). This **is** the operator app; there is no separate one	Company KYC, buses, seat layouts, routes, fares, trips, bookings, earnings, driver rules and reports
Driver and conductor	Driver app (Flutter, Android)	Check passengers in, start and end the trip, send GPS, raise SOS, send reports, pass the pre-trip checklist
Anyone on Google	Public website (Next.js)	Route and city pages, offers, help, track-my-bus, contact, partner enquiry; booking hands over to the customer app
External agents	Partner app and partner API (last)	Book through the API

1.2 The flow from start to payout

1. **Admin sets up the catalogue:** cities and places, bus types and amenities, required documents, KYC questions, banners, settings.
2. **An operator joins:** admin invites, or the owner signs in with Google. The owner fills the KYC questions and uploads documents. Admin reviews each document and approves it, or rejects it with a reason. The owner fixes only the rejected items. Approval makes the company active. Each bus gets its own KYC, and only after the company.
3. **Commercial terms:** admin sets the commission rule (agreed privately with the operator) and the payout mode (automatic split with Razorpay Route, or manual settlement).

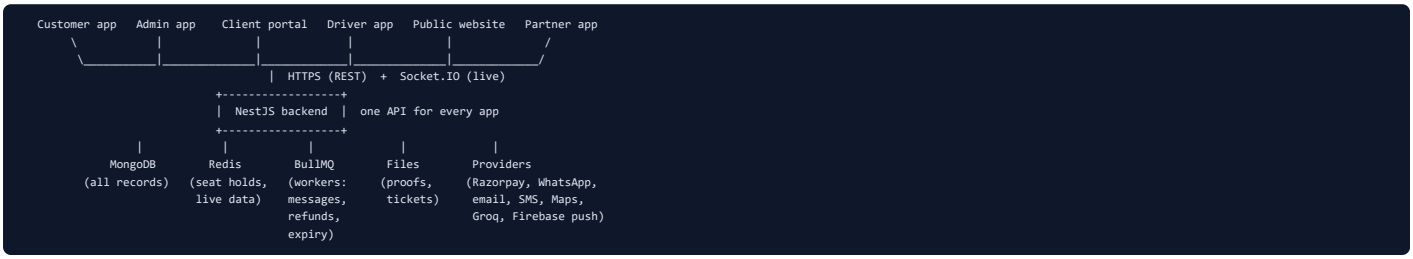
4. **Supply:** the operator adds buses with a seat layout, routes with timed stops and boarding and dropping points, fares (one price per trip, or per seat type and stop pair), and single or recurring trips, then publishes. No per-trip approval: company KYC and bus KYC are the only gates.
5. **Booking:** the customer searches without logging in, picks seats (10 minute hold, at most 6 seats), logs in at Continue, chooses boarding and dropping points, enters passengers (email is required together with the mobile number), applies a coupon, and pays with Razorpay.
6. **After payment:** the server confirms, issues the ticket (PNR, QR, PDF), sends WhatsApp and email messages, notifies the operator.
7. **Travel day:** the driver passes the pre-trip checklist, checks passengers in by QR, starts the trip and sends GPS. The customer sees live tracking and has SOS and sharing. Admin and operator see the same bus on their maps.
8. **After the trip:** cancellation and refund (slabs set by admin), settlement to the operator (commission deducted), reports and exports.

1.3 Who sees whose data

- An operator and their staff see **only their own company's** data. This is enforced on the server.
- Admin roles see what the access matrix allows: None, View, Edit, Edit+Delete per page. Only the Super Admin edits the matrix.
- Customers see only their own bookings, support tickets and messages.
- Drivers see only the trips assigned to them and the rules their own operator set.

Part 2. Architecture and decisions

2.1 The system in one picture



2.2 Technology

Layer	Choice
Mobile and web apps	Flutter (Android, iOS, web), shared packages in `packages/shared_*`
Public website	Next.js
Backend	NestJS + TypeScript, one API
Database	MongoDB (Atlas). Money is stored in paise (integers).
Cache, locks, queues	Redis with BullMQ
Live data	Socket.IO (GPS, SOS, notifications, seat changes)
Payments	Razorpay: orders, webhooks, Route (automatic split)
Messages	WhatsApp provider, email provider, SMS (DLT), Firebase push (project ids are in the project memory notes)
Assistant	Groq, called only from the backend; admin picks the model and key
Maps	Placeholder until the provider and key are chosen (Part 7)

2.3 Decisions already taken

Topic	Decision
Commission	Agreed privately between admin and operator. A rule per operator, per route or per bus type; a percentage or a flat amount; versioned. Only admin edits; the operator sees it read-only. Each booking stores the rule that applied at booking time.
Fares	The operator chooses: one price per trip, or a price per seat type and per stop pair. Both are supported.
Trip approval	None. Company KYC and bus KYC are the only gates.
Payouts	Both modes supported; admin picks the mode per operator.
Bus location	From the driver app.
Seat hold	10 minutes on the server. At most 6 seats per booking, enforced on the server.
Guests	Search without login; login only at Continue to book and on personal screens.
Customer contact	Mobile and email are both required. Google and Apple sign-in are offered next to the WhatsApp code.
Languages	All 8 in the first release: English, Telugu, Hindi, Kannada, Tamil, Malayalam, Marathi, Bengali. Language names are always shown in English next to the native script.
Proofs	Every sensitive admin action asks for a proof file (picture, PDF or text, up to 5 files, 600 KB each) that Abra and the person concerned can view and download.
Number cards	Every number card opens the records behind it. A card that says 1 must show that one record.
UI	The real UI is the same as the demo in every app. Nothing is redesigned, reworded or rearranged while connecting the backend. Any change to a screen is agreed with the user first and the demo is updated at the same time.
Operator app	The client portal is the operator app. One app, not two.
Driver rules	The operator maintains pre-trip and end-of-trip checklists and the report types. Abra provides default templates and a locked minimum that operators cannot remove.
Driver reports	Daily and weekly numbers come from the system and are frozen when the driver sends the report. The driver only adds what the system cannot know.
Database separation	This project uses its own database (`abra_travels`) on the Atlas cluster the user shared. Keep it separate from the databases of other Abra projects.

Part 3. The file rule: three files per feature

> **Rule:** Every feature is built as a small, complete package: **one screen file**, **one backend file**, **one service file**. Nothing is split into many small files. A feature is finished from the screen to the database before the next feature starts.

3.1 The three files

#	File	Where	What is inside it
1	`<feature>_<app>_screen.dart`	`apps/<app>/lib/features/<feature>/`	Everything the user sees for this feature, in this order inside the one file: (1) data models, (2) state and controller, (3) API calls through the shared client, (4) every widget the screen needs (cards, tables, dialogs, forms, charts), (5) the screen itself. **No widgets in other files.**

#	File	Where	What is inside it
2	`<feature>_<app>.ts`	`backend/src/modules/<module>/`	Everything the API needs for this feature, in this order: (1) Mongoose schema, (2) request and response types with validation, (3) the controller with every endpoint, (4) the Nest module wiring. **No other backend files for the feature.** If the service needs the schema, the schema sits at the top of the service file instead, so the two files never import each other in a circle.
3	`<feature>_<app>.service.ts`	`backend/src/modules/<module>/`	All business logic: rules, calculations, permission checks, events, notifications, queries. The controller only calls the service.

`<app>` is the audience: `admin_manage` for the admin portal, `client_manage` for the operator portal, `customer` for the customer app, `driver` for the driver app, `public` for the website. The SOS feature therefore has, for the admin: `sos_admin_manage_screen.dart`, `sos_admin_manage.ts`, `sos_admin_manage.service.ts`; for the customer: `sos_customer_screen.dart`, `sos_customer.ts`, `sos_customer.service.ts`; for the driver: the same with `driver`.

3.2 What is allowed to be shared

- **Flutter:** the shared packages already in `packages/` (network client, authentication, permissions, theme, UI kit, localization, validation, storage, notifications). A feature never copies them.
- **Backend:** authentication guards, the permission decorator, the audit service, the notification service, the files service, the Mongoose connection, and shared helper functions (money, dates, pagination). Written once, imported by every feature.
- **Collections:** the schema of a collection lives in the file of the feature that owns it (the `*_admin_manage.ts` file). Other audiences import it.
- **Tests** live in `test/` folders and are not counted as feature files.

3.3 Rules for the files

- Section banners inside each file (`// ===== 1. MODELS =====`) keep a long file easy to read. A file can be long; it must stay in order.
- The existing backend modules (cities, buses, trips, onboarding and so on) keep their present structure. The three-file rule applies to everything new. When an existing module needs new endpoints, they go into the new `<feature>_<app>.ts`.
- Every endpoint in the `.ts` file carries the permission it needs, the audit event it writes, and the notification it sends, written in a comment line above it so a reader sees the whole behaviour in one place.
- Every screen file shows loading, empty, error, no-permission and offline states. These are part of the screen, not extras.
- Money is shown in rupees, stored in paise. Dates are shown in India time.
- **The screen file reproduces the demo screen exactly:** the same sections in the same order, the same labels, buttons, filters, table columns, dialogs and form fields that Appendices D to H list, the same table / list toggle, the same horizontal / vertical layout option, the same number-card behaviour, the same 8-language switching and the same proof attachment box. If the demo and the screen file differ, the screen file is wrong.
- Naming: feature names are lower snake case and are the same in all three files and in Appendix J.

3.4 Example: the SOS feature for the admin

```
apps/admin_app/lib/features/sos/sos_admin_manage_screen.dart     <- models, controller, API calls, every widget, the screen
backend/src/modules/women-safety/sos_admin_manage.ts             <- schema, DTOs, controller (list, detail, acknowledge,
                                                                    escalate, resolve, false alarm, note), module
backend/src/modules/women-safety/sos_admin_manage.service.ts     <- rules: status changes, contact alerts, escalation timers,
                                                                    audit, notifications to customer, driver, operator
```

Part 4. The phases

Each phase builds backend and screens together for the features inside it, so every phase can be tested end to end. The customer app is finished inside Phases 3 to 5; the admin portal and the client portal are built page by page in the phase of their feature and completed in Phases 7 and 8; the driver app starts in Phase 5 and is completed in Phase 9; the public website is Phase 10.

Phase	Name	Apps	Features
0	Foundation	backend, shared Flutter packages	8
1	Catalogue, operators and KYC	backend, admin app, client portal (sign-up and KYC), customer app (banners)	7
2	Supply and commercial terms	backend, admin app, client portal	7
3	Customer booking journey	backend, customer app, admin app (bookings, payments, offers), client portal (bookings, offers)	6
4	After booking	backend, customer app, admin app (refunds, tickets, customers, content), client portal	5
5	Safety, driver GPS and live tracking	backend, driver app (new), customer app, admin app, client portal	5
6	Money out: settlements and payouts	backend, admin app, client portal	1
7	Rest of the admin portal	admin app, backend	3
8	Rest of the client (operator) portal	client portal, backend	0
9	Driver app in full: rules, reports and checklists	driver app, client portal, admin app, backend	2
10	Public website, then partner app	public website, backend, partner app	2
11	Languages, hardening and launch	all	0

> **Note:** Why this order and not "all backend first": a mistake in an early endpoint stays hidden until a screen uses it. Building each feature with its screen finds mistakes the same day. The order still follows your plan: foundation and backend first, customer app, then admin, client, driver, website.

4.1 Customer app: what to check before building (found by comparing the Flutter code with the demo)

Item in the demo	In the Flutter app today	Phase
Search, trip details, seats, boarding, passengers, review, payment, ticket, My Trips, cancel, refund, tracking, trip sharing, women safety, notifications and offers	Screens exist; their data comes from `Mock...Repository` classes with `Api...` stubs waiting	3, 4, 5
Home banners, city lookup, trip search	Real (call the backend today)	done
Messages page (copies of WhatsApp and email)	Not found in `features/`; check	3
Google and Apple sign-in buttons; email required together with mobile	Check the login and passenger screens (login files must not be edited without asking)	3
Per-seat-type prices on the seat map and in the fare breakup	Check	3
Running-trip banner on Home with Track and SOS	Check	5
SOS button on the tracking screen	Check	5
Support tickets with proof files and a reply thread	`support_screen.dart` looks static; check	4
Notification settings, delete account, 8 languages, AI assistant	Check each	4, 11

Phase 0. Foundation

Goal	One working backend on real MongoDB and Redis, with sign-in, access rules, audit, notifications, files and seed data. Nothing else can start without this.
Apps touched	backend, shared Flutter packages
Done when	Backend starts clean; health check shows MongoDB and Redis up; a Super Admin can sign in; seed script loads the demo data; one notification travels from the backend to a phone and to the admin bell.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Sign-in and access control (`auth_access`)	Customer: login sheet Driver: login	Partly built (authentication, sessions, rbac, users)	users, sessions	0 new / 1 existing	WhatsApp code, password, Google, Apple, MFA, sessions are built. Access matrix (None / View / Edit / Edit+Delete per role per page, Super Admin only, audited) built 2026-10-12 as access_matrix_admin_manage; waiting for a database test. Still to add: Apple sign-in.
Audit log (`audit`)	Admin: audit	Partly built (audit-logs)	audit	2 new / 0 existing	Read endpoint exists. Add the write hook every sensitive action must call, filters, detail view and export.
Notifications and event router (`notifications`)	Admin: notifications Client: notifications Customer: notifs, nset Driver: alerts	Partly built (notifications)	notifs, inbox	12 new / 12 existing	In-app, push and live delivery exist. Add the event router that turns a business event into the right notification for the right person (see Appendix B).
File storage and proof attachments (`files`)	backend only	Done (files)	-	0 new / 0 existing	Upload and download exist. Proofs use the same service: picture, PDF or text, up to 5 files and 600 KB each, viewable and downloadable by Abra and the person concerned.
WhatsApp and email messages (`messages`)	Admin: messages Customer: msgs	Partly built (whatsapp, notification-templates, notifications)	messages	6 new / 0 existing	BUILT 2026-10-12 as messages_admin_manage and messages_customer: the `messages` log, admin list, filters, read and resend with audit, customer "Messages sent to me", and MessageSenderService for other features to send and log WhatsApp and email. 10 unit and 11 end-to-end tests pass. Still to add: per-language templates and the event handlers that call the sender (Phase 3 onward).
Integrations, feature flags and maintenance (`settings`)	Admin: settings, flags	Partly built (system-settings, feature-flags, maintenance)	settings, flags, maint	8 new / 2 existing	Email and WhatsApp settings exist. Razorpay, Maps, SMS, Push and the assistant settings BUILT 2026-10-12 as integrations_admin_manage (16 unit and 9 end-to-end tests). Feature flags and maintenance BUILT 2026-10-12 as flags_admin_manage: the nine switches of the demo, five maintenance scopes with a message and a window, a global guard that enforces both on every request (staff with maintenance.bypass and the Super Admin are never locked out), public GET /features and GET /maintenance/status, audit on every change (31 unit and 15 end-to-end tests). No endpoint uses @RequireFeature yet: each feature adds it when it is built. Apps must send the X-Abra-App header.
AI assistant (Groq) (`assistant`)	Admin: ai Customer: assistant Driver: assistant	New (chatbot)	-	5 new / 0 existing	BUILT 2026-10-12 as assistant_chat: POST /assistant/chat for every app. Groq runs through the backend with the key and model from Admin settings. It answers only from tools the person's role allows (profile, access and notifications today; each feature adds its own tools). Proposed changes wait for confirm or decline and are audited. 20 questions a minute, thumbs up or down, and a development assistant when no AI key is set. 23 unit and 16 end-to-end tests pass. Not built: streaming the answer word by word, and tools for bookings, trips and the like (added with those features).
Seed data and live updates (`seed`)	backend only	New (database, events)	-	0 new / 0 existing	BUILT 2026-10-12: the live-update channel (live_updates_channel, Appendix B.3) and the demo seed (`npm run seed:demo` , backend/src/seed, fixture backend/seed/demo-data.json taken from the demos' store). The seed loads 20 cities, 52 places, 6 bus types, 14 companies and 31 buses; it is development-only, adds only what is missing (never changes an existing record), supports --dry-run and --only=catalogue|companies, and prints created, already there and skipped per collection. 18 unit tests check that every demo record becomes a valid real document. Not seeded yet: routes, trips, fares, bookings, customers, tickets and the rest, which are added to the seed in the phase that creates each collection.

Phase 0: your test checklist

- Start Docker Desktop and wait for "running", then start the backend (`npm run start` in the backend folder) and open `http://localhost:4000/api/docs` .
- **A. Health:** `GET /health/ready` shows ready, MongoDB up, Redis up.
- **B. Sign in:** `POST /api/v1/auth/login` with `{ "identifier": "<Super Admin email>", "password": "...", "device": { "clientType": "MOBILE" } }` (use MOBILE so the refresh token is in the answer). Copy `accessToken` into Authorize. `GET /me` shows your name and `userType` STAFF inside `user` .
- **C. Access matrix:** `GET /access-matrix?audience=ADMIN` shows 38 pages, the Super Admin row locked and FULL. Create a role with `POST /access-matrix/roles` , set Trips to VIEW, try to change SUPER_ADMIN (must be refused with 403), then delete the test role.
- **D. Audit:** `GET /audit-logs` shows created, access changed and deleted, each with who and why.
- **E. Messages:** `GET /admin/messages` is an empty list (correct until bookings exist).
- **F. Integrations:** `GET /settings/integrations` lists 7 cards. Set Maps to `osm` , test it. Save a test SMS secret: only "set" and the last four characters come back. Clear it.
- **G. Assistant:** `POST /assistant/chat` with "who am i?" answers with your role.
- **H. Feature flags:** `GET /features` (no sign-in) shows 9 switches, 4 on. Switch `live_tracking` off, see it false, switch it back on.
- **I. Maintenance:** switch the customer scope on with a message, see `/maintenance/status?app=customer` active, see a customer-app request answered 503 with the message, see the admin portal still works, then **switch it off**.
- **J. Sample data: skipped on purpose.** No sample data is loaded into Atlas; you create the real data through the apps. (On a throw-away local database only: `npm run seed:demo` , then 20 cities show.)
- **K. Sign out:** `POST /auth/logout` with `{ "refreshToken": "..." }` from the sign-in answer; the old token then answers 401.
- All of this can be run by script: `CHECKLIST_EMAIL=... CHECKLIST_PASSWORD=... node scripts/phase0-checklist.mjs` (26 steps on Atlas; step J is skipped).

Phase 1. Catalogue, operators and KYC

Goal	Everything admin sets up before a bus owner can join: places, bus types, required documents, KYC questions, invitations, KYC review, operators, banners.
Apps touched	backend, admin app, client portal (sign-up and KYC), customer app (banners)
Done when	Admin invites an operator; the owner signs in, fills the KYC, admin approves or rejects document by document; the owner fixes only the rejected items.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Cities and places (`places`)	Admin: places Client: places Customer: city	Partly built (cities, locations)	cities, places, placeReqs	8 new / 3 existing	Cities and locations exist. Add operator place requests with approval, import, usage counts and delete protection.
Bus types and amenities (`bus_types`)	Admin: bustypes	Partly built (bus-types)	busTypes, amenities	5 new / 1 existing	Bus types exist. Add amenities.
Required documents and KYC questionnaire (`kyc_config`)	Admin: documents, questions	Partly built (document-types, onboarding)	docTypes, questions	12 new / 2 existing	Document types exist. Add the questionnaire builder, preview and export.
Operator invitations (`invitations`)	Admin: invitations	Done (onboarding)	invitations	4 new / 2 existing	Send, resend, revoke, accept exist. Add edit, delete, export.
KYC review and company onboarding (`kyc`)	Admin: kyc Client: kyc, company	Done (onboarding)	submissions	4 new / 2 existing	Submissions, per-document review, approve and reject exist. Add history, export, bus-level KYC from the client portal.
Operators (`operators`)	Admin: operators	Partly built (operators)	operators	6 new / 2 existing	List, read, update, status exist. Add suspend and reactivate with reason and proof, message to owner, bank verification.
Banners (`banners`)	Admin: banners Customer: home banner Website: home	Done (admin-banners)	banners	5 new / 5 existing	Done. Add click and impression counts and the archive job.

Phase 1: your test checklist

- Admin, **Cities and places**: the list loads; try the filters (Search, City, Used for, Status); press Export CSV, Save, Import, View, Delete, Approve, Reject; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Places**: the list loads; try the filters (Search, City); press Send request, View, Export CSV; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Bus types and amenities**: the list loads; press Save, Export CSV, View, Delete, Add, Toggle Charging point; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Required documents**: the list loads; try the filters (Search, For); press Save, Export CSV, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **KYC questionnaire**: the list loads; try the filters (Search); press Export CSV, Save, Export config, View, Delete, Show checklist; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.

- Admin, **Invitations**: the list loads; try the filters (Search, Status); press Send invitation, Export CSV, Resend, Revoke, View, Save changes, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **KYC reviews**: the list loads; try the filters (Search, Status, Type); press Export CSV, View, History; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **KYC verification**: the list loads; press each button; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Company profile**: the list loads; try the filters (Company name *, Owner name *, Email *, Support phone); press Save, Save bank details; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Operators**: the list loads; try the filters (Search, Status, Payout mode); press Export CSV, View, Save changes, Delete, Suspend, Send, Save; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Banners**: the list loads; try the filters (Search, Status); press Export CSV, Save draft, Publish, Unpublish, Duplicate, Archive, View; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Website, **Home**: Search; Banners / offers / popular routes.

Phase 2. Supply and commercial terms

Goal	Buses, seat layouts, crew, routes, fares, schedules, trips, seat inventory, commission rules and payout settings.
Apps touched	backend, admin app, client portal
Done when	An operator runs the whole supply side alone and publishes a week of trips; admin sets commission and payout mode; the customer price is calculated on the server.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Buses and seat layouts (`buses`)	Admin: buses Client: buses, layouts	Partly built (buses, seat-layouts)	buses, busList	5 new / 3 existing	Buses exist. Layouts (versioned, decks, rows, seater and sleeper, ladies and blocked seats) are new.
Crew: drivers and conductors (`crew`)	Client: crew	New (operator-staff, drivers)	opStaff	4 new / 0 existing	Driver and conductor records with licence, expiry, phone and the trips assigned.
Routes, stops and boarding points (`routes`)	Client: routes	Partly built (routes, boarding-points)	routes	2 new / 2 existing	Routes exist. Add timed stops, boarding and dropping points, and copy-from-public-route.
Fares and schedules (`fares`)	Client: fares, schedules	New (fares, schedules)	fares, schedules	6 new / 0 existing	Two styles chosen by the operator: one price per trip, or per seat type and stop pair. Recurring schedules generate trips.
Trips and seat inventory (`trips`)	Admin: trips, inventory Client: trips	Partly built (trips, inventory)	trips	12 new / 3 existing	Create, publish and cancel exist. Add delay, driver change, seat blocking, release of expired holds, manifest.
Commission rules (`commission`)	Admin: commission Client: commission	New (commissions)	commissions, commissionDefault, commissionHist	8 new / 0 existing	Rule per operator, route or bus type; percent or flat; versioned; admin edits only; the operator sees it read-only; each booking stores the rule that applied.
Payout and fee settings (`payout_cfg`)	Admin: payout	New (settlements, payments)	payoutCfg, cancelSlabs	8 new / 0 existing	Default payout mode, cycle, minimum payout, hold days, convenience fee, refund slabs, per-operator mode and bank verification.

Phase 2: your test checklist

- Admin, **Buses**: the list loads; try the filters (Search, Status, Operator, Type); press Import buses, Export CSV, Mark maintenance, Retire bus, View, Save changes, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Buses**: the list loads; try the filters (Search, Status); each number card opens its records; press Send for approval, Import, Export CSV, Mark maintenance, View; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Seat layouts**: the list loads; try the filters (Bus); press Save layout; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Drivers and crew**: the list loads; try the filters (Search, Role, Status); each number card opens its records; press Add, Export CSV, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Routes and stops**: the list loads; try the filters (Search); press Add, Export CSV, View, Save, Copy as my route; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Fares**: the list loads; try the filters (Search, Status); press Save fare, Export CSV, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Schedules**: the list loads; try the filters (Search); press Save, Export CSV; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Trips**: the list loads; try the filters (Search, Status, Operator, Route); press Create draft, Export CSV, View, Save changes, Delete, 1A, Block; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Seat inventory**: the list loads; try the filters (Search, Operator, Occupancy); each number card opens its records; press Release expired holds, Export CSV, View, Save changes, Delete, Delay and notify, Save; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Trips**: the list loads; try the filters (Search, Status, When); each number card opens its records; press Create trip, Export CSV, Download manifest, View, Save changes; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Commission**: the list loads; try the filters (Search, Applies to, Status); press Save, Export CSV, Save rule, Save as new version, Deactivate, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Commission**: the list loads; try the filters (Your fare per seat (Rs), Route, Seat type); press Calculate; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Payout and fees**: the list loads; try the filters (Default mode for new operators, Settlement cycle (days), Minimum payout (Rs), Hold days after trip); press Save changes, Add, Remove, View, Delete, Export CSV, Save; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.

Phase 3. Customer booking journey

Goal	Search, trip details, seat map with a 10 minute hold, boarding and dropping, passengers, coupons, Razorpay payment, ticket, WhatsApp and email messages.
Apps touched	backend, customer app, admin app (bookings, payments, offers), client portal (bookings, offers)
Done when	A real customer books a real seat and pays in Razorpay test mode; the ticket, the messages and the operator notification all arrive; 500 concurrent requests for one seat give exactly one winner.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Search and trip details (`discovery`)	Customer: results, trip Website: home, results, routes	Partly built (trips (public search))	-	4 new / 4 existing	City lookup, search, filters and trip details exist. Add per-seat-type prices and the commission-inclusive price.
Seat selection and hold (`seats`)	Customer: seats	New (inventory)	-	3 new / 0 existing	10 minute hold on the server, at most 6 seats, atomic lock in Redis, expiry worker.
Booking, passengers and bookings lists (`booking`)	Admin: bookings Client: bookings Customer: points, pax, review	New (bookings, passengers)	bookings	16 new / 0 existing	Server-calculated amount, idempotent create, GST details, email required with the mobile number.
Offers and coupons (`offers`)	Admin: offers Client: offers Customer: offers Website: offers	New (coupons, offers)	offers	12 new / 0 existing	Admin owns offers; an operator can request one; who funds the discount is stored per offer.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Payments (`payments`)	Admin: payments Customer: pay	New (payments)	payments	8 new / 0 existing	Razorpay orders, verification, webhooks, late payment recovery, reconciliation tab.
Ticket, PNR and QR (`tickets_pdf`)	Customer: conf, ticket	New (tickets, check-in)	-	5 new / 0 existing	PNR, QR, PDF, calendar file, share, resend from admin.

Phase 3: your test checklist

- Customer app, **Search results**: Day chips; Sort / Filters / Apply; Trip card. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **Trip details**: Select seats; Bus no longer available. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Website, **Home**: Search; Banners / offers / popular routes.
- Website, **Search results**: Book in app.
- Website, **Popular routes**: Route page.
- Customer app, **Select seats**: Tap a seat; Continue (hold); Back / timer end; Live seat changes. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Admin, **Bookings**: the list loads; try the filters (PNR, name, phone, route, Status, Operator, Route); press Export CSV, View, Save changes, Cancel and refund, Resend ticket, Save; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Bookings**: the list loads; try the filters (Search, Status, Route); each number card opens its records; press Export CSV, View, Save changes; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Boarding and dropping**: Boarding / Dropping tabs; Open in maps; Continue. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **Passenger details**: Use my profile; Continue. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **Review booking**: Apply coupon; Remove coupon; Fare breakup; Proceed to pay. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Admin, **Offers and coupons**: the list loads; try the filters (Search, Status, Funded by); press Export CSV, Save offer, Pause, View, Delete, Resume, Check; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Offers**: the list loads; try the filters (Search); press Send to Abra, Export CSV, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Offers and promotions**: List; Terms. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Website, **Offers**: List.
- Admin, **Payments**: the list loads; try the filters (Search, Status, Mode); each number card opens its records; press Export CSV, View, Mark reconciled, Check with gateway, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Payment**: Pay now; Payment result; Retry / Change method. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **Booking confirmed**: Copy PNR; View ticket; Add to calendar; View the messages. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **E-ticket**: Call operator; Download PDF; Share trip. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.

Phase 4. After booking

Goal	My Trips, cancellation and refund, support tickets, profile, notification settings, account deletion, languages, information pages.
Apps touched	backend, customer app, admin app (refunds, tickets, customers, content), client portal
Done when	A cancelled booking refunds the right amount through the admin slabs and the seat returns to sale; a ticket raised with a proof file is answered by admin and seen by the customer.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
My Trips (`my_trips`)	Customer: trips tab, bk	New (bookings)	-	3 new / 0 existing	Upcoming, completed, cancelled, booking details.
Cancellation and refunds (`refunds`)	Admin: refunds Customer: cancel, refund	New (refunds)	refunds	6 new / 0 existing	Refund quote from the admin slabs, partial cancellation, refund worker, bulk approve, retry, reject with proof.
Support tickets (`support`)	Admin: tickets Customer: support, tk Website: help	New (support)	tickets, replies	12 new / 0 existing	Ticket with proof files, assignment, canned replies, time limit, reopen.
Customers and profile (`customers`)	Admin: customers Customer: profile, pedit, lang, del, me tab	New (customers)	customers	10 new / 1 existing	Profile, email required, block and unblock with proof, sign-out of all devices, account deletion.
Information pages and FAQs (`content`)	Admin: content Customer: info Website: help, contact	New (content)	faqs, pages, contact	11 new / 0 existing	Terms, privacy, about, FAQs and contact details edited by admin, shown in every app and on the website.

Phase 4: your test checklist

- Customer app, **Booking details**: Track; Cancel booking; Refund status. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Admin, **Refunds**: the list loads; try the filters (Search, Status, Paid by); each number card opens its records; press Export CSV, View, Mark paid; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Cancel booking**: Refund estimate; Cancel. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **Refund status**: Open. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Admin, **Support tickets**: the list loads; try the filters (Search, Status, Category, Priority); each number card opens its records; press Export CSV, View, Save changes, Delete, Assign, Mark resolved, Send reply; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Support (tab) and ticket**: FAQs; New ticket; My tickets; Reply; Contact support. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Website, **Help and FAQs**: FAQs.
- Admin, **Customers**: the list loads; try the filters (Name, phone, email, Status, Sign-in method); press Export CSV, View, Save changes, Delete, Block, Sign out all devices; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Profile (tab), edit, notification settings**: Save profile; Notification settings; Language; Sign out; Delete account. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Admin, **Help content and contact**: the list loads; try the filters (Search, Status, Category); press Export CSV, Save, Unpublish, View, Delete, Publish, Save draft; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Information pages**: Open page. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Website, **Help and FAQs**: FAQs.
- Website, **Contact us**: Send.

Phase 5. Safety, driver GPS and live tracking

Goal	The driver app (sign-in, trips, check-in, start and end, delay, GPS, SOS), live tracking for customer, admin and operator, SOS board, trip sharing, incidents, safety settings.
Apps touched	backend, driver app (new), customer app, admin app, client portal
Done when	A driver trip shows live on the customer, admin and operator maps; SOS reaches the safety team, the operator, the driver and the customer's contacts; sharing expires after the trip.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Driver app: trips and check-in (`driver_core`)	Driver: home, trip, profile	New (drivers, trip-operations, check-in)	opStaff, trips	10 new / 1 existing	Driver sign-in, assigned trips, passenger list, QR check-in, board, start, end, delay.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Live tracking (`tracking`)	Admin: tracking Client: live Customer: track Driver: gps Website: track	New (live-tracking)	live	4 new / 0 existing	GPS from the driver app every few seconds, stale-position handling, replay.
SOS and women safety (`sos`)	Admin: sos, safety Customer: safety Driver: sos	New (women-safety)	sos, safetyCfg	13 new / 0 existing	SOS raised by customer or driver, acknowledge, escalate, resolve or false alarm with proof, contacts alerted by SMS and WhatsApp.
Trip sharing (`sharing`)	Customer: share	New (women-safety)	sharing	4 new / 0 existing	Share link by WhatsApp, SMS or copy; stops at trip end; admin can revoke.
Incidents (`incidents`)	Admin: incidents	New (women-safety)	incidents	6 new / 0 existing	Complaints logged by support; severity, status, proof.

Phase 5: your test checklist

- Driver app, **Home (tab)**: Trip cards; Reports to send banner; SOS.
- Driver app, **Your trip (tab)**: Passenger list; Check in; Board; Start (opens pre-trip checklist); Delay.
- Driver app, **Profile**: Sign out.
- Admin, **Live tracking**: the list loads; try the filters (Operator, State); press TN38 EF 7890, AP16 CD 7939, TS07 EF 3887, AP39 AB 9725, KA01 AB 3469; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Live buses and safety**: the list loads; press each button; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Live tracking**: Live position; SOS and safety; Call driver or operator. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Website, **Track my bus**: Track.
- Admin, **SOS board**: the list loads; press Acknowledge, Save, Escalate, Resolve, Mark false alarm; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Safety settings**: the list loads; try the filters (Acknowledge within (minutes), Escalate to safety lead after (minutes), Police control room number); press Save, Add, Remove; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Customer app, **Women safety and SOS**: Hold SOS; I am safe; Add / remove contact; Status updates. Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Customer app, **Trip sharing**: Start sharing; WhatsApp / SMS / Copy; Stop sharing; Family view (public link). Everything shown comes from the real backend; nothing is mock, and it looks the same as the demo.
- Admin, **Incidents and sharing**: the list loads; try the filters (Search, Status, Severity); press Export CSV, Log, Save, View, Save changes, Revoke link, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.

Phase 6. Money out: settlements and payouts

Goal	Booking financials, settlement cycles, adjustments, holds, Razorpay Route automatic split and manual payout, GST statement, payout file, operator earnings screen.
Apps touched	backend, admin app, client portal
Done when	Settlement totals equal payments minus commission minus refunds, to the paisa, for every operator in the seed data.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Settlements and earnings (`settlements`)	Admin: settlements Client: earnings	New (settlements, reconciliation)	settlements	12 new / 0 existing	Generate, approve, hold, release, pay, adjust, query and reply, statements, payout file, GST statement.

Phase 6: your test checklist

- Admin, **Settlements**: the list loads; try the filters (Search, Status, Operator, Mode); each number card opens its records; press Generate, Payout file, GST statement, Export CSV, View, Release hold, Download statement; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Earnings and settlements**: the list loads; try the filters (Search); each number card opens its records; press Export CSV, Download statement, View, Save changes; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.

Phase 7. Rest of the admin portal

Goal	Dashboard, reports, announcements, popular routes, staff and roles, global search, AI assistant, demo-to-real clean-up. Every admin page is now live and no mock remains.
Apps touched	admin app, backend
Done when	All 38 admin pages pass the click-through against the real backend; reports match the transactions.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Dashboards and reports (`dashboard`)	Admin: dashboard, reports Client: dashboard	New (reports, analytics)	series	7 new / 0 existing	Every number card opens the records behind it. Reports render on the page and export to CSV, Excel and PDF.
Announcements and popular routes (`announce`)	Admin: announce, popular Website: home	New (content, popular-routes)	announce, popular	11 new / 1 existing	Broadcast to customers, operators or drivers; ordered popular routes.
Staff, roles and access matrix (`staff`)	Admin: staff Client: staff	Partly built (rbac, users)	staff, opUsers	9 new / 1 existing	Staff invitations, roles, and the matrix editable by Super Admin only. Operator staff are limited to their own company.

Phase 7: your test checklist

- Admin, **Dashboard**: the list loads; each number card opens its records; press Today, 7 days, 30 days, Export; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Reports**: the list loads; try the filters (Search); press 7 days, 30 days, 90 days, Export CSV, Excel (CSV), Export these rows, View; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Dashboard**: the list loads; each number card opens its records; press each button; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Announcements**: the list loads; try the filters (Search, Status); press Save draft, Send, Export CSV, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Admin, **Popular routes**: the list loads; press Add, Export CSV, View, Delete; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Website, **Home**: Search; Banners / offers / popular routes.
- Admin, **Staff and roles**: the list loads; try the filters (Search, Role, Status); press Export CSV, Send invitation, Save, Deactivate, View, Activate, Add role; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Staff and roles**: the list loads; try the filters (Search); press Export CSV, Send invitation, View, Add role; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.

Phase 8. Rest of the client (operator) portal

Goal	Operator dashboard, staff and roles, earnings, live view, offers requests, anything left. The portal is the operator app: there is no separate operator app.
Apps touched	client portal, backend

Done when	All 20 client pages pass the click-through; an operator can only ever see their own data.

Phase 8: your test checklist

- Backend only: Claude shows you the results of the automatic tests; there is nothing to press.

Phase 9. Driver app in full: rules, reports and checklists

Goal	Operator-maintained pre-trip and end-of-trip checklists, daily and weekly reports built from system numbers, incident, bus problem and expense reports, review by the operator, minimum rules set by Abra.
Apps touched	driver app, client portal, admin app, backend
Done when	A driver cannot start a trip past a blocking check; a daily report shows the system numbers and the owner approves or sends it back.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Driver rules and trip checklists (`driver_rules`)	Admin: drules Client: drules Driver: pre-trip checklist, end-trip checklist	New (drivers, trip-operations)	ruleDefaults, driverRules, tripChecks	25 new / 0 existing	Abra defaults and a locked minimum; each operator edits its own copy; blocking, warning and automatic checks.
Driver reports (`driver_reports`)	Admin: dreports Client: dreports Driver: reports tab	New (drivers, reports)	driverReports	9 new / 0 existing	Daily and weekly reports use numbers computed by the system and frozen at send; the driver adds fuel, expenses, incidents and notes; the owner approves or sends back.

Phase 9: your test checklist

- Admin, **Driver rules (defaults)**: the list loads; press Push the minimum to every operator, Push, Export CSV, Add, Save, Remove, Add this question; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Driver rules**: the list loads; press Reset to Abra defaults, Reset, Export CSV, Add, Save, Remove, Add this question; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Driver app, **Before you start**: Automatic checks; Answer items; Start trip.
- Driver app, **End of trip**: End trip.
- Admin, **Driver reports**: the list loads; try the filters (Driver or report, Status, Report, Driver); each number card opens its records; press Export CSV, Tell the operator; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.
- Client portal, **Driver reports**: the list loads; try the filters (Driver or report, Status, Report, Driver); each number card opens its records; press Export CSV, Approve, Remind the driver; export downloads; as a view-only role the buttons are hidden and the server refuses; the screen looks the same as the demo page.

Phase 10. Public website, then partner app

Goal	Next.js route and city pages with SEO, search hand-off to the customer app, help, offers, track-my-bus, contact and partner enquiry. The partner app and partner API come last.
Apps touched	public website, backend, partner app
Done when	Route pages are indexed and booking opens in the customer app; partner and customer bookings cannot take the same seat.

Feature	Screens in the demo	Backend today	Collections	Endpoints (new / existing)	What to build
Public website (`website`)	Website: about, partner, contact, footer pages	New (public API, content)	-	3 new / 0 existing	Next.js pages with SEO; partner and contact enquiries land in admin.
Partner app and partner API (`partner`)	backend only	New (partners, api-clients, partner-webhooks)	-	0 new / 0 existing	Last: API clients, scopes, signed requests, webhooks.

Phase 10: your test checklist

- Website, **Partner with us**: Send.
- Website, **Contact us**: Send.
- Website, **About, Terms, Privacy**: Open.

Phase 11. Languages, hardening and launch

Goal	All 8 languages reviewed by native speakers, load and security tests, backup restore test, staging run, store release.
Apps touched	all
Done when	Acceptance criteria in the functional PDF are met and signed off.

Phase 11: your test checklist

- Backend only: Claude shows you the results of the automatic tests; there is nothing to press.

Part 5. Rules that apply to every feature

Area	Rule
UI matches the demo	Every screen is built to look and behave like its demo page: layout, wording, colours, spacing, icons, dialogs, states and responsive behaviour at phone, tablet and desktop widths. The demo files in `prototypes/` are opened side by side with the real screen during the check. Only real data replaces the sample data.
Access	Server-side permission check on every endpoint using the matrix (None, View, Edit, Edit+Delete). A view-only user can open and copy but not change; the buttons are hidden **and** the server refuses.
Audit	Every create, edit, delete, status change, block, refund, payout and permission change writes an audit entry: who, what, when, from where, before and after.
Proofs	SOS resolve, false alarm, refund reject, trip cancel, booking cancel, customer block, settlement hold, incident, ticket replies, delay and driver override need a proof (file or note) as the demos show. Stored with the record.
Money	Integers in paise. One rounding function used everywhere. Customer price = fare + commission + convenience fee + GST, always calculated on the server, never taken from the app. Commission changes never alter bookings already made.
Time	India time everywhere; trips across midnight; cut-off before departure; travel date versus departure date.
Idempotency	Every create that costs money or changes stock takes an idempotency key. A double tap returns the same booking.
Seat stock	Held, sold and blocked seats are separate; 10 minute hold; expiry worker; the booked count always equals the real bookings.
Lists	Server-side paging, stable sorting, the filters shown in the demo, search, export (CSV, Excel, PDF where the demo has it) limited by the same filters and permissions. Table and list view are two views of the same data.
Import	Excel import with a template download, a dry run, row-by-row errors, then save (places, buses, schedules, fares).
Delete	Soft delete wherever a record is referenced. Deleting a place used by trips is refused.
Notifications	Created by the event service, never by the app. A provider failure never fails the business action. No duplicates for one event. See Appendix B for every event, recipient and message.
Messages	WhatsApp and email after booking, cancellation, refund, payment failure, delay and boarding; logged; resendable by admin.
Languages	All text through one translation system; 8 languages; names shown in English; the assistant answers in the language the user used.

Area	Rule
Number cards	Each card links to the filtered list behind it; the count and the list always agree.
Maintenance and switches	Enforced by one global guard on every request. Every app sends the header `X-Abra-App: customer | operator | driver | partner | admin | website` so a request without a token can be placed; signed-in people are placed by their account. `GET /maintenance/status` and `GET /features` are public and always work, so an app can show the message and hide what is off. Staff with `maintenance.bypass` (and the Super Admin) are never locked out of the admin portal. A feature that can be switched off carries `@RequireFeature('key')`.
Realtime	GPS, SOS, notifications, seat changes and report status arrive by Socket.IO; the screen updates without a reload.
Privacy	Phone numbers and emails are masked in lists for roles that only need to see the record. Account deletion keeps past financial records but removes personal data.
Errors	One error format with a code and a human sentence; the app shows the sentence, not the code.
Security	Secrets only in git-ignored environment files; rate limits on OTP and login; webhooks verified by signature; files served by short-lived signed links; no trust of amounts from the app.

Part 6. Testing plan

Layer	What	Who
Unit	Pricing, commission, cancellation and refund maths, settlement totals, driver report numbers, rule evaluation	Claude, automatic
Endpoint	Every endpoint in Appendices D to H: allowed role works, forbidden role is refused, bad input gives a clear error, an audit entry is written	Claude, automatic (Jest end-to-end)
Concurrency	500 parallel requests for one seat give exactly one winner; hold expiry frees seats	Claude, automatic
Payments	Duplicate webhook, failed, pending, expired-then-paid, amount mismatch	Claude, automatic
Apps	`flutter analyze` clean and existing tests green	Claude, automatic
Screen walk	The same click-through that passes on the demo (every page, every button) must pass on the real app. The demo test scripts are the acceptance spec.	Claude runs a minimum; **you test with the checklist**
Visual	Every new screen is compared side by side with its demo page (layout, wording, colours, dialogs, states) and checked against the design system at phone, tablet and desktop widths. A screen that differs from the demo is not done.	Claude, then you
User test	At the end of each phase you receive a short checklist in plain language: what to open, what to press, what you should see	You

> **Rule:** To keep effort and cost sensible, Claude runs only the minimum self-tests and hands you the checklist; you do the full manual test of each phase. Nothing moves to the next phase until you say so.

Definition of done for one feature

1. The three files exist and nothing else was needed.
2. Every button in the demo page has a working endpoint (Appendices D to H show them).
3. Permissions, audit, notifications and messages work as described.
4. Loading, empty, error, no-permission and offline states show.
5. The screen matches the demo page side by side: same layout, wording, buttons, colours, dialogs and fields.
6. Automatic tests pass; analyzer clean.
7. You tested the checklist and said it is good.
8. Appendix J is updated (status, date).

Part 7. Open items and who owns them

Needed	Phase	From
Redis running (Docker is installed)	0	Claude to set up; you confirm Docker Desktop is running
New MongoDB Atlas password (the current one was shown in a screenshot and in chat) and your IP allowed in Atlas	0	You
Razorpay account: test and live keys, webhook secret, Route enabled for splits	3, 6	You
Google Maps (or Mappls) provider decision and key, from the Google Cloud project with billing and a budget alert	5	You
Groq API key for the assistant	0 or 7	You
WhatsApp provider credentials, SMS provider (DLT registration), email provider	3 onward	You
Cancellation slabs and refund percentages (defaults are in the demo; operators may override)	4	You and operators
Who funds coupon discounts: platform or operator	3, 6	You
Hosting provider and a real storage bucket for documents	11 (staging earlier)	You
Safety escalation contacts and legal review	5	You
Inter font files and official logos (Google, Apple, WhatsApp, UPI, banks) and the Abra logo as transparent PNG or SVG	any	You
Native-speaker review of Kannada, Tamil, Malayalam, Marathi and Bengali (machine-written so far); Telugu and Hindi also reviewed	11	You
Decision: where proof files live (MongoDB GridFS or a cloud bucket such as S3)	0	You
Report reminders on a timer (the demo only reminds on a button press)	9	You to confirm

Part 8. How we work

- One phase at a time. At the end of each phase you get the checklist; the next phase starts when you say so. This document is updated after each phase with the status and the date.
- You prefer short, simple English. When you say "just tell me", Claude explains and does not code. When you give a clear build instruction, Claude builds.
- You test; Claude keeps self-testing to the minimum and gives you the checklist.
- The login files of the customer app are not edited without asking.
- Secrets live in git-ignored files, never in code, documents or chat.
- No fake or sample data ships to production; development seed data stays behind a development switch.
- The demos in `prototypes/` stay as the specification and are not deleted. `prototypes/_build/build_client.py` is stale and must not be run.
- Keep the three-file rule (Part 3) in every new feature.

Part 9. Progress log

Kept up to date as work is done. Last updated 13 October 2026. Unit tests: 730 (729 pass; one queue test fails now and then). End-to-end tests: every new suite passes; the 15 older failures are fixed. Phase 0 checklist: 27 of 27.

9.1 Phase 0 (main folder, not committed yet)

Phase	What	Status	How it was checked	Where
0	MongoDB Atlas connected; the database accepts only the two PC addresses	Done	Connection tested from the backend; the open "anywhere" entry was removed	backend/.env.development (git-ignored)
0	Redis in Docker, backend starts with MongoDB, Redis, queues and outbox up	Done	/health/ready answers ready	infrastructure/docker-compose.development.yml
0	Super Admin account created	Done	Signed in over HTTP; the bootstrap lines were then deleted from the env file	npm run bootstrap:super-admin
0	Access matrix (None, View, Edit, Edit and delete per role per page), Super Admin only, audited	Built and tested	19 unit tests; tested against Atlas: create, change and delete a role, the Super Admin row refuses changes	rbac/access_matrix_admin_manage*.ts, rbac/permissions/access-pages.ts
0	Messages log: admin list, filters, read, resend; customer "Messages sent to me"; the sender other features call	Built and tested	10 unit and 11 end-to-end tests	messages/messages_admin_manage*.ts, messages_customer*.ts
0	Integration settings: Razorpay, Maps, SMS, Push, AI assistant (secrets encrypted, never returned), connection tests	Built and tested	16 unit and 9 end-to-end tests; real provider calls tested only against a fake network	system-settings/integrations_admin_manage*.ts
0	Assistant endpoint (Groq through the backend, tools by role, proposed changes need confirmation, development mode without a key)	Built and tested	23 unit and 16 end-to-end tests; real Groq not tested (no key)	chatbot/assistant_chat*.ts
0	Live-update channel (sockets): token check, rooms, followable topics, publisher for API and worker, re-check every minute	Built and tested	29 unit and 13 end-to-end tests with real socket connections	realtime/live_updates_channel*.ts
0	Feature flags (9 switches) and maintenance (5 scopes, message, window) enforced by one global guard	Built and tested	31 unit and 15 end-to-end tests	feature-flags/flags_admin_manage*.ts
0	Demo seed: 20 cities, 52 places, 6 bus types, 14 companies, 31 buses	Built, tested locally; NOT loaded into Atlas by decision	18 unit tests; run for real on a local database and twice (adds nothing the second time). Decision of 13 October 2026: no sample data goes into Atlas; real data is created by testing through the apps. The seed stays a tool for throw-away local databases only	npm run seed:demo, backend/src/seed, backend/seed/demo-data.json
0	15 end-to-end tests that were failing before any of this work	Fixed	Cause: test operators had no active company record, which the login guard now requires. Test helper fixed; one test now checks the real data-scope protection	rbac/tests/rbac-e2e.harness.ts, authorization-isolation.e2e-spec.ts
0	Phase 0 checklist (27 steps) run end to end over HTTP	All 27 pass	On a temporary local database with a throw-away admin, then deleted (step J, the sample data, is skipped on Atlas). Rerun on Atlas with your password	backend/scripts/phase0-checklist.mjs

9.2 Phase 1 (other session, branch `phase1-catalogue-kyc`, worktree `D:/New_project/abra-wt-p1`, not merged)

Phase	What	Status	How it was checked	Where
1	Step 0 and 1: permission keys for delete and export, shared soft delete and restore, audit of delete and restore, "in use" refusals	Built and tested (branch)	catalogue-shared end-to-end suite passes (33)	modules/catalogue-shared/catalogue_shared_manage*.ts
1	Step 2: banner duplicate, archive and reorder; banners shown higher-order first with audience and colour style as in the demo; read-one routes; invitation edit and expiry	Built and tested (branch)	banners end-to-end 28 pass; invitations suite passes; admin app 144 tests pass	banners_admin_manage*, invitations_admin_manage*
1	Step 3: server-side export (CSV and Excel) for 8 lists; import with template and dry run for cities and places	Built and tested (branch)	export 15 and import 10 end-to-end tests pass	catalogue-shared; exceljs added to package.json
1	Step 4: amenities (add, switch on and off, bus forms use the active ones)	Built and tested (branch)	amenities 8 end-to-end tests pass; admin app 150 tests pass	modules/amenities/amenities_admin_manage*.ts

Still to do in Phase 1:

- Step 5: place requests (operator asks, admin approves or rejects) and the three-tab Places page
- Step 6: operator detail page with tabs; suspend with reason and proof; reactivate; message the owner
- Step 7: payout mode and bank details with verification
- Step 8: company profile and bank page in the client portal
- Step 9: bus KYC from the client portal, KYC history
- Step 10: compare the KYC questionnaire with the demo; export of the questionnaire
- Import with a dry run for bus types, document types and banners (they still use the old on-device CSV import)
- Merge the Phase 1 branch (worktree abra-wt-p1) with the Phase 0 work in the main folder

9.3 What is not done or not proven

Item	Detail
Real outside services	Razorpay, Google Maps, Groq, SMS, Firebase push and WhatsApp were only tested against a fake network. Keys are needed (Part 7).
Atlas for the newer features	Messages, integration settings, assistant, feature flags, maintenance and live updates were tested on a local database. The checklist script can run them against Atlas with your password.
Event router and notification demo	The plan's Phase 0 goal "a notification travels from the backend to a phone and to the admin bell" is not shown yet: the notification and socket machinery exists and is tested, but no business event feeds it until bookings, trips and refunds exist.
Audit log page	The read endpoint exists. Filters, the detail view and export from the demo are still to build (Phase 7).
Proof files	The storage decision (existing files service, local in development, a bucket before staging) is recommended but not yet built into the proof boxes.
Apple sign-in	Not built.
Flutter apps	No app calls any of the Phase 0 endpoints yet. The admin app is being wired to Phase 1 on its own branch.
Assistant details	Answers arrive whole, not word by word. Feature tools (bookings, trips) are added with those features. Behaviour with the real Groq model, including attempts to break its rules, is untested.

9.4 Known problems

Problem	Detail
Docker Desktop is unreliable on this PC	It stopped twice and once crashed during the Redis-restart test. Database tests need it. Start it first and wait for "running".
Two tests are timing-sensitive	A Super Admin bootstrap test that starts several runs at once (passes about 2 of 3 times alone) and a queue test (passes about 3 of 4). Neither is in code changed in Phase 0.
Redis-restart test	It stops and starts the Redis container itself and can bring Docker down. Run it alone, or leave it out of a full run.
Branch merge	Both branches changed the same shared files (permission list, audit action list, module list, eslint config). Keep both sides; the cross-session notes list each line. The Dart permission parity test needs messages.read, messages.resend and maintenance.bypass.
customer_app analyze	Shows 3 informational lines on the untouched main branch.
Atlas password	The database password was shown in a screenshot and in chat. It still needs resetting; then update backend/.env.development.

9.5 What happens next

1. You run the Phase 0 checklist (Part 4) or tell Claude to run it against Atlas with your password.
2. No sample data is loaded into Atlas (your decision): the first real data is created through the apps while you test.
3. You say "Phase 0 is good". Then the two branches are merged and Phase 1 continues with step 5.

4. You reset the Atlas password and supply the keys listed in Part 7 when each phase needs them.

Appendix A. Collections (the database)

The demo store has 62 keys. This table maps each to a real MongoDB collection. Collection names are plural snake case. Every collection also gets `_id`, `createdAt`, `updatedAt`, `createdBy`, and `deletedAt` where soft delete applies. Operator-owned collections carry `operatorId` and every query for an operator user is filtered by it on the server. Money fields are integers in paise. "Demo fields" are the fields of the first record in the demo store, with their types.

Demo key	Collection	Backend today	Phase	Owner module	Purpose	Relations	Indexes
users	`users`	exists	0	users	Every person who can sign in: admin staff, operator users, drivers, customers (identity only).	roles, sessions	email unique, phone unique
(sessions)	`sessions`	exists	0	sessions	Signed-in devices; used by "sign out of all devices".	users	userId, expiresAt (TTL)
customers	`customers`	new	4	customers	Traveller profile: name, email (required), phone, sign-in method, status, language, emergency contacts, notification settings.	users, bookings	phone, email, status
operators	`operators`	exists	1	operators	Bus companies: owner, contact, GST, bank details (with verification), payout mode, status, KYC state, rating.	buses, trips, settlements, commissions	status, name text
opUsers	`operator_users`	new	8	operator-staff	Operator portal users and their roles.	operators, users	operatorId+email
opStaff	`operator_crew`	new	2	operator-staff, drivers	Drivers and conductors: licence, licence expiry, phone, status.	operators, trips	operatorId+role, licExp
cities	`cities`	exists	1	cities	Cities with state, popular flag, Telugu and Hindi names and the other language names.	places	name text, active
places	`places (locations)`	exists	1	locations	Boarding and dropping places with address, landmark, latitude, longitude, phone and instructions.	cities, routes	cityId, type, active
placeReqs	`place_requests`	new	1	locations	An operator asks Abra to add a place; approve or reject with a reason.	operators, places	status
busTypes	`bus_types`	exists	1	bus-types	AC Sleeper, Seater and so on, with a permanent key.	buses	key unique
amenities	`amenities`	new	1	bus-types	Charging point, blanket, water and so on.	buses	name unique
docTypes	`document_types`	exists	1	document-types	Required KYC documents for operator or bus: file types, size, count, expiry, instructions, order.	kyc_submissions	applies, active
questions	`kyc_questions`	new	1	onboarding	KYC questionnaire items with answer type and options.	kyc_submissions	applies, order
rules	`kyc_document_rules`	new	1	onboarding	Conditions that make a document required (for example GST when turnover is above a limit).	document_types, questions	active
invitations	`operator_invitations`	exists	1	onboarding	Invitation to an operator: company, owner, email, status, expiry.	operators	email, status
submissions	`kyc_submissions`	exists	1	onboarding	Company or bus KYC: documents with review state and rejection reason, answers, history.	operators, buses, files	operatorId, status, type
buses	`buses`	exists	2	buses	Registered buses: registration, type, seats, layout, insurance expiry, status (pending, active, maintenance, retired).	operators, seat_layouts, trips	operatorId, reg unique, status
(layouts)	`seat_layouts`	new	2	seat-layouts	Versioned layout: decks, rows, columns, seater and sleeper, ladies seats, blocked seats, seat numbers and seat category.	buses, trips	busId+version
routes	`routes`	exists	2	routes, boarding-points	Route with timed stops, boarding and dropping points, distance and hours.	places, trips	operatorId, from+to
fares	`fares`	new	2	fares	Fare rules per route and seat type: weekday, weekend, festival, date ranges, stop pairs.	routes, operators	operatorId+routeId
schedules	`schedules`	new	2	schedules	Recurring trips: route, bus, time, days, date range, fare; generates trips.	routes, buses, trips	operatorId, status
trips	`trips`	exists	2	trips	A dated departure: bus, driver, conductor, status, delay, fares by seat category, counts.	routes, buses, seat_inventory, bookings	depart, status, operatorId, routeId
(inventory)	`seat_inventory`	new	2	inventory	One record per seat per trip: free, held, sold, blocked; holds also live in Redis with a 10 minute expiry.	trips, bookings	tripId+seat unique
commissions	`commission_rules`	new	2	commissions	Rules per operator, route or bus type; percent or flat; version; effective date; agreed note.	operators	scope+target, status
commissionDefault	`commission_default (in payout_settings)`	new	2	commissions	The fallback rule.	-	-
commissionHist	`commission_history`	new	2	commissions	Every change to a rule: who, when, before, after.	commission_rules	ruleId
payoutCfg	`payout_settings`	new	2	settlements	Default payout mode, cycle days, minimum payout, hold days, convenience fee, GST percent, coupon funding.	-	singleton
cancelSlabs	`cancellation_slabs`	new	2	refunds	Refund percent by hours before departure.	refunds	hours
bookings	`bookings`	new	3	bookings, passengers	A booking with passengers, seats, boarding and dropping points, amounts in paise (fare, commission, fee, GST, discount), commission rule snapshot, contact, GST details, status.	customers, trips, payments, refunds	pnr unique, customerId, tripId, operatorId, status
payments	`payments`	new	3	payments	Gateway order and payment: mode, status, gateway ids, failure reason, settled flag, reconciliation state.	bookings	bookingId, gatewayOrderId unique, status
offers	`offers`	new	3	coupons, offers	Coupons and offers: code, type, value, maximum discount, minimum amount, validity, limits, funder (platform or operator), terms.	bookings	code unique, status
refunds	`refunds`	new	4	refunds	Refund of a cancellation: amount, percent, bearer (platform or operator), status, gateway reference, reason, proof.	bookings, payments	bookingId, status
tickets	`support_tickets`	new	4	support	Customer support tickets with category, priority, assignee, time limit and the message thread with proof files.	customers, bookings	status, assignee, customerId
faqs	`faqs`	new	4	content	Help centre questions.	-	status
pages	`cms_pages`	new	4	content	Terms, privacy, about and other pages with versions.	-	slug unique
contact	`contact_info (in system_settings)`	new	4	content	Support phone, WhatsApp, email, hours, emergency number.	-	singleton
banners	`banners`	exists	1	admin-banners	Home banners with window, placement, audience, priority, impressions, clicks.	offers	status, placement
announce	`announcements`	new	7	content	Broadcasts to customers, operators or drivers.	-	status
popular	`popular_routes`	new	7	popular-routes	Ordered routes shown on home and the website with search and booking counts.	routes	order
sos	`sos_alerts`	new	5	women-safety	SOS raised by a customer or a driver: trip, bus, driver, operator, location, contacts alerted, status log, assignee, proof on resolve.	trips, customers	status, tripId, at
incidents	`incidents`	new	5	women-safety	Complaints logged by support: severity, against, status, notes, proof.	trips, customers	status, severity
sharing	`trip_shares`	new	5	women-safety	Live trip links shared with family: token, opens, expiry, status.	bookings	token unique, expiresAt (TTL)
safetyCfg	`safety_settings`	new	5	women-safety	Acknowledge and escalate minutes, police number, safety team, SMS and WhatsApp switches, SOS hold seconds.	-	singleton

Demo key	Collection	Backend today	Phase	Owner module	Purpose	Relations	Indexes
live	`live_positions (latest in Redis, history in Mongo)`	new	5	live-tracking	Bus position, speed, progress, ETA, checked-in count.	trips	tripId+at
settlements	`settlements`	new	6	settlements	Per operator per cycle: gross, commission, refunds, adjustments, net, mode, status, lines, query thread, hold reason, payment reference.	operators, bookings	operatorId+cycle unique, status
staff	`admin_staff (users with admin roles)`	exists	7	users, rbac	Admin team: role, MFA state, last sign-in, overrides to the matrix.	users	email
(matrix)	`roles (fields pageLevels, operatorId)`	exists	0	rbac	Level per role per page: None, View, Edit, Edit+Delete, stored on the role and turned into its permission keys. Editable by Super Admin only. Built 2026-10-12.	roles	role+page unique
audit	`audit_logs`	exists	0	audit-logs	Who did what, when, from where, with before and after.	users	at, who, target
flags	`feature_flags`	new	0	feature-flags	Switches such as maintenance scopes and optional features.	-	key unique
maint	`maintenance (in system_settings)`	new	0	maintenance	Maintenance message, window and which apps it affects.	-	singleton
settings	`system_settings`	exists	0	system-settings	Company, booking, integrations and security settings. Secrets are encrypted.	-	key unique
notifs / inbox	`notifications`	exists	0	notifications	In-app notifications for admin, operator, customer and driver; read state; deep link.	users	userId+read, at
messages	`messages`	new	0	whatsapp, notifications	Log of WhatsApp and email messages: channel, address, subject, body, booking, status, resend.	bookings, customers	bookingId, at
ruleDefaults / driverRules	`driver_rule_sets`	new	9	drivers	Abra default checklists and report types plus one editable copy per operator; locked items cannot be removed.	operators	operatorId unique
driverReports	`driver_reports`	new	9	drivers, reports	A driver's report: type, period, answers, files, frozen system snapshot, status, review comment.	operators, opStaff	operatorId+driver+type+period unique, status
tripChecks	`trip_checks`	new	9	trip-operations	Pre-trip and end-of-trip checklist results with items, odometer, warnings and override with reason and proof.	trips, opStaff	tripId+kind
series	`(not stored)`	new	7	reports, analytics	Dashboard series: computed from bookings and payments, cached in Redis.	-	-
(files)	`files`	exists	0	files	Uploaded files and proof attachments with owner, type, size and access list.	any	ownerId

A.2 Demo fields per collection

Demo key	Fields (name:type)
cities (20 in the demo)	id:string, name:string, state:string, active:boolean, popular:boolean, te:string, hi:string, places:number
places (52 in the demo)	id:string, city:string, name:string, address:string, landmark:string, lat:string, lng:string, phone:string, type:string, source:string, active:boolean, trips:number
placeReqs (6 in the demo)	id:string, operator:string, city:string, name:string, address:string, status:string, at:string, reason:string
busTypes (6 in the demo)	id:string, name:string, key:string, ac:boolean, berth:string, active:boolean, buses:number
amenities (10 in the demo)	id:string, name:string, active:boolean
operators (14 in the demo)	id:string, name:string, owner:string, email:string, phone:string, city:string, status:string, buses:number, trips:number, joined:string, payout:string, gst:string, rating:string, bank:object, kyc:string, fund:string
invitations (6 in the demo)	id:string, company:string, owner:string, email:string, status:string, sentAt:string, expiresAt:string, email_status:string
submissions (6 in the demo)	id:string, operatorId:string, operator:string, type:string, bus:string, status:string, submittedAt:string, docs:array, answers:array, history:array
buses (31 in the demo)	id:string, operatorId:string, operator:string, reg:string, name:string, type:string, ac:boolean, seats:number, status:string, insExp:string, layout:string, added:string, decks:number, ladies:array
routes (12 in the demo)	id:string, from:string, to:string, name:string, km:number, hours:number, operatorId:null, status:string, stops:array
trips (64 in the demo)	id:string, routeId:string, route:string, operatorId:string, operator:string, busId:string, bus:string, busName:string, type:string, depart:string, status:string, seats:number, booked:number, held:number, blocked:number, fare:number, driver:string, conductor:string, taken:array, checkedIn:array, ladies:number, blockedSeats:array
customers (36 in the demo)	id:string, name:string, email:string, phone:string, status:string, joined:string, bookings:number, spent:number, method:string, city:string, lastActive:string
bookings (326 in the demo)	id:string, pnr:string, customerId:string, customer:string, phone:string, tripId:string, route:string, operator:string, operatorId:string, travelDate:string, seatList:string, seatsN:number, amount:number, commission:number, gst:number, fee:number, discount:number, status:string, mode:string, at:string, coupon:string
payments (326 in the demo)	id:string, bookingId:string, pnr:string, customer:string, operator:string, amount:number, mode:string, status:string, gw:string, at:string, reason:string, settled:boolean
refunds (5 in the demo)	id:string, bookingId:string, pnr:string, customer:string, operator:string, amount:number, pct:number, status:string, utr:string, at:string, bearer:string, reason:string
settlements (11 in the demo)	id:string, operatorId:string, operator:string, cycle:string, gross:number, commission:number, refunds:number, adj:number, net:number, status:string, mode:string, lines:number, paidAt:null, query:string
commissions (7 in the demo)	id:string, scope:string, target:string, mode:string, value:number, note:string, from:string, version:number, status:string, by:string
commissionDefault (2 in the demo)	mode:string, value:number
commissionHist (4 in the demo)	0:string, 1:string, 2:string, 3:string
payoutCfg (8 in the demo)	defaultMode:string, cycleDays:number, minPayout:number, holdDays:number, razorpayRoute:boolean, gstPct:number, fee:number, couponFunding:string
cancelSlabs (5 in the demo)	id:string, hours:number, pct:number
docTypes (12 in the demo)	id:string, key:string, name:string, applies:string, required:boolean, expires:boolean, types:string, maxMb:number, maxFiles:number, active:boolean, order:number, instr:string
questions (6 in the demo)	id:string, key:string, label:string, applies:string, type:string, opts:string, required:boolean, active:boolean, order:number
rules (5 in the demo)	id:string, name:string, doc:string, cond:string, required:boolean, active:boolean
tickets (10 in the demo)	id:string, customer:string, customerId:string, subject:string, category:string, priority:string, status:string, assignee:string, at:string, sla:string, bookingId:string, msgs:array
faqs (8 in the demo)	id:string, q:string, cat:string, status:string, a:string, updated:string, views:number
pages (5 in the demo)	id:string, title:string, status:string, version:number, updated:string
contact (5 in the demo)	phone:string, whatsapp:string, email:string, hours:string, emergency:string
banners (6 in the demo)	id:string, title:string, sub:string, code:string, status:string, style:number, from:string, to:string, placement:string, audience:string, priority:number, imp:number, clicks:number, link:string
offers (8 in the demo)	id:string, code:string, desc:string, type:string, value:number, maxDisc:number, minAmt:number, status:string, funder:string, perUser:number, limit:number, used:number, from:string, to:string, scope:string, newOnly:boolean, terms:array
announce (4 in the demo)	id:string, title:string, body:string, audience:string, status:string, channel:string, at:string, reach:number
popular (8 in the demo)	id:string, route:string, order:number, active:boolean, searches:number, bookings:number
sos (5 in the demo)	id:string, customer:string, phone:string, trip:string, route:string, bus:string, driver:string, operator:string, status:string, at:string, note:string, x:number, y:number, contacts:array, log:array, assignee:string
incidents (5 in the demo)	id:string, title:string, customer:string, against:string, severity:string, status:string, at:string, trip:string, notes:string
sharing (10 in the demo)	id:string, customer:string, trip:string, link:string, opened:number, created:string, expires:string, status:string
safetyCfg (8 in the demo)	ackMinutes:number, escalateMinutes:number, police:string, team:array, smsOn:boolean, whatsOn:boolean, notifyOperator:boolean, sosHold:number
live (5 in the demo)	tripId:string, route:string, bus:string, operator:string, driver:string, state:string, ping:string, speed:number, progress:number, eta:string, x:number, y:number, pax:number, checkedIn:number, stops:array
staff (8 in the demo)	id:string, name:string, title:string, role:string, status:string, email:string, mfa:boolean, last:string, overrides:number

Demo key	Fields (name:type)
audit (60 in the demo)	id:string, at:string, who:string, role:string, action:string, target:string, ip:string, sensitive:boolean
flags (9 in the demo)	key:string, name:string, on:boolean, desc:string
maint (4 in the demo)	scopes:object, msg:string, from:string, to:string
settings (4 in the demo)	company:object, booking:object, integrations:object, security:object
notifs (6 in the demo)	id:string, t:string, b:string, i:string, go:string, at:string, read:boolean
opStaff (70 in the demo)	id:string, operatorId:string, name:string, role:string, phone:string, licence:string, licExp:string, status:string, joined:string, pin:string
fares (43 in the demo)	id:string, operatorId:string, routeId:string, route:string, seatType:string, weekday:number, weekend:number, festival:number, from:string, status:string
schedules (8 in the demo)	id:string, operatorId:string, routeId:string, route:string, busId:string, bus:string, time:string, days:array, from:string, to:string, status:string, fare:number
inbox (3 in the demo)	id:string, to:string, t:string, b:string, go:string, at:string, read:boolean
opUsers (56 in the demo)	id:string, operatorId:string, name:string, email:string, role:string, status:string, last:string
ruleDefaults (3 in the demo)	pre:array, end:array, reports:array
driverRules (14 in the demo)	operatorId:string, updated:string, pre:array, end:array, reports:array
driverReports (4 in the demo)	id:string, operatorId:string, driver:string, typeId:string, typeName:string, period:string, status:string, answers:object, files:object, submittedAt:string, dueAt:string, late:boolean, reviewComment:string, reviewedBy:string, snapshot:object
tripChecks (12 in the demo)	id:string, operatorId:string, tripId:string, route:string, driver:string, kind:string, at:string, items:array, odometer:number, result:string, warnings:array, override:null
messages (18 in the demo)	id:string, at:string, to:string, name:string, address:string, channel:string, kind:string, subject:string, body:string, bookingId:string, status:string

Appendix B. Events, notifications and messages

These are the rules the demo's event router applies. In the real build the backend event service does exactly this, once per event, and never from the app. "Go to" is the page or screen the notification opens.

Event	Who is told	Title	Message	Go to
Operator sends KYC documents (company or bus)	Admin	New KYC submission	<operator> sent company documents / documents for bus <reg>	KYC detail
Admin approves KYC	Operator	Company approved / Bus approved	Verified; buses, trips and staff are open / you can schedule trips on the bus	Dashboard / Buses
Admin rejects KYC	Operator	Documents need fixing	Each rejected document with its reason	KYC
Operator resends corrected documents	Admin	KYC corrections sent	<operator> replaced the rejected documents	KYC detail
Operator requests a place	Admin	Place request	<operator> wants to add <place>, <city>	Places
Admin approves or rejects a place request	Operator	Place request approved / rejected	Can be used in routes / reason	Routes
Settlement status changes	Operator	Settlement <status>	<id> for <cycle>	Earnings
Operator raises a settlement query	Admin	Settlement query	<operator>: <query>	Settlement detail
Admin answers the query	Operator	Query answered	Abra replied to your query on <id>	Earnings
Operator accepts the invitation	Admin	New operator signed up	<company> accepted the invitation	Operator detail
Operator suspended or status changes	Operator	Account suspended / status changed	Trips hidden from customers; contact Abra support	Dashboard
Bus marked for maintenance	Operator	Bus marked for maintenance	<reg> cannot be used for trips until it returns to service	Buses
Driver assigned or changed on a trip	Driver, Operator	New trip assigned / Driver changed	<trip> and time / <driver> now drives <trip>	Trip
Trip cancelled	Customers (booked and cancelled), Driver, Operator	Trip cancelled	Route and date; full refund has started	My trips / Home / Trips
Trip delayed	Customers, Admin, Operator	Bus running late / Trip delayed	Route, minutes, reason	My trips / Tracking / Trip
Trip boarding	Customers	Boarding started	Your bus <reg> is boarding now	My trips
Trip starts	Customers	Your trip has started	Track your bus live	My trips
Trip arrives	Customers	You have arrived	Thanks for travelling; rate your trip	My trips
Booking created	Operator	New booking	<PNR> - seats on <route>	Bookings
Booking cancelled	Operator, Customer	Booking cancelled	<PNR> on <route>; your refund has started	Bookings / My trips
Refund status changes	Customer	Refund <status>	Refund of Rs <amount> is <status>	My trips
SOS raised by a customer	Admin, Operator, Driver	SOS from <name> / SOS on your bus	Note and route; "stop safely and check, Abra safety team is calling"	SOS board / Live / Trip
SOS raised by a driver	Admin, Operator	SOS from Driver <name>	Note and route	SOS board / Live
SOS status changes (acknowledged, resolved, false alarm)	Customer or Driver who raised it	SOS <status>	Our safety team has your alert and is calling you / your SOS is now <status>	SOS
New support ticket	Admin	New support ticket	<id> - <subject>	Ticket detail
Agent replies / customer replies	Customer / Admin	Support replied / Customer replied	<subject>	Support / Ticket detail
Operator asks for an offer	Admin	Offer request	<code> from an operator needs approval	Offers
Driver sends a report	Operator	Driver report received	<driver> sent the <type> for <period> (late if late)	Driver reports
Driver sends a returned report again	Operator	Driver report sent again	<driver> fixed the <type>	Driver reports
Operator approves or returns a report	Driver	Report approved / Report sent back	Type and period / the review note	Reports
Trip started past a failed check	Operator, Admin	Trip started with a failed check	<driver> started <trip> after overriding: <checks>	Driver reports
Trip started with warnings	Operator	Trip checklist has warnings	<driver> started <trip> with: <warnings>	Driver reports
Missing report reminder (button)	Operator or Driver	Driver report missing / Please send your report	Which report and period	Driver reports / Reports

B.2 WhatsApp and email messages to the customer

Event	Channels	Subject	Body (in the customer's language)
Booking confirmed	WhatsApp and email, to the booking's contact (not only the account)	Booking confirmed: <PNR>	Route, date and time, seats, boarding point, amount, how to track, link to the ticket
Booking cancelled	WhatsApp and email	Booking cancelled: <PNR>	Cancelled; the refund starts and arrives in 5 to 7 working days
Refund paid	WhatsApp and email	Refund processed: <PNR>	Amount and reference number
Payment failed	WhatsApp and email	Payment failed	Amount and reason; seats are held for a short time; try again
Bus delayed	WhatsApp and email	Bus delayed: <PNR>	Minutes and reason; track live in the app
Boarding started	WhatsApp and email	Boarding now: <PNR>	Bus number, reach your boarding point

> **Note:** The demo stores English text only. The real build uses templates per language (Appendix K lists the 8 languages) and logs every message in the `messages` collection.

B.3 Live events (the socket channel)

Built 2026-10-12 (`backend/src/modules/realtime/live_updates_channel*`). One socket event, `live`, carries every "something changed" message: `{ type, topic?, data, at }`. `data` holds ids and the new state only, never a record; the app fetches the rest with its own permissions. A client connects with its access token (`auth: { token }`), receives `ready`, or `unauthorized` and a closed socket. It joins by itself: its own room, `staff` (admin portal), `operator:<id>` (operator staff and drivers). It may follow a topic such as `trip:<id>` by sending `subscribe`; each feature registers who may follow its topic. The connection is checked again every minute, so signing out or a change of access ends it within a minute. The bell keeps its own event, `notification`.

Event type	Published to	Data	When
sos.raised	staff, operator:<id>, user:<driver>	id, tripId	A customer or a driver raises SOS
sos.updated	staff, operator:<id>, user:<who raised it>	id, status	Acknowledged, escalated, resolved, false alarm
trip.status-changed	topic trip:<id>, staff, operator:<id>	tripId, status	Boarding, started, arrived, cancelled
trip.delayed	topic trip:<id>, staff, operator:<id>	tripId, minutes	A delay is reported
tracking.position	topic tracking:<tripId>	lat, lng, speed	About every 5 seconds while a trip runs (not stored in the socket)
seat.held, seat.released, seat.sold	topic trip:<id>	seat	Another person holds, frees or buys a seat
booking.created, booking.cancelled	operator:<id>, staff, user:<customer>	bookingId	A booking is made or cancelled
refund.updated	user:<customer>, staff	refundId, status	A refund moves on
report.submitted	operator:<id>	reportId	A driver sends a report
report.reviewed	user:<driver>	reportId, status	The owner approves or returns a report
trip-check.warning	operator:<id>, staff	checkId	A trip starts with a warning or an override
settlement.updated	operator:<id>, staff	settlementId, status	A settlement changes state or gets a query
ticket.replied	user:<customer>, staff	ticketId	A support reply arrives
kyc.updated	staff, operator:<id>	submissionId, status	Documents are sent, approved or rejected

Topic kind	Who may follow it (registered by the feature that owns it)
trip:<id>	A customer with a booking on the trip, staff of the operator that runs it, the assigned driver, admins who may read trips
tracking:<tripId>	The same people as trip, only while the trip runs

Appendix C. Access matrix (starting values)

Levels: None, View, Edit, Edit+Delete. These are the demo's starting values. The Super Admin can change them in Staff and roles. Operators have their own matrix for their staff.

C.1 Admin portal

Page	Super Admin	Admin	Support	Accounts
Dashboard (`dashboard`)	Edit+Delete	Edit+Delete	View	View
Invitations (`invitations`)	Edit+Delete	Edit+Delete	None	None
KYC reviews (`kyc`)	Edit+Delete	Edit+Delete	None	None
Operators (`operators`)	Edit+Delete	Edit+Delete	View	View
Commission (`commission`)	Edit+Delete	Edit+Delete	None	View
Payout and fees (`payout`)	Edit+Delete	View	None	View
Cities and places (`places`)	Edit+Delete	Edit+Delete	View	None
Bus types and amenities (`bustypes`)	Edit+Delete	Edit+Delete	None	None
Required documents (`documents`)	Edit+Delete	Edit+Delete	None	None
KYC questionnaire (`questions`)	Edit+Delete	Edit+Delete	None	None
Buses (`buses`)	Edit+Delete	Edit+Delete	View	None
Trips (`trips`)	Edit+Delete	Edit+Delete	View	None
Seat inventory (`inventory`)	Edit+Delete	Edit+Delete	View	None
Live tracking (`tracking`)	Edit+Delete	Edit+Delete	View	None
Bookings (`bookings`)	Edit+Delete	Edit+Delete	View	View
Payments (`payments`)	Edit+Delete	Edit+Delete	None	Edit
Refunds (`refunds`)	Edit+Delete	Edit+Delete	None	Edit
Settlements (`settlements`)	Edit+Delete	Edit+Delete	None	Edit
Customers (`customers`)	Edit+Delete	Edit+Delete	Edit	None
Support tickets (`tickets`)	Edit+Delete	Edit+Delete	Edit	None
Help content and contact (`content`)	Edit+Delete	Edit+Delete	Edit	None
Banners (`banners`)	Edit+Delete	Edit+Delete	None	None
Offers and coupons (`offers`)	Edit+Delete	Edit+Delete	None	None
Announcements (`announce`)	Edit+Delete	Edit+Delete	Edit	None
Popular routes (`popular`)	Edit+Delete	Edit+Delete	None	None
SOS board (`sos`)	Edit+Delete	Edit+Delete	Edit	None
Incidents and sharing (`incidents`)	Edit+Delete	Edit+Delete	Edit	None
Safety settings (`safety`)	Edit+Delete	Edit+Delete	None	None
Reports (`reports`)	Edit+Delete	Edit+Delete	None	View
Settings and integrations (`settings`)	Edit+Delete	View	None	None
Staff and roles (`staff`)	Edit+Delete	None	None	None
Feature flags and maintenance (`flags`)	Edit+Delete	Edit+Delete	None	None
Audit log (`audit`)	Edit+Delete	View	None	None
Notifications (`notifications`)	Edit+Delete	Edit+Delete	Edit	Edit
AI assistant (`ai`)	Edit+Delete	Edit	View	View
Messages sent (`messages`)	Edit+Delete	Edit+Delete	View	None
Driver rules (defaults) (`drules`)	Edit+Delete	Edit+Delete	None	None
Driver reports (`dreports`)	Edit+Delete	Edit+Delete	View	None

C.2 Client (operator) portal

Page	Owner	Manager	Booking staff	Accounts
Dashboard (`dashboard`)	Edit+Delete	Edit+Delete	View	View
KYC verification (`kyc`)	Edit+Delete	Edit+Delete	View	View
Company profile (`company`)	Edit+Delete	Edit+Delete	View	View
Buses (`buses`)	Edit+Delete	Edit+Delete	View	None
Seat layouts (`layouts`)	Edit+Delete	Edit+Delete	None	None
Drivers and crew (`crew`)	Edit+Delete	Edit+Delete	View	None
Routes and stops (`routes`)	Edit+Delete	Edit+Delete	View	None
Places (`places`)	Edit+Delete	Edit+Delete	None	None
Fares (`fares`)	Edit+Delete	Edit+Delete	None	View
Schedules (`schedules`)	Edit+Delete	Edit+Delete	None	None
Trips (`trips`)	Edit+Delete	Edit+Delete	Edit	None
Bookings (`bookings`)	Edit+Delete	Edit+Delete	Edit	View
Offers (`offers`)	Edit+Delete	Edit+Delete	None	None
Earnings and settlements (`earnings`)	Edit+Delete	Edit+Delete	None	Edit
Commission (`commission`)	Edit+Delete	Edit+Delete	None	View
Live buses and safety (`live`)	Edit+Delete	Edit+Delete	View	None
Notifications (`notifications`)	Edit+Delete	Edit+Delete	Edit	Edit
Staff and roles (`staff`)	Edit+Delete	None	None	None
Driver rules (`drules`)	Edit+Delete	Edit+Delete	None	None
Driver reports (`dreports`)	Edit+Delete	Edit+Delete	View	None

Appendix D. Admin portal: every page, button and endpoint

How to read each page: first its facts (feature, base path, files), then what is on the screen, then one row per button. "Where" says whether the button is on the page, on a tab, in a dialog or on a detail page. "Opens" lists the dialog and its fields (a star marks a required field). "Status" is **Exists** when the backend already has the route, **New** when it must be built, **UI** when no server call is needed. Row actions (View, Edit, Delete) exist on every table as in the demo; where changing a field would be unsafe (money, KYC status) the server refuses and says why.

> **Rule:** Build the real screen to look and behave exactly like the demo page of the same name. This appendix lists what the demo page contains; the demo file itself (`prototypes/admin_demo_model.html`) is the visual reference.

D.1 Dashboard (`dashboard`)

Item	Value
Menu group	Overview
Feature / phase	Dashboards and reports / Phase 7
API base	`/admin/dashboard`
Files	`dashboard_admin_manage_screen.dart` , `dashboard_admin_manage.ts` , `dashboard_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: View

On the screen	Contents
Sections of the page	Needs your attention | Bookings over time (chart, today / 7 days / 30 days) | Booking status (chart) | Revenue and commission (chart) | Top routes by bookings | Top operators by sales | Latest activity
Number cards (each opens the records behind it)	Bookings | Gross sales | ommission earned | Active operators2 joined this month | Active trips | Buses live now | Refunds pending | Average occupancy

Where	Button	Opens	Endpoint	Status	Notes
Page	Today		GET /admin/dashboard?range=today|7d|30d	New	
Page	7 days		GET /admin/dashboard?range=today|7d|30d	New	
Page	30 days		GET /admin/dashboard?range=today|7d|30d	New	
Page	Export		GET /admin/dashboard/export	New	

D.2 Invitations (`invitations`)

Item	Value
Menu group	Operators
Feature / phase	Operator invitations / Phase 1
API base	`/onboarding/invitations`
Files	`invitations_admin_manage_screen.dart` , `invitations_admin_manage.ts` , `invitations_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Filters	Search | Status [All / Pending / Accepted / Expired / Revoked]
Table columns	ID | Company | Owner | Email | Status | Sent ▼ | Expires | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Invite operator	Invite operator: Company name*; Owner name*; Email*; Phone (optional)	No server call	UI	opens the invite form
Page > dialog	Send invitation		POST /onboarding/invitations	Exists	sends the email and WhatsApp link
Page	Export CSV		GET /onboarding/invitations/export? format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Invitation INV-205 (CompanyPearl TravelsOwnerJose ThomasEmailjose@pearl.exampleStatusPendingEmail de)	GET /onboarding/invitations/:id	New	detail
Page > dialog	Copy link		No server call	UI	copies text to the clipboard
Page	Resend		POST /onboarding/invitations/:id/resend	Exists	
Page	Revoke	Revoke invitation (The link will stop working. The operator will not be able to join with it.)	No server call	UI	opens the revoke confirmation
Page > dialog	Revoke		DELETE /onboarding/invitations/:id	New	confirm of a delete or revoke; audit
Page	View	Invitation INV-205 (CompanyPearl TravelsOwnerJose ThomasEmailjose@pearl.exampleStatusPendingEmail de)	GET /onboarding/invitations/:id	New	detail
Page	Edit	Edit INV-205: Company*; Owner*; Email*; Status [Pending / Accepted / Expired / Revoked]; Sent*; Expires*	No server call	UI	opens the edit form, filled from GET /onboarding/invitations/:id
Page > dialog	Save changes		PATCH /onboarding/invitations/:id	New	save changes; audit before and after
Page	Delete	Delete INV-205? (This removes it from every list. You can undo for a few seconds.)	DELETE /onboarding/invitations/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /onboarding/invitations/:id/restore
Page > dialog	Delete		DELETE /onboarding/invitations/:id	New	confirm of a delete or revoke; audit

D.3 KYC reviews (`kyc`)

Item	Value
Menu group	Operators
Feature / phase	KYC review and company onboarding / Phase 1
API base	`/onboarding/submissions`
Files	`kyc_admin_manage_screen.dart` , `kyc_admin_manage.ts` , `kyc_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Filters	Search | Status [All / Pending / Approved / Rejected] | Type [All / Company / Bus]
Table columns	ID | Operator | Type | Status | Documents | Waiting ▲ | Actions
Detail page #/kyc/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /onboarding/submissions/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/kyc/:id	GET /onboarding/submissions/:id	Exists	
Page	View	Opens page #/kyc/:id	GET /onboarding/submissions/:id	Exists	
Page	Edit	Edit SUB-1006: Operator*; Type [Company / Bus]; Status [Pending / Rejected / Approved]; Waiting*	No server call	UI	opens the edit form, filled from GET /onboarding/submissions/:id
Page > dialog	Save changes		No server call	UI	not allowed to edit KYC status; use review, approve or reject
Detail page #/kyc/:id	History	History (Submitted10 Oct 2026, 02:37 amApproved10 Oct 2026, 05:37 pm)	GET /onboarding/submissions/:id/history	New	new
Detail page #/kyc/:id	View	GST certificate (GST_scan.pdfDocument preview placeholder. The real portal shows the uploaded PDF)	GET /files/:id/content	Exists	

D.4 Operators (`operators`)

Item	Value
Menu group	Operators
Feature / phase	Operators / Phase 1
API base	`/catalogue/operators`
Files	`operators_admin_manage_screen.dart` , `operators_admin_manage.ts` , `operators_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: View

On the screen	Contents
Filters	Search | Status [All / Active / Pending approval / Suspended] | Payout mode [All / Automatic / Manual]
Table columns	Operator | Status | KYC | Buses | Trips | Payout | Rating | Joined | Actions
Detail page #/operators/:id	tabs: Overview, Buses, Commission and payout, Bookings, Activity. cards: Buses, Trips, Bookings, Sales

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /catalogue/operators/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/operators/:id	GET /catalogue/operators/:id	Exists	plus tabs Overview, Buses, Bookings, Commission and payout
Page	View	Opens page #/operators/:id	GET /catalogue/operators/:id	Exists	plus tabs Overview, Buses, Bookings, Commission and payout
Page	Edit	Edit Kaveri Express: Operator*; Status [Active / Suspended / Pending approval]; KYC [Approved / Pending / Rejected / Draft]; Buses*; Trips*; Payout [Auto / Manual]; Rating*; Joined*	No server call	UI	opens the edit form, filled from GET /catalogue/operators/:id
Page > dialog	Save changes		PATCH /catalogue/operators/:id	New	save changes; audit before and after
Page	Delete	Delete Kaveri Express? (This removes it from every list. You can undo for a few seconds.)	DELETE /catalogue/operators/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/operators/:id/restore
Page > dialog	Delete		DELETE /catalogue/operators/:id	New	confirm of a delete or revoke; audit
Detail page #/operators/:id	Suspend	Suspend Kaveri Express: Reason (shown to the owner)*	No server call	UI	opens reason form (proof allowed)
Detail page #/operators/:id > dialog	Suspend		POST /catalogue/operators/:id/status	Exists	body status SUSPENDED + reason + proof; stops new sales, keeps existing bookings; notifies operator
Detail page #/operators/:id	Message owner	Message owner: Message*	No server call	UI	opens message form
Detail page #/operators/:id > dialog	Send		POST /admin/operators/:id/messages	New	in-app and WhatsApp to the owner
Detail page #/operators/:id / tab Overview	Suspend	Suspend Kaveri Express: Reason (shown to the owner)*	No server call	UI	opens reason form (proof allowed)
Detail page #/operators/:id / tab Overview > dialog	Suspend		POST /catalogue/operators/:id/status	Exists	body status SUSPENDED + reason + proof; stops new sales, keeps existing bookings; notifies operator
Detail page #/operators/:id / tab Overview	Message owner	Message owner: Message*	No server call	UI	opens message form
Detail page #/operators/:id / tab Overview > dialog	Send		POST /admin/operators/:id/messages	New	in-app and WhatsApp to the owner
Detail page #/operators/:id / tab Buses	Suspend	Suspend Kaveri Express: Reason (shown to the owner)*	No server call	UI	opens reason form (proof allowed)
Detail page #/operators/:id / tab Buses > dialog	Suspend		POST /catalogue/operators/:id/status	Exists	body status SUSPENDED + reason + proof; stops new sales, keeps existing bookings; notifies operator
Detail page #/operators/:id / tab Buses	Message owner	Message owner: Message*	No server call	UI	opens message form
Detail page #/operators/:id / tab Buses > dialog	Send		POST /admin/operators/:id/messages	New	in-app and WhatsApp to the owner
Detail page #/operators/:id / tab Buses	Export CSV		GET /catalogue/operators/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/operators/:id / tab Buses	Table		No server call	UI	table or list view of the same data
Detail page #/operators/:id / tab Buses	List		No server call	UI	table or list view of the same data

Where	Button	Opens	Endpoint	Status	Notes
Detail page #/operators/:id / tab Buses	View	Kaveri Prime 39 (IdBUS1OperatoridOP1OperatorKaveri ExpressRegistrationKA01 AB 3469NameKaveri Prim)	GET /catalogue/operators/:id	Exists	read one record for the detail window
Detail page #/operators/:id / tab Buses > dialog	Edit		No server call	UI	opens the edit form, filled from GET /catalogue/operators/:id
Detail page #/operators/:id / tab Buses > dialog	Delete		DELETE /catalogue/operators/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/operators/:id/restore
Detail page #/operators/:id / tab Commission and payout	Suspend	Suspend Kaveri Express: Reason (shown to the owner)*	No server call	UI	opens reason form (proof allowed)
Detail page #/operators/:id / tab Commission and payout > dialog	Suspend		POST /catalogue/operators/:id/status	Exists	body status SUSPENDED + reason + proof; stops new sales, keeps existing bookings; notifies operator
Detail page #/operators/:id / tab Commission and payout	Message owner	Message owner: Message*	No server call	UI	opens message form
Detail page #/operators/:id / tab Commission and payout > dialog	Send		POST /admin/operators/:id/messages	New	in-app and WhatsApp to the owner
Detail page #/operators/:id / tab Commission and payout	Change payout mode	Payout mode - Kaveri Express: Payout mode [Automatic - Razorpay Route splits each payment / Manual - Abra collects and settles each cycle]	No server call	UI	opens payout mode form
Detail page #/operators/:id / tab Commission and payout > dialog	Save		PUT /admin/operators/:id/payout-mode	New	AUTO_SPLIT or MANUAL; audit
Detail page #/operators/:id / tab Bookings	Suspend	Suspend Kaveri Express: Reason (shown to the owner)*	No server call	UI	opens reason form (proof allowed)
Detail page #/operators/:id / tab Bookings > dialog	Suspend		POST /catalogue/operators/:id/status	Exists	body status SUSPENDED + reason + proof; stops new sales, keeps existing bookings; notifies operator
Detail page #/operators/:id / tab Bookings	Message owner	Message owner: Message*	No server call	UI	opens message form
Detail page #/operators/:id / tab Bookings > dialog	Send		POST /admin/operators/:id/messages	New	in-app and WhatsApp to the owner
Detail page #/operators/:id / tab Bookings	Export CSV		GET /catalogue/operators/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/operators/:id / tab Bookings	Table		No server call	UI	table or list view of the same data
Detail page #/operators/:id / tab Bookings	List		No server call	UI	table or list view of the same data
Detail page #/operators/:id / tab Bookings	View	Opens page #/bookings/:id	GET /admin/bookings/:id	New	detail with passengers, payment, refund, messages
Detail page #/operators/:id / tab Bookings	Edit	Edit ABRK1563A: PNR*; Customer*; Route*; Amount (Rs)*; Status [Confirmed / Completed / Cancelled]	No server call	UI	opens the edit form, filled from GET /catalogue/operators/:id
Detail page #/operators/:id / tab Bookings > dialog	Save changes		PATCH /catalogue/operators/:id	New	save changes; audit before and after
Detail page #/operators/:id / tab Activity	Suspend	Suspend Kaveri Express: Reason (shown to the owner)*	No server call	UI	opens reason form (proof allowed)
Detail page #/operators/:id / tab Activity > dialog	Suspend		POST /catalogue/operators/:id/status	Exists	body status SUSPENDED + reason + proof; stops new sales, keeps existing bookings; notifies operator
Detail page #/operators/:id / tab Activity	Message owner	Message owner: Message*	No server call	UI	opens message form
Detail page #/operators/:id / tab Activity > dialog	Send		POST /admin/operators/:id/messages	New	in-app and WhatsApp to the owner

D.5 Driver rules (defaults) (`drules`)

Item	Value
Menu group	Operators
Feature / phase	Driver rules and trip checklists / Phase 9
API base	`/admin/driver-rules`
Files	`driver_rules_admin_manage_screen.dart` , `driver_rules_admin_manage.ts` , `driver_rules_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Tabs	Before the trip | After the trip | Reports
Table columns	Item | Rule | Photo | Applies to | On
Tab "Before the trip"	columns: Item, Rule, Photo, Applies to, On
Tab "After the trip"	columns: Item, Rule, Photo, Applies to, On
Tab "Reports"	columns: Report, How often, Due, Questions, On

Where	Button	Opens	Endpoint	Status	Notes
Page	Push the minimum to every operator	Push the minimum to every operator (Every operator gets the locked items (added if missing). Their own extra items s)	POST /admin/driver-rules/push-minimum	New	adds the locked items to every operator's copy; audit
Page > dialog	Push		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Page	Export CSV		GET /admin/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add an item	Add an item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]; Abra minimum (operators cannot remove it) [No / Yes, locked]	No server call	UI	opens item form
Page > dialog	Add		POST /admin/driver-rules/defaults/items or PATCH .../items/:id	New	defaults every new operator starts from
Page	Table		No server call	UI	table or list view of the same data

Where	Button	Opens	Endpoint	Status	Notes
Page	List		No server call	UI	table or list view of the same data
Page	(icon)		PATCH /admin/driver-rules/defaults/items/:id/active	New	locked items cannot be switched off
Page	(icon)		PATCH /admin/driver-rules/defaults/order	New	
Page	Edit	Edit item: Item (what the driver reads)*; If it is a problem [Blocks the trip / Warns only]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]; Abra minimum (operators cannot remove it) [No / Yes, locked]	No server call	UI	opens item form
Page > dialog	Save		POST /admin/driver-rules/defaults/items or PATCH .../items/:id	New	defaults every new operator starts from
Page	Remove	Remove this item (Drivers will no longer be asked: Your driving licence is valid. Old checklists k)	DELETE /admin/driver-rules/defaults/items/:id	New	locked items cannot be removed
Page > dialog	Remove		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Before the trip	Push the minimum to every operator	Push the minimum to every operator (Every operator gets the locked items (added if missing). Their own extra items s)	POST /admin/driver-rules/push-minimum	New	adds the locked items to every operator's copy; audit
Tab: Before the trip > dialog	Push		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Before the trip	Export CSV		GET /admin/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Before the trip	Add an item	Add an item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]; Abra minimum (operators cannot remove it) [No / Yes, locked]	No server call	UI	opens item form
Tab: Before the trip > dialog	Add		POST /admin/driver-rules/defaults/items or PATCH .../items/:id	New	defaults every new operator starts from
Tab: Before the trip	Table		No server call	UI	table or list view of the same data
Tab: Before the trip	List		No server call	UI	table or list view of the same data
Tab: Before the trip	(icon)		PATCH /admin/driver-rules/defaults/items/:id/active	New	locked items cannot be switched off
Tab: Before the trip	(icon)		PATCH /admin/driver-rules/defaults/order	New	
Tab: Before the trip	Edit	Edit item: Item (what the driver reads)*; If it is a problem [Blocks the trip / Warns only]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]; Abra minimum (operators cannot remove it) [No / Yes, locked]	No server call	UI	opens item form
Tab: Before the trip > dialog	Save		POST /admin/driver-rules/defaults/items or PATCH .../items/:id	New	defaults every new operator starts from
Tab: Before the trip	Remove	Remove this item (Drivers will no longer be asked: Your driving licence is valid. Old checklists k)	DELETE /admin/driver-rules/defaults/items/:id	New	locked items cannot be removed
Tab: Before the trip > dialog	Remove		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: After the trip	Push the minimum to every operator	Push the minimum to every operator (Every operator gets the locked items (added if missing). Their own extra items s)	POST /admin/driver-rules/push-minimum	New	adds the locked items to every operator's copy; audit
Tab: After the trip > dialog	Push		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: After the trip	Export CSV		GET /admin/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: After the trip	Add an item	Add an item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]; Abra minimum (operators cannot remove it) [No / Yes, locked]	No server call	UI	opens item form
Tab: After the trip > dialog	Add		POST /admin/driver-rules/defaults/items or PATCH .../items/:id	New	defaults every new operator starts from
Tab: After the trip	Table		No server call	UI	table or list view of the same data
Tab: After the trip	List		No server call	UI	table or list view of the same data
Tab: After the trip	(icon)		PATCH /admin/driver-rules/defaults/items/:id/active	New	locked items cannot be switched off
Tab: After the trip	(icon)		PATCH /admin/driver-rules/defaults/order	New	
Tab: After the trip	Edit	Edit item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]; Abra minimum (operators cannot remove it) [No / Yes, locked]	No server call	UI	opens item form
Tab: After the trip > dialog	Save		POST /admin/driver-rules/defaults/items or PATCH .../items/:id	New	defaults every new operator starts from
Tab: After the trip	Remove	Remove this item (Drivers will no longer be asked: End odometer reading (km). Old checklists keep)	DELETE /admin/driver-rules/defaults/items/:id	New	locked items cannot be removed
Tab: After the trip > dialog	Remove		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Reports	Push the minimum to every operator	Push the minimum to every operator (Every operator gets the locked items (added if missing). Their own extra items s)	POST /admin/driver-rules/push-minimum	New	adds the locked items to every operator's copy; audit
Tab: Reports > dialog	Push		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Reports	Export CSV		GET /admin/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Reports	Add a report	Add a report: Report name*; How often [Every day / Every week / When needed]; Due time (daily and weekly); Due day (weekly) [Mon / Tue / Wed / Thu / Fri / Sat ...]	No server call	UI	opens report type form

Where	Button	Opens	Endpoint	Status	Notes
Tab: Reports > dialog	Add		POST /admin/driver-rules/defaults/reports or PATCH .../reports/:id	New	
Tab: Reports	Table		No server call	UI	table or list view of the same data
Tab: Reports	List		No server call	UI	table or list view of the same data
Tab: Reports	(icon)		PATCH /admin/driver-rules/defaults/reports/:id/active	New	
Tab: Reports	Questions	Questions: Question; Answer type [Text / Number / Yes or no / Choose one / Photo / File]; Choices (for "Choose one"); Required [Optional / Required]	No server call	UI	opens the question list of a report type
Tab: Reports > dialog	Add this question		POST /admin/driver-rules/defaults/reports/:id/questions	New	
Tab: Reports	Edit	Edit report: Report name*; How often [Every day / Every week / When needed]; Due time (daily and weekly); Due day (weekly) [Mon / Tue / Wed / Thu / Fri / Sat ...]	No server call	UI	opens report type form
Tab: Reports > dialog	Save		POST /admin/driver-rules/defaults/reports or PATCH .../reports/:id	New	
Tab: Reports	Remove	Remove this report (Drivers will no longer see Daily report. Reports already sent are kept.)	DELETE /admin/driver-rules/defaults/reports/:id	New	
Tab: Reports > dialog	Remove		DELETE /admin/driver-rules/:id	New	confirm of a delete or revoke; audit

D.6 Driver reports (`dreports`)

Item	Value
Menu group	Operators
Feature / phase	Driver reports / Phase 9
API base	`/admin/driver-reports`
Files	`driver_reports_admin_manage_screen.dart` , `driver_reports_admin_manage.ts` , `driver_reports_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	Reports due or late20 overdue | Waiting for review | Sent back to drivers | Trips started past a failed check
Tabs	Reports | Missing | Trip checklists
Filters	Driver or report | Status [All / Waiting for review / Approved / Sent back] | Report [All / Daily report / Weekly report] | Driver [All / Ramesh k / Sunil y] | Operator [All / Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours / Tungabhadra Express / Konkan Rider / Vindhya Travels]
Table columns	Report | Driver | Operator | Sent ▼ | Status | Files
Tab "Reports"	cards: Reports due or late20 overdue, Waiting for review, Sent back to drivers, Trips started past a failed check. filters: Driver or report, Status [All / Waiting for review / Approved / Sent back], Report [All / Daily report / Weekly report], Driver [All / Ramesh k / Sunil y], Operator [All / Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours / Tungabhadra Express / Konkan Rider / Vindhya Travels]. columns: Report, Driver, Operator, Sent ▼, Status, Files
Tab "Missing"	cards: Reports due or late20 overdue, Waiting for review, Sent back to drivers, Trips started past a failed check. filters: Status [All / Overdue / Due soon]. columns: Driver, Operator, Report, Due, Status
Tab "Trip checklists"	cards: Reports due or late20 overdue, Waiting for review, Sent back to drivers, Trips started past a failed check. filters: Search, Result [All / Passed / Warnings / Started anyway], When [All / Before the trip / After the trip], Operator [All / Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours / Tungabhadra Express / Konkan Rider / Vindhya Travels]. columns: Trip, Driver, Operator, When, Time ▼, Odometer, Result

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Daily report - Ramesh K (DriverRamesh KOperatorKaveri ExpressReport11 Oct 2026Sent11 Oct 2026, 07:42 pmSt)	GET /admin/driver-reports/:id	New	read-only: system numbers (frozen at send) and the driver's answers
Tab: Reports	Export CSV		GET /admin/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Reports	Table		No server call	UI	table or list view of the same data
Tab: Reports	List		No server call	UI	table or list view of the same data
Tab: Reports	Row (open the record)	Daily report - Ramesh K (DriverRamesh KOperatorKaveri ExpressReport11 Oct 2026Sent11 Oct 2026, 07:42 pmSt)	GET /admin/driver-reports/:id	New	read-only: system numbers (frozen at send) and the driver's answers
Tab: Missing	Export CSV		GET /admin/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Missing	Table		No server call	UI	table or list view of the same data
Tab: Missing	List		No server call	UI	table or list view of the same data
Tab: Missing	Tell the operator		POST /admin/driver-reports/remind	New	tells the operator which report is missing
Tab: Trip checklists	Export CSV		GET /admin/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Trip checklists	Table		No server call	UI	table or list view of the same data
Tab: Trip checklists	List		No server call	UI	table or list view of the same data
Tab: Trip checklists	Row (open the record)	After the trip: TRP-3018 (DriverRamesh KTripTRP-3018 Vijayawada to VisakhapatnamWhen11 Oct 2026, 07:42 amO)	GET /admin/trip-checks/:id	New	read-only checklist result

D.7 Commission (`commission`)

Item	Value
Menu group	Commercial
Feature / phase	Commission rules / Phase 2
API base	`/admin/commission/rules`
Files	`commission_admin_manage_screen.dart` , `commission_admin_manage.ts` , `commission_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: View

On the screen	Contents
Tabs	Rules | Price calculator | Change history
Filters	Search | Applies to [All / Operator / Route / Bus type] | Status [All / Active / Inactive]
Table columns	ID | Applies to | Target | Commission | Effective from | Version | Status | Agreed note | Actions
Tab "Rules"	filters: Search, Applies to [All / Operator / Route / Bus type], Status [All / Active / Inactive]. columns: ID, Applies to, Target, Commission, Effective from, Version, Status, Agreed note, Actions
Tab "Price calculator"	filters: Operator fare per seat (Rs), Seats, Operator [Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours / Tungabhadra Express / Konkan Rider / Vindhya Travels / Malabar Lines], Route [Hyderabad to Bengaluru / Bengaluru to Hyderabad / Hyderabad to Chennai / Chennai to Coimbatore / Vijayawada to Visakhapatnam / Hyderabad to Vijayawada / Bengaluru to Mysuru / Chennai to Madurai / Mumbai to Pune / Hyderabad to Tirupati], Bus type [AC Sleeper / Non-AC Sleeper / AC Seater / Non-AC Seater / Volvo Multi-Axle AC / Semi-Sleeper AC]

Where	Button	Opens	Endpoint	Status	Notes
Page	Change	Default commission: Type [Percentage / Flat per seat]; Value (percent, or rupees per seat)	No server call	UI	opens default form
Page > dialog	Save		PUT /admin/commission/default	New	audit
Page	Export CSV		GET /admin/commission/rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	New rule	New commission rule: Applies to [One operator / One route / One bus type]; Which one [Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours ...]; Type [Percentage / Flat per seat]; Value; Effective from; Agreed note	No server call	UI	opens the rule form
Page > dialog	Save rule		POST /admin/commission/rules	New	new rule, or POST /admin/commission/rules/:id/versions for "Save as new version"; history row written; never changes past bookings
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Edit	Edit rule CR-1: Applies to [One operator / One route / One bus type]; Which one [Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours ...]; Type [Percentage / Flat per seat]; Value; Effective from; Agreed note	No server call	UI	opens the rule form
Page > dialog	Save as new version		POST /admin/commission/rules	New	new rule, or POST /admin/commission/rules/:id/versions for "Save as new version"; history row written; never changes past bookings
Page	Deactivate		PATCH /admin/commission/rules/:id/active	New	activate or deactivate; audit
Page	View	CR-1 (IDCR-1Applies toOperatorTargetKaveri ExpressModePercentCommission1000Agreed note)	GET /admin/commission/rules/:id	New	read one record for the detail window
Page > dialog	Delete		DELETE /admin/commission/rules/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/commission/rules/:id/restore
Page	Activate		PATCH /admin/commission/rules/:id/active	New	activate or deactivate; audit
Tab: Rules	Change	Default commission: Type [Percentage / Flat per seat]; Value (percent, or rupees per seat)	No server call	UI	opens default form
Tab: Rules > dialog	Save		PUT /admin/commission/default	New	audit
Tab: Rules	Export CSV		GET /admin/commission/rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Rules	New rule	New commission rule: Applies to [One operator / One route / One bus type]; Which one [Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours ...]; Type [Percentage / Flat per seat]; Value; Effective from; Agreed note	No server call	UI	opens the rule form
Tab: Rules > dialog	Save rule		POST /admin/commission/rules	New	new rule, or POST /admin/commission/rules/:id/versions for "Save as new version"; history row written; never changes past bookings
Tab: Rules	Table		No server call	UI	table or list view of the same data
Tab: Rules	List		No server call	UI	table or list view of the same data
Tab: Rules	Edit	Edit rule CR-1: Applies to [One operator / One route / One bus type]; Which one [Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours ...]; Type [Percentage / Flat per seat]; Value; Effective from; Agreed note	No server call	UI	opens the rule form
Tab: Rules > dialog	Save as new version		POST /admin/commission/rules	New	new rule, or POST /admin/commission/rules/:id/versions for "Save as new version"; history row written; never changes past bookings
Tab: Rules	Deactivate		PATCH /admin/commission/rules/:id/active	New	activate or deactivate; audit
Tab: Rules	View	CR-1 (IDCR-1Applies toOperatorTargetKaveri ExpressModePercentCommission1000Agreed note)	GET /admin/commission/rules/:id	New	read one record for the detail window
Tab: Rules > dialog	Delete		DELETE /admin/commission/rules/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/commission/rules/:id/restore
Tab: Rules	Activate		PATCH /admin/commission/rules/:id/active	New	activate or deactivate; audit
Tab: Price calculator	Calculate		POST /admin/commission/calculate	New	price calculator: fare in, customer price and operator payable out (same function the booking uses)

D.8 Payout and fees (`payout`)

Item	Value
Menu group	Commercial
Feature / phase	Payout and fee settings / Phase 2
API base	`/admin/payout/refund-slabs`
Files	`payout_cfg_admin_manage_screen.dart` , `payout_cfg_admin_manage.ts` , `payout_cfg_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: View; Support: None; Accounts: View

On the screen	Contents
Filters	Default mode for new operators [Automatic (Razorpay Route) / Manual settlement] | Settlement cycle (days) | Minimum payout (Rs) | Hold days after trip | GST on fare (%) | Convenience fee (Rs) | Who funds coupon discounts [Abra Travels / The operator / Shared 50/50] | Search | Mode [All / Automatic / Manual]
Table columns 1	Cancel at least (hours before) | Refund | Actions
Table columns 2	Operator | Mode | Bank account | Verification | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Save changes		PUT /admin/payout/settings	New	cycle days, minimum payout, hold days, convenience fee, GST percent, coupon funding; audit
Page	(icon)		PUT /admin/payout/settings/default-mode	New	automatic (Razorpay Route) or manual
Page	Add slab	Add refund slab: Hours before departure*; Refund percent*	No server call	UI	opens slab form
Page > dialog	Add		POST /admin/payout/refund-slabs	New	audit
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Remove		DELETE /admin/payout/refund-slabs/:id	New	audit
Page	View	CS1 (IdCS1Cancel at least (hours before)24Refund75)	GET /admin/payout/refund-slabs/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/payout/refund-slabs/:id
Page > dialog	Delete		DELETE /admin/payout/refund-slabs/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/payout/refund-slabs/:id/restore
Page	Export CSV		GET /admin/payout/refund-slabs/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Change mode	Payout mode - Kaveri Express: Payout mode [Automatic - Razorpay Route splits each payment / Manual - Abra collects and settles each cycle]	No server call	UI	opens payout mode form
Page > dialog	Save		PUT /admin/operators/:id/payout-mode	New	AUTO_SPLIT or MANUAL; audit
Page	Verify bank		POST /admin/operators/:id/bank/verify	New	penny-drop or manual check; audit

D.9 Cities and places (`places`)

Item	Value
Menu group	Catalogue
Feature / phase	Cities and places / Phase 1
API base	`/catalogue/locations`
Files	`places_admin_manage_screen.dart` , `places_admin_manage.ts` , `places_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Tabs	Places | Cities | Operator requests
Filters	Search | City [All / Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore / Kochi / Mumbai / Pune] | Used for [All / Boarding / Dropping / Both] | Status [All / Active / Inactive]
Table columns	Place | City | Used for | Added by | Trips using it | Status | Actions
Tab "Places"	filters: Search, City [All / Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore / Kochi / Mumbai / Pune], Used for [All / Boarding / Dropping / Both], Status [All / Active / Inactive]. columns: Place, City, Used for, Added by, Trips using it, Status, Actions
Tab "Cities"	filters: Search, State [All / Telangana / Karnataka / Tamil nadu / Andhra pradesh / Kerala / Maharashtra / Goa / Gujarat], Status [All / Active / Inactive]. columns: City, State, Places, Popular, Status, Actions
Tab "Operator requests"	filters: Search, Status [All / Pending / Approved / Rejected]. columns: Requested place, City, Operator, Requested, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /catalogue/locations/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add place	Add place: Place name*; City [Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore ...]; Address*; Landmark; Used for [Boarding only / Dropping only / Both]; Latitude; Longitude; Contact phone; Instructions for passengers	No server call	UI	opens place form
Page > dialog	Save		POST /catalogue/locations or PUT /catalogue/locations/:id	Exists	
Page	Import	Import places (Upload a CSV file with these columns: name, city, address, landmark. Excel files)	POST /catalogue/locations/import?dryRun=true then false	New	upload, row-by-row errors, then save; GET /catalogue/locations/import-template for the file
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Edit	Edit place: Place name*; City [Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore ...]; Address*; Landmark; Used for [Boarding only / Dropping only / Both]; Latitude; Longitude; Contact phone; Instructions for passengers	No server call	UI	opens place form
Page	View	Hyderabad Bus Stand (IdPL1CityHyderabadPlaceHyderabad Bus StandAddress12, Main Road, HyderabadLandmar)	GET /catalogue/locations/:id	Exists	read one record for the detail window
Page > dialog	Delete		DELETE /catalogue/locations/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/locations/:id/restore
Tab: Places	Export CSV		GET /catalogue/locations/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Places	Add place	Add place: Place name*; City [Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore ...]; Address*; Landmark; Used for [Boarding only / Dropping only / Both]; Latitude; Longitude; Contact phone; Instructions for passengers	No server call	UI	opens place form
Tab: Places > dialog	Save		POST /catalogue/locations or PUT /catalogue/locations/:id	Exists	
Tab: Places	Import	Import places (Upload a CSV file with these columns: name, city, address, landmark. Excel files)	POST /catalogue/locations/import?dryRun=true then false	New	upload, row-by-row errors, then save; GET /catalogue/locations/import-template for the file
Tab: Places	Table		No server call	UI	table or list view of the same data
Tab: Places	List		No server call	UI	table or list view of the same data
Tab: Places	Edit	Edit place: Place name*; City [Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore ...]; Address*; Landmark; Used for [Boarding only / Dropping only / Both]; Latitude; Longitude; Contact phone; Instructions for passengers	No server call	UI	opens place form
Tab: Places	View	Hyderabad Bus Stand (IdPL1CityHyderabadPlaceHyderabad Bus StandAddress12, Main Road, HyderabadLandmar)	GET /catalogue/locations/:id	Exists	read one record for the detail window
Tab: Places > dialog	Delete		DELETE /catalogue/locations/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/locations/:id/restore

Where	Button	Opens	Endpoint	Status	Notes
Tab: Cities	Export CSV		GET /catalogue/locations/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Cities	Add city	Add city: City name*; State*; Telugu name; Hindi name; Status [Active / Inactive]; Show as popular [Yes / No]	No server call	UI	opens city form
Tab: Cities > dialog	Save		POST /catalogue/cities or PUT /catalogue/cities/:id	Exists	
Tab: Cities	Import	Import cities (Upload a CSV file with these columns: name, state. Excel files can be saved as C)	POST /catalogue/locations/import?dryRun=true then false	New	upload, row-by-row errors, then save; GET /catalogue/locations/import-template for the file
Tab: Cities	Table		No server call	UI	table or list view of the same data
Tab: Cities	List		No server call	UI	table or list view of the same data
Tab: Cities	Edit	Edit city: City name*; State*; Telugu name; Hindi name; Status [Active / Inactive]; Show as popular [Yes / No]	No server call	UI	opens city form
Tab: Cities	View	Hyderabad (IdCT1CityHyderabadStateTelanganaStatusYesPopularYesTeహైదరాబాద్HiహైదరాబాదPlaces2)	GET /catalogue/locations/:id	Exists	read one record for the detail window
Tab: Cities > dialog	Delete		DELETE /catalogue/locations/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/locations/:id/restore
Tab: Operator requests	Export CSV		GET /catalogue/locations/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Operator requests	Table		No server call	UI	table or list view of the same data
Tab: Operator requests	List		No server call	UI	table or list view of the same data
Tab: Operator requests	Approve		POST /admin/place-requests/:id/approve	New	creates the place for every operator; notifies the operator
Tab: Operator requests	Reject	Reject request: Reason (the operator reads it)*	No server call	UI	opens reason form
Tab: Operator requests > dialog	Reject		POST /admin/place-requests/:id/reject	New	reason; notifies the operator
Tab: Operator requests	View	Hosur Bus Stand Gate 2 (IdPR-100OperatorKaveri ExpressCityHosurRequested placeHosur Bus Stand Gate 2Addr)	GET /catalogue/locations/:id	Exists	read one record for the detail window
Tab: Operator requests > dialog	Edit		No server call	UI	opens the edit form, filled from GET /catalogue/locations/:id
Tab: Operator requests > dialog	Delete		DELETE /catalogue/locations/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/locations/:id/restore

D.10 Bus types and amenities (`bustypes`)

Item	Value
Menu group	Catalogue
Feature / phase	Bus types and amenities / Phase 1
API base	`/catalogue/bus-types`
Files	`bus_types_admin_manage_screen.dart` , `bus_types_admin_manage.ts` , `bus_types_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Table columns	Bus type | Key | AC | Seating | Buses | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Add bus type	Add bus type: Name*; Key (cannot change later)*; Air conditioned [Yes / No]; Seating [Seater / Sleeper / Semi-sleeper]	No server call	UI	opens bus type form
Page > dialog	Save		POST /catalogue/bus-types or PUT /catalogue/bus-types/:id	Exists	
Page	Export CSV		GET /catalogue/bus-types/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Edit	Edit bus type: Name*; Key (cannot change later)*; Air conditioned [Yes / No]; Seating [Seater / Sleeper / Semi-sleeper]	No server call	UI	opens bus type form
Page	View	AC Sleeper (IdBT1Bus typeAC SleeperKeyAc sleeperACYesSeatingSleeperStatusYesBuses2)	GET /catalogue/bus-types/:id	New	read one record for the detail window
Page > dialog	Delete		DELETE /catalogue/bus-types/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/bus-types/:id/restore
Page	Add amenity	Add amenity: Amenity name*	No server call	UI	opens amenity form
Page > dialog	Add		POST /admin/amenities	New	new
Page	Toggle Charging point		PATCH /admin/amenities/:id/active	New	new

D.11 Required documents (`documents`)

Item	Value
Menu group	Catalogue
Feature / phase	Required documents and KYC questionnaire / Phase 1
API base	`/catalogue/document-types`
Files	`kyc_config_admin_manage_screen.dart` , `kyc_config_admin_manage.ts` , `kyc_config_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Filters	Search | For [All / Operator / Bus]

On the screen	Contents
Table columns	Document | For | Required | Has expiry | File types | Max size | Max files | Active | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Add document type	Add document type: Name*; Key (permanent)*; For [Company / Bus]; Required [Required / Optional]; Has expiry date [Yes / No]; Accepted file types [PDF / PDF, JPG / PDF, JPG, PNG / JPG, PNG]; Largest file (MB, 1 to 25); Most files (1 to 10); Instructions shown to the owner	No server call	UI	opens document type form
Page > dialog	Save		POST /catalogue/document-types or PUT /catalogue/document-types/:id	Exists	
Page	Export CSV		GET /catalogue/document-types/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	(icon)		POST /catalogue/document-types/:id/active	Exists	
Page	Edit	Edit document type: Name*; Key (permanent)*; For [Company / Bus]; Required [Required / Optional]; Has expiry date [Yes / No]; Accepted file types [PDF / PDF, JPG / PDF, JPG, PNG / JPG, PNG]; Largest file (MB, 1 to 25); Most files (1 to 10); Instructions shown to the owner	No server call	UI	opens document type form
Page	View	GST certificate (IdDT1KeygstDocumentGST certificateForOperatorRequiredYesHas expiryNoFile typesPD)	GET /catalogue/document-types/:id	New	read one record for the detail window
Page > dialog	Delete		DELETE /catalogue/document-types/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/document-types/:id/restore

D.12 KYC questionnaire (`questions`)

Item	Value
Menu group	Catalogue
Feature / phase	Required documents and KYC questionnaire / Phase 1
API base	`/admin/kyc/questions`
Files	`kyc_config_admin_manage_screen.dart` , `kyc_config_admin_manage.ts` , `kyc_config_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Tabs	Questions | Document rules | Preview checklist
Filters	Search
Table columns	Question | Answer type | Required | Active | Actions
Tab "Questions"	filters: Search. columns: Question, Answer type, Required, Active, Actions
Tab "Document rules"	columns: Rule, When, Document asked, Required, Active, Actions
Tab "Preview checklist"	filters: Business type [Company / Individual / Partnership], GST registered [Yes / No], Services [Seat bookings / Charters / Parcel]

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/kyc/questions/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add question	Add question: Question text*; Key (permanent)*; Answer type [TEXT / NUMBER / BOOLEAN / SINGLE_CHOICE / MULTI_CHOICE]; Options (comma separated); Required [Required / Optional]	No server call	UI	opens question form
Page > dialog	Save		POST /admin/kyc/questions or PATCH /admin/kyc/questions/:id	New	new
Page	Export config		GET /admin/kyc/config/export	New	questions and document rules as a file
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	(icon)		PATCH /admin/kyc/questions/:id/active	New	new
Page	Edit	Edit question: Question text*; Key (permanent)*; Answer type [TEXT / NUMBER / BOOLEAN / SINGLE_CHOICE / MULTI_CHOICE]; Options (comma separated); Required [Required / Optional]	No server call	UI	opens question form
Page	View	Q1 (IdQ1Keybusiness_typeQuestionWhat is your business type?AppliesOperatorAnswer typ)	GET /admin/kyc/questions/:id	New	read one record for the detail window
Page > dialog	Delete		DELETE /admin/kyc/questions/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/kyc/questions/:id/restore
Tab: Questions	Export CSV		GET /admin/kyc/questions/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Questions	Add question	Add question: Question text*; Key (permanent)*; Answer type [TEXT / NUMBER / BOOLEAN / SINGLE_CHOICE / MULTI_CHOICE]; Options (comma separated); Required [Required / Optional]	No server call	UI	opens question form
Tab: Questions > dialog	Save		POST /admin/kyc/questions or PATCH /admin/kyc/questions/:id	New	new
Tab: Questions	Export config		GET /admin/kyc/config/export	New	questions and document rules as a file
Tab: Questions	Table		No server call	UI	table or list view of the same data
Tab: Questions	List		No server call	UI	table or list view of the same data
Tab: Questions	(icon)		PATCH /admin/kyc/questions/:id/active	New	new
Tab: Questions	Edit	Edit question: Question text*; Key (permanent)*; Answer type [TEXT / NUMBER / BOOLEAN / SINGLE_CHOICE / MULTI_CHOICE]; Options (comma separated); Required [Required / Optional]	No server call	UI	opens question form
Tab: Questions	View	Q1 (IdQ1Keybusiness_typeQuestionWhat is your business type?AppliesOperatorAnswer typ)	GET /admin/kyc/questions/:id	New	read one record for the detail window
Tab: Questions > dialog	Delete		DELETE /admin/kyc/questions/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/kyc/questions/:id/restore
Tab: Document rules	Export CSV		GET /admin/kyc/questions/export?format=csv|xlsx	New	same filters, search and permissions as the list

Where	Button	Opens	Endpoint	Status	Notes
Tab: Document rules	Add rule	Add document rule: Rule name*; Document [GST certificate / PAN card / Company registration / Owner ID proof / Bank proof / Interstate permit]; Question [What is your business type? / Do you own the buses you operate? / Which services do you offer? / Are you GST registered? / How many buses do you run? / Routes you plan to operate]; Answer that triggers it*; Required [Required / Optional]	No server call	UI	opens rule form
Tab: Document rules > dialog	Save		POST /admin/kyc/document-rules or PATCH /admin/kyc/document-rules/:id	New	new
Tab: Document rules	Table		No server call	UI	table or list view of the same data
Tab: Document rules	List		No server call	UI	table or list view of the same data
Tab: Document rules	(icon)		PATCH /admin/kyc/document-rules/:id/active	New	new
Tab: Document rules	View	GST certificate for GST-registered operators (IdR1RuleGST certificate for GST-registered operatorsDocument askedgstWhengst_reg)	GET /admin/kyc/questions/:id	New	read one record for the detail window
Tab: Document rules > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/kyc/questions/:id
Tab: Document rules > dialog	Delete		DELETE /admin/kyc/questions/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/kyc/questions/:id/restore
Tab: Preview checklist	Show checklist		POST /admin/kyc/preview	New	evaluates which documents a sample answer set would require

D.13 Buses (`buses`)

Item	Value
Menu group	Operations
Feature / phase	Buses and seat layouts / Phase 2
API base	`/catalogue/buses`
Files	`buses_admin_manage_screen.dart` , `buses_admin_manage.ts` , `buses_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Filters	Search | Status [All / Active / Pending approval / Maintenance / Retired] | Operator [All / Kaveri express / Godavari lines / Deccan voyager / Sahyadri coaches / Nilgiri travels / Charminar tours / Tungabhadra express / Konkan rider / Vindhya travels] | Type [All / Ac sleeper / Non-ac sleeper / Ac seater / Non-ac seater / Volvo multi-axle ac / Semi-sleeper ac] | Documents [All / Documents expired]
Table columns	Registration | Operator | Type | Seats | Layout | Insurance | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Import buses	Import buses (Upload a CSV file with these columns: registration, name, type, seats. Excel fil)	POST /catalogue/buses/import?dryRun=true then false	New	upload, row-by-row errors, then save; GET /catalogue/buses/import-template for the file
Page	Export CSV		GET /catalogue/buses/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	KA01 AB 3469 - Kaveri Prime 39 (OperatorKaveri ExpressTypeSemi-Sleeper ACAir conditionedYesSeats45Seat layout2+2)	GET /catalogue/buses/:id	Exists	
Page > dialog	Mark maintenance		POST /catalogue/buses/:id/status	Exists	MAINTENANCE or back to ACTIVE; notifies operator
Page > dialog	Retire bus		POST /catalogue/buses/:id/status	Exists	RETIRED
Page	View	KA01 AB 3469 - Kaveri Prime 39 (OperatorKaveri ExpressTypeSemi-Sleeper ACAir conditionedYesSeats45Seat layout2+2)	GET /catalogue/buses/:id	Exists	
Page	Edit	Edit Kaveri Prime 39: Registration*; Operator*; Type*; Seats*; Layout*; Insurance*; Status [Active / Maintenance / Pending approval]	No server call	UI	opens the edit form, filled from GET /catalogue/buses/:id
Page > dialog	Save changes		PATCH /catalogue/buses/:id	New	save changes; audit before and after
Page	Delete	Delete Kaveri Prime 39? (This removes it from every list. You can undo for a few seconds.)	DELETE /catalogue/buses/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/buses/:id/restore
Page > dialog	Delete		DELETE /catalogue/buses/:id	New	confirm of a delete or revoke; audit

D.14 Trips (`trips`)

Item	Value
Menu group	Operations
Feature / phase	Trips and seat inventory / Phase 2
API base	`/catalogue/trips`
Files	`trips_admin_manage_screen.dart` , `trips_admin_manage.ts` , `trips_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Filters	Search | Status [All / Draft / Scheduled / Boarding / Departed / In transit / Delayed / Arrived / Cancelled] | Operator [All / Kaveri express / Godavari lines / Deccan voyager / Sahyadri coaches / Nilgiri travels / Charminar tours / Tungabhadra express / Konkan rider / Vindhya travels] | Route [All / Hyderabad to bengaluru / Bengaluru to hyderabad / Hyderabad to chennai / Chennai to coimbatore / Vijayawada to visakhapatnam / Hyderabad to vijayawada / Bengaluru to mysuru / Chennai to madurai / Mumbai to pune] | When [All / Today / Upcoming / Past]
Table columns	Trip | Route | Bus | Departs ▼ | Status | Seats | From | Actions

On the screen	Contents
Detail page #/trips/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	Add trip for operator	Add trip for an operator: Operator [Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Tungabhadra Express ...]; Bus [KA01 AB 3469 - Kaveri Express / KA01 AB 4945 - Kaveri Express / KA01 AB 8721 - Godavari Lines / TS09 TX 1487 - Deccan Voyager / AP39 TX 6954 - Deccan Voyager / AP39 UB 4422 - Deccan Voyager ...]; Route [Hyderabad to Bengaluru / Bengaluru to Hyderabad / Hyderabad to Chennai / Chennai to Coimbatore / Vijayawada to Visakhapatnam / Hyderabad to Vijayawada ...]; Date*; Departure time*; Lowest fare (Rs)*	No server call	UI	opens the trip form
Page > dialog	Create draft		POST /catalogue/trips	Exists	creates a draft for an operator
Page	Export CSV		GET /catalogue/trips/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/trips/:id	GET /catalogue/trips/:id	Exists	plus seat map GET /admin/trips/:id/seats
Page	View	Opens page #/trips/:id	GET /catalogue/trips/:id	Exists	plus seat map GET /admin/trips/:id/seats
Page	Edit	Edit Hyderabad to Chennai: Route*; Bus*; Departs*; Status [Arrived / Scheduled / Delayed / In transit / Draft / Cancelled]; From (Rs)*	No server call	UI	opens the edit form, filled from GET /catalogue/trips/:id
Page > dialog	Save changes		PATCH /catalogue/trips/:id	New	save changes; audit before and after
Page	Delete	Delete Hyderabad to Chennai? (This removes it from every list. You can undo for a few seconds.)	DELETE /catalogue/trips/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /catalogue/trips/:id/restore
Page > dialog	Delete		DELETE /catalogue/trips/:id	New	confirm of a delete or revoke; audit
Detail page #/trips/:id	1A	Block seat 1A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id > dialog	Block		DELETE /catalogue/trips/:id	New	confirm of a delete or revoke; audit
Detail page #/trips/:id	1B	Block seat 1B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	1C	Block seat 1C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	1D	Block seat 1D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2A	Block seat 2A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2B	Block seat 2B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2C	Block seat 2C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2D	Block seat 2D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3A	Block seat 3A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3B	Block seat 3B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3C	Block seat 3C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3D	Block seat 3D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4A	Block seat 4A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4B	Block seat 4B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4C	Block seat 4C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4D	Block seat 4D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5A	Block seat 5A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5B	Block seat 5B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5C	Block seat 5C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it

Where	Button	Opens	Endpoint	Status	Notes
Detail page #/trips/:id	5D	Block seat 5D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6A	Block seat 6A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6B	Block seat 6B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6C	Block seat 6C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6D	Block seat 6D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7A	Block seat 7A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7B	Block seat 7B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7C	Block seat 7C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7D	Block seat 7D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8A	Block seat 8A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8B	Block seat 8B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8C	Block seat 8C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8D	Block seat 8D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9A	Block seat 9A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9B	Block seat 9B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9C	Block seat 9C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9D	Block seat 9D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10A	Block seat 10A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10B	Block seat 10B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10C	Block seat 10C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10D	Block seat 10D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	Export CSV		GET /catalogue/trips/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/trips/:id	Table		No server call	UI	table or list view of the same data
Detail page #/trips/:id	List		No server call	UI	table or list view of the same data

D.15 Seat inventory (`inventory`)

Item	Value
Menu group	Operations
Feature / phase	Trips and seat inventory / Phase 2
API base	`/admin/inventory`
Files	`trips_admin_manage_screen.dart` , `trips_admin_manage.ts` , `trips_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Sections of the page	Cards: trips on sale, seats sold, seats held right now, low-occupancy trips; table of trips with seat counts
Number cards (each opens the records behind it)	Trips on sale | Seats sold | Seats held right nowHold time is 10 minutes | ow occupancy, leaving in 2 daysCheck pricing
Filters	Search | Operator [All / Kaveri express / Godavari lines / Deccan voyager / Sahyadri coaches / Nilgiri travels / Charminar tours / Tungabhadra express / Konkan rider / Vindhya travels] | Occupancy [All / Under 25% / Over 80%]
Table columns	Trip | Route | Departs | Sold | Held | Blocked | Free | Occupancy | Actions

On the screen	Contents
Detail page #/trips/:id	columns: PNR, Customer, Seats, Amount, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Release expired holds		POST /admin/inventory/release-expired-holds	New	frees holds past 10 minutes (also runs by itself every minute)
Page	Export CSV		GET /admin/inventory/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/trips/:id	GET /catalogue/trips/:id	Exists	plus seat map GET /admin/trips/:id/seats
Page	View	Opens page #/trips/:id	GET /catalogue/trips/:id	Exists	plus seat map GET /admin/trips/:id/seats
Page	Edit	Edit Hyderabad to Chennai: Route*; Departs*; Held*; Blocked*	No server call	UI	opens the edit form, filled from GET /admin/inventory/:id
Page > dialog	Save changes		PATCH /admin/trips/:id/inventory	New	held and blocked seats (generic row edit)
Page	Delete	Delete Hyderabad to Chennai? (This removes it from every list. You can undo for a few seconds.)	No server call	UI	not allowed: inventory is derived from bookings; the server refuses
Page > dialog	Delete		DELETE /admin/inventory/:id	New	confirm of a delete or revoke; audit
Detail page #/trips/:id	Delay	Delay trip: Delay (minutes)*; Reason*	No server call	UI	opens delay form
Detail page #/trips/:id > dialog	Delay and notify		POST /admin/trips/:id/delay	New	minutes, reason, proof; notifies customers, operator, admin
Detail page #/trips/:id	Change driver	Change driver: Driver [Ramesh K / Sunil Y / Anwar S / Prakash M / Naveen B]	No server call	UI	opens driver form
Detail page #/trips/:id > dialog	Save		PUT /admin/trips/:id/driver	New	notifies the new driver and the operator
Detail page #/trips/:id	Cancel trip	Cancel trip TRP-3002: Reason (customers see it)*	No server call	UI	opens reason form (proof)
Detail page #/trips/:id > dialog	Cancel trip and refund		POST /catalogue/trips/:id/cancel	Exists	refunds all bookings, notifies customers, operator and driver; audit
Detail page #/trips/:id	1A	Seat 1A is booked (PNRABRM31016BPassengerSneha JoshiPhone+91 9352721339All seats in this booking6D,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id > dialog	Open booking		No server call	UI	opens the booking from the trip
Detail page #/trips/:id	1B	Seat 1B is booked (PNRABRK31025APassengerAarav SharmaPhone+91 9622428883All seats in this booking10)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	1C	Seat 1C is held (A customer is paying for this seat right now. It cannot be changed until the hol)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	1D	Seat 1D is booked (PNRABRT31023DPassengerHarish ChandraPhone+91 9906697386All seats in this booking)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2A	Block seat 2A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id > dialog	Block		DELETE /admin/inventory/:id	New	confirm of a delete or revoke; audit
Detail page #/trips/:id	2B	Seat 2B is booked (PNRABRM31021BPassengerImran KhanPhone+91 9519223342All seats in this booking2B,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2C	Block seat 2C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	2D	Seat 2D is booked (PNRABRQ31017CPassengerSanjay GuptaPhone+91 9377021306All seats in this booking2D)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3A	Seat 3A is booked (PNRABRK31025APassengerAarav SharmaPhone+91 9622428883All seats in this booking10)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3B	Seat 3B is booked (PNRABRT9609CPassengerAnjali DasPhone+91 9477076400All seats in this booking3B, 5)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3C	Seat 3C is booked (PNRABRT31023DPassengerHarish ChandraPhone+91 9906697386All seats in this booking)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	3D	Block seat 3D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4A	Seat 4A is booked (PNRABRM31021BPassengerImran KhanPhone+91 9519223342All seats in this booking2B,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4B	Block seat 4B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4C	Seat 4C is booked (PNRABRT31018DPassengerFatima BegumPhone+91 9886431792All seats in this booking4C)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	4D	Seat 4D is booked (PNRABRK31025APassengerAarav SharmaPhone+91 9622428883All seats in this booking10)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5A	Seat 5A is booked (PNRABRT9609CPassengerAnjali DasPhone+91 9477076400All seats in this booking3B, 5)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5B	Seat 5B is booked (PNRABRX31024EPassengerBhavana SPhone+91 9496509971All seats in this booking5B, 7)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5C	Block seat 5C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	5D	Seat 5D is booked (PNRABRM31021BPassengerImran KhanPhone+91 9519223342All seats in this booking2B,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6A	Block seat 6A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6B	Seat 6B is booked (PNRABRX31019EPassengerYash ThakurPhone+91 9188923686All seats in this booking6B,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6C	Seat 6C is booked (PNRABRM31026BPassengerKavya MenonPhone+91 9192297973All seats in this booking6CA)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	6D	Seat 6D is booked (PNRABRM31016BPassengerSneha JoshiPhone+91 9352721339All seats in this booking6D,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it

Where	Button	Opens	Endpoint	Status	Notes
Detail page #/trips/:id	7A	Seat 7A is booked (PNRABRX31024EPassengerBhavana SPhone+91 9496509971All seats in this booking5B, 7)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7B	Block seat 7B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7C	Seat 7C is booked (PNRABRQ31022CPassengerLakshmi NarayanPhone+91 9851157211All seats in this bookin)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	7D	Block seat 7D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8A	Seat 8A is booked (PNRABRX31019EPassengerYash ThakurPhone+91 9188923686All seats in this booking6B,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8B	Block seat 8B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8C	Seat 8C is booked (PNRABRM31016BPassengerSneha JoshiPhone+91 9352721339All seats in this booking6D,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	8D	Seat 8D is booked (PNRABRX31024EPassengerBhavana SPhone+91 9496509971All seats in this booking5B, 7)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9A	Block seat 9A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9B	Seat 9B is booked (PNRABRQ31022CPassengerLakshmi NarayanPhone+91 9851157211All seats in this bookin)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9C	Block seat 9C (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	9D	Seat 9D is booked (PNRABRX31019EPassengerYash ThakurPhone+91 9188923686All seats in this booking6B,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10A	Block seat 10A (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10B	Seat 10B is booked (PNRABRM31016BPassengerSneha JoshiPhone+91 9352721339All seats in this booking6D,)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10C	Seat 10C is booked (PNRABRK31025APassengerAarav SharmaPhone+91 9622428883All seats in this booking10)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	10D	Block seat 10D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	11A	Seat 11A is booked (PNRABRT31023DPassengerHarish ChandraPhone+91 9906697386All seats in this booking)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	11B	Block seat 11B (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	11C	Seat 11C is booked (PNRABRK31020APassengerPriya NairPhone+91 9950575426All seats in this booking11CA)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	11D	Block seat 11D (The seat is removed from sale. Use it for a counter sale or a damaged seat. Cust)	POST /admin/trips/:id/seats/:seat/block (or /unblock) for a free seat; GET /admin/bookings/:id for a booked seat	New	audit; a held seat only explains who holds it
Detail page #/trips/:id	Export CSV		GET /admin/inventory/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/trips/:id	Table		No server call	UI	table or list view of the same data
Detail page #/trips/:id	List		No server call	UI	table or list view of the same data
Detail page #/trips/:id	View	Opens page #/bookings/:id	GET /admin/bookings/:id	New	detail with passengers, payment, refund, messages
Detail page #/trips/:id	Edit	Edit ABRT9609C: PNR*; Customer*; Seats*; Amount (Rs)*; Status [Confirmed / Completed / Cancelled]	No server call	UI	opens the edit form, filled from GET /admin/inventory/:id
Detail page #/trips/:id > dialog	Save changes		PATCH /admin/trips/:id/inventory	New	held and blocked seats (generic row edit)

D.16 Live tracking (`tracking`)

Item	Value
Menu group	Operations
Feature / phase	Live tracking / Phase 5
API base	`/admin/tracking`
Files	`tracking_admin_manage_screen.dart` , `tracking_admin_manage.ts` , `tracking_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Sections of the page	Map with every running bus, list of buses with state, speed, progress and ETA, filter by operator and state, route replay
Filters	Operator [All operators / Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels / Charminar Tours / Tungabhadra Express / Konkan Rider / Vindhya Travels] | State [All states / Live / Delayed / Signal lost]

Where	Button	Opens	Endpoint	Status	Notes
Page	TN38 EF 7890		GET /admin/tracking/trips/:id	New	live card; position by Socket.IO tracking:{tripId}
Page	AP16 CD 7939		GET /admin/tracking/trips/:id	New	live card; position by Socket.IO tracking:{tripId}
Page	TS07 EF 3887		GET /admin/tracking/trips/:id	New	live card; position by Socket.IO tracking:{tripId}
Page	AP39 AB 9725		GET /admin/tracking/trips/:id	New	live card; position by Socket.IO tracking:{tripId}
Page	KA01 AB 3469		GET /admin/tracking/trips/:id	New	live card; position by Socket.IO tracking:{tripId}
Page	Row (open the record)		GET /admin/tracking/trips/:id	New	live card; position by Socket.IO tracking:{tripId}

D.17 Bookings (`bookings`)

Item	Value
Menu group	Sales
Feature / phase	Booking, passengers and bookings lists / Phase 3

Item	Value
API base	`/admin/bookings`
Files	`booking_admin_manage_screen.dart` , `booking_admin_manage.ts` , `booking_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: View

On the screen	Contents
Filters	PNR, name, phone, route | Status [All / Confirmed / Completed / Cancelled] | Operator [All / Kaveri express / Godavari lines / Deccan voyager / Sahyadri coaches / Nilgiri travels / Charminar tours / Tungabhadra express / Konkan rider / Vindhya travels] | Route [All / Hyderabad to bengaluru / Bengaluru to hyderabad / Hyderabad to chennai / Chennai to coimbatore / Vijayawada to visakhapatnam / Hyderabad to vijayawada / Bengaluru to mysuru / Chennai to madurai / Mumbai to pune] | Payment mode [All / Upi / Card / Netbanking / Wallet] | Booked [All / Last 7 days / Last 14 days]
Table columns	PNR | Customer | Route | Travel | Seats | Amount | Paid by | Status | Actions
Detail page #/bookings/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/bookings/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/bookings/:id	GET /admin/bookings/:id	New	detail with passengers, payment, refund, messages
Page	View	Opens page #/bookings/:id	GET /admin/bookings/:id	New	detail with passengers, payment, refund, messages
Page	Edit	Edit ABRX7058A: PNR*; Customer*; Route*; Travel*; Seats*; Amount (Rs)*; Paid by [Wallet / Upi / Card / Netbanking]; Status [Confirmed / Completed / Cancelled]	No server call	UI	opens the edit form, filled from GET /admin/bookings/:id
Page > dialog	Save changes		PATCH /admin/bookings/:id	New	only notes and contact fields are editable; amounts and seats are refused
Detail page #/bookings/:id	Cancel booking	Cancel booking ABRX7058A: Refund percent (admin override); Reason*	No server call	UI	opens cancel form (refund percent override, reason, proof)
Detail page #/bookings/:id > dialog	Cancel and refund		POST /admin/bookings/:id/cancel	New	frees the seats, starts the refund, messages the customer; audit
Detail page #/bookings/:id	Resend ticket		POST /admin/bookings/:id/resend-ticket	New	WhatsApp and email again
Detail page #/bookings/:id	Add note	Add internal note: Note (only staff see it)*	No server call	UI	opens note form
Detail page #/bookings/:id > dialog	Save		POST /admin/bookings/:id/notes	New	staff-only note

D.18 Payments (`payments`)

Item	Value
Menu group	Sales
Feature / phase	Payments / Phase 3
API base	`/admin/payments`
Files	`payments_admin_manage_screen.dart` , `payments_admin_manage.ts` , `payments_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: Edit

On the screen	Contents
Number cards (each opens the records behind it)	Successful | Failed | Pending | ollected
Tabs	All payments | Failed and stuck | Reconciliation
Filters	Search | Status [All / Success / Failed / Pending / Refunded] | Mode [All / Upi / Card / Netbanking / Wallet]
Table columns	Payment | PNR | Customer | Amount | Mode | Status | Time ▼ | Actions
Tab "All payments"	cards: Successful, Failed, Pending, ollected. filters: Search, Status [All / Success / Failed / Pending / Refunded], Mode [All / Upi / Card / Netbanking / Wallet]. columns: Payment, PNR, Customer, Amount, Mode, Status, Time ▼, Actions
Tab "Failed and stuck"	cards: Successful, Failed, Pending, ollected. filters: Search, Status [All / Failed / Pending]. columns: Payment, PNR, Customer, Amount, Mode, Status, Time, Actions
Tab "Reconciliation"	cards: Successful, Failed, Pending, ollected. filters: Search, Status [All / Open / Done]. columns: Case, Payment, PNR, Problem, Amount, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/payments/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Payment PAY-9652 (BookingBK-8153PNRABRK5152ACustomerRahul VermaOperatorKaveri ExpressAmount₹2,044M)	GET /admin/payments/:id	New	detail with gateway ids
Page	View	Payment PAY-9652 (BookingBK-8153PNRABRK5152ACustomerRahul VermaOperatorKaveri ExpressAmount₹2,044M)	GET /admin/payments/:id	New	detail with gateway ids
Tab: All payments	Export CSV		GET /admin/payments/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: All payments	Table		No server call	UI	table or list view of the same data
Tab: All payments	List		No server call	UI	table or list view of the same data
Tab: All payments	Row (open the record)	Payment PAY-9652 (BookingBK-8153PNRABRK5152ACustomerRahul VermaOperatorKaveri ExpressAmount₹2,044M)	GET /admin/payments/:id	New	detail with gateway ids
Tab: All payments	View	Payment PAY-9652 (BookingBK-8153PNRABRK5152ACustomerRahul VermaOperatorKaveri ExpressAmount₹2,044M)	GET /admin/payments/:id	New	detail with gateway ids
Tab: Failed and stuck	Export CSV		GET /admin/payments/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Failed and stuck	Table		No server call	UI	table or list view of the same data
Tab: Failed and stuck	List		No server call	UI	table or list view of the same data
Tab: Failed and stuck	Row (open the record)	Payment PAY-9004 (BookingBK-7004PNRABRT9471ECustomerNeha BhatOperatorTungabhadra ExpressAmount₹2,7)	GET /admin/payments/:id	New	detail with gateway ids
Tab: Failed and stuck	View	Payment PAY-9004 (BookingBK-7004PNRABRT9471ECustomerNeha BhatOperatorTungabhadra ExpressAmount₹2,7)	GET /admin/payments/:id	New	detail with gateway ids
Tab: Reconciliation	Export CSV		GET /admin/payments/export?format=csv|xlsx	New	same filters, search and permissions as the list

Where	Button	Opens	Endpoint	Status	Notes
Tab: Reconciliation	Table		No server call	UI	table or list view of the same data
Tab: Reconciliation	List		No server call	UI	table or list view of the same data
Tab: Reconciliation	Mark reconciled		POST /admin/payments/reconciliation/:id/resolve	New	marks reconciled; audit
Tab: Reconciliation	Check with gateway		POST /admin/payments/:id/verify	New	asks Razorpay for the real status and updates
Tab: Reconciliation	View	ABRX7058A (CaseRC-100PaymentPAY-9001PNRABRX7058AProblemAmount differs by Rs 49Amount₹1,317.)	GET /admin/payments/:id	New	read one record for the detail window
Tab: Reconciliation > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/payments/:id
Tab: Reconciliation > dialog	Delete		DELETE /admin/payments/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/payments/:id/restore

D.19 Refunds (`refunds`)

Item	Value
Menu group	Sales
Feature / phase	Cancellation and refunds / Phase 4
API base	`/admin/refunds`
Files	`refunds_admin_manage_screen.dart` , `refunds_admin_manage.ts` , `refunds_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: Edit

On the screen	Contents
Number cards (each opens the records behind it)	Waiting for approval | Processing | ompleted | Failed
Filters	Search | Status [All / Initiated / Processing / Completed / Failed] | Paid by [All / Operator / Platform]
Table columns	Refund | Customer | Amount | Reason | Proof | Paid by | Status | Started ▼ | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/refunds/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	ABRX7013B (RefundRF-4001BookingidBK-7011PnrABRX7013BCustomerAnjali DasOperatorVindhya Trave)	GET /admin/refunds/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/refunds/:id
Page	Mark paid		POST /admin/refunds/:id/mark-paid	New	for manual refunds; reference required; messages the customer

D.20 Settlements (`settlements`)

Item	Value
Menu group	Sales
Feature / phase	Settlements and earnings / Phase 6
API base	`/admin/settlements`
Files	`settlements_admin_manage_screen.dart` , `settlements_admin_manage.ts` , `settlements_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: Edit

On the screen	Contents
Number cards (each opens the records behind it)	Still to pay | Paid this cycle | Waiting for approval | On hold
Filters	Search | Status [All / Draft / Approved / Paid / On hold] | Operator [All / Kaveri express / Godavari lines / Deccan voyager / Sahyadri coaches / Nilgiri travels / Charminar tours / Tungabhadra express / Konkan rider / Vindhya travels] | Mode [All / Automatic / Manual]
Table columns	Settlement | Operator | Gross | Commission | Refunds | Net payable | Mode | Status | Actions
Detail page #/settlements/:id	columns: PNR, Route, Fare, Commission, Operator gets, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Generate settlements	Generate settlements: Cycle [Last 7 days / Last 14 days / Custom range]; Operator [All operators / Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels ...]	No server call	UI	opens generate form
Page > dialog	Generate		POST /admin/settlements/generate	New	cycle and operator; gross minus commission minus refunds plus adjustments
Page	Payout file		GET /admin/settlements/payout-file?cycle=	New	bank file for manual mode
Page	GST statement		GET /admin/settlements/gst-statement?cycle=	New	file
Page	Export CSV		GET /admin/settlements/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/settlements/:id	GET /admin/settlements/:id	New	with booking lines
Page	View	Opens page #/settlements/:id	GET /admin/settlements/:id	New	with booking lines
Page	Edit	Edit ST-6001: Operator*; Gross (Rs)*; Commission (Rs)*; Refunds (Rs)*; Net payable (Rs)*; Mode [Auto / Manual]; Status [Draft / Paid / Approved / On hold]	No server call	UI	opens the edit form, filled from GET /admin/settlements/:id
Page > dialog	Save changes		No server call	UI	not allowed to edit totals; use adjustments
Detail page #/settlements/:id	Release hold		POST /admin/settlements/:id/release-hold	New	audit

Where	Button	Opens	Endpoint	Status	Notes
Detail page #/settlements/:id	Download statement		GET /admin/settlements/:id/statement.pdf	New	
Detail page #/settlements/:id	Add adjustment	Add adjustment: Amount (Rs, negative to deduct)*; Reason*	No server call	UI	opens adjustment form
Detail page #/settlements/:id > dialog	Save		POST /admin/settlements/:id/adjustments	New	signed amount and reason; audit
Detail page #/settlements/:id	Export CSV		GET /admin/settlements/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/settlements/:id	Table		No server call	UI	table or list view of the same data
Detail page #/settlements/:id	List		No server call	UI	table or list view of the same data
Detail page #/settlements/:id	View	ABRK1563A (IdBK-7075PNRABRK1563ACustomeridCU8CustomerKavya MenonPhone+91 9192297973TripidTR)	GET /admin/settlements/:id	New	read one record for the detail window
Detail page #/settlements/:id > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/settlements/:id

D.21 Customers (`customers`)

Item	Value
Menu group	Customers
Feature / phase	Customers and profile / Phase 4
API base	`/admin/customers`
Files	`customers_admin_manage_screen.dart` , `customers_admin_manage.ts` , `customers_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: None

On the screen	Contents
Filters	Name, phone, email | Status [All / Active / Suspended / Deactivated] | Sign-in method [All / Google / Whatsapp otp / Apple / Password]
Table columns	Customer | Phone | Signs in with | Status | Bookings | Spent | Joined | Actions
Detail page #/customers/:id	columns: PNR, Route, Travel, Amount, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/customers/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/customers/:id	GET /admin/customers/:id	New	with bookings, tickets, SOS, devices
Page	View	Opens page #/customers/:id	GET /admin/customers/:id	New	with bookings, tickets, SOS, devices
Page	Edit	Edit Aarav Sharma: Customer*; Phone*; Signs in with*; Status [Active / Suspended / Deactivated]; Bookings*; Spent (Rs)*; Joined*	No server call	UI	opens the edit form, filled from GET /admin/customers/:id
Page > dialog	Save changes		PATCH /admin/customers/:id	New	save changes; audit before and after
Page	Delete	Delete Aarav Sharma? (This removes it from every list. You can undo for a few seconds.)	DELETE /admin/customers/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/customers/:id/restore
Page > dialog	Delete		DELETE /admin/customers/:id	New	confirm of a delete or revoke; audit
Detail page #/customers/:id	Block account	Block account: Reason*	No server call	UI	opens block form (reason, proof)
Detail page #/customers/:id > dialog	Block		POST /admin/customers/:id/block	New	revokes sessions; audit
Detail page #/customers/:id	Sign out all devices		POST /admin/customers/:id/sessions/revoke	New	sign out of all devices
Detail page #/customers/:id	Export CSV		GET /admin/customers/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/customers/:id	Table		No server call	UI	table or list view of the same data
Detail page #/customers/:id	List		No server call	UI	table or list view of the same data
Detail page #/customers/:id	View	Opens page #/bookings/:id	GET /admin/bookings/:id	New	detail with passengers, payment, refund, messages
Detail page #/customers/:id	Edit	Edit ABRX7158B: PNR*; Route*; Travel*; Amount (Rs)*; Status [Confirmed / Completed / Cancelled]	No server call	UI	opens the edit form, filled from GET /admin/customers/:id
Detail page #/customers/:id > dialog	Save changes		PATCH /admin/customers/:id	New	save changes; audit before and after

D.22 Support tickets (`tickets`)

Item	Value
Menu group	Customers
Feature / phase	Support tickets / Phase 4
API base	`/admin/tickets`
Files	`support_admin_manage_screen.dart` , `support_admin_manage.ts` , `support_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	Open | In progress | Past time limit | Resolved
Filters	Search | Status [All / Open / In progress / Resolved / Closed] | Category [All / Booking / Payment / Refund / Safety / App] | Priority [All / Urgent / High / Medium / Low] | Assigned [All / Unassigned / Divya / Kiran]

On the screen	Contents
Table columns	Ticket | Customer | Category | Priority | Status | Assigned to | Time limit | Actions
Detail page #/tickets/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/tickets/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/tickets/:id	GET /admin/tickets/:id	New	thread
Page	View	Opens page #/tickets/:id	GET /admin/tickets/:id	New	thread
Page	Edit	Edit Refund not received: Customer*; Category [Refund / Booking / Payment / App / Safety]; Priority [High / Medium / Urgent / Low]; Status [Resolved / In progress / Open / Closed]; Assigned to*; Time limit*	No server call	UI	opens the edit form, filled from GET /admin/tickets/:id
Page > dialog	Save changes		PATCH /admin/tickets/:id	New	save changes; audit before and after
Page	Delete	Delete Refund not received? (This removes it from every list. You can undo for a few seconds.)	DELETE /admin/tickets/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/tickets/:id/restore
Page > dialog	Delete		DELETE /admin/tickets/:id	New	confirm of a delete or revoke; audit
Detail page #/tickets/:id	Assign	Assign ticket: Assign to [Nobody / Divya (Support) / Kiran (Support)]	No server call	UI	opens assign form
Detail page #/tickets/:id > dialog	Assign		POST /admin/tickets/:id/assign	New	notifies the assignee
Detail page #/tickets/:id	Mark resolved		POST /admin/tickets/:id/resolve	New	audit
Detail page #/tickets/:id	Send reply		POST /admin/tickets/:id/replies	New	reply with proof files; notifies the customer

D.23 Help content and contact (`content`)

Item	Value
Menu group	Customers
Feature / phase	Information pages and FAQs / Phase 4
API base	`/admin/content`
Files	`content_admin_manage_screen.dart` , `content_admin_manage.ts` , `content_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: None

On the screen	Contents
Tabs	FAQs | Policy pages | Contact and hours
Filters	Search | Status [All / Published / Draft] | Category [All / Bookings / Refunds / Payments / Safety / Trips / Offers]
Table columns	Question | Category | Status | Views | Updated | Actions
Tab "FAQs"	filters: Search, Status [All / Published / Draft], Category [All / Bookings / Refunds / Payments / Safety / Trips / Offers]. columns: Question, Category, Status, Views, Updated, Actions
Tab "Policy pages"	columns: Page, Status, Version, Updated, Actions
Tab "Contact and hours"	filters: Support phone, WhatsApp number, Support email, Working hours, Emergency number

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/content/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add FAQ	Add FAQ: Question*; Answer*; Category [Bookings / Refunds / Payments / Safety / Trips / Offers]; Status [DRAFT / PUBLISHED]	No server call	UI	opens FAQ form
Page > dialog	Save		POST /admin/faqs or PATCH /admin/faqs/:id	New	
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Edit	Edit FAQ: Question*; Answer*; Category [Bookings / Refunds / Payments / Safety / Trips / Offers]; Status [DRAFT / PUBLISHED]	No server call	UI	opens FAQ form
Page	Unpublish		POST /admin/faqs/:id/publish or /unpublish	New	
Page	View	FAQ1 (IdFAQ1QuestionHow do I cancel a booking?CategoryBookingsStatusDraftAThis is the)	GET /admin/content/:id	New	read one record for the detail window
Page > dialog	Delete		DELETE /admin/content/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/content/:id/restore
Page	Publish		POST /admin/faqs/:id/publish or /unpublish	New	
Tab: FAQs	Export CSV		GET /admin/content/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: FAQs	Add FAQ	Add FAQ: Question*; Answer*; Category [Bookings / Refunds / Payments / Safety / Trips / Offers]; Status [DRAFT / PUBLISHED]	No server call	UI	opens FAQ form
Tab: FAQs > dialog	Save		POST /admin/faqs or PATCH /admin/faqs/:id	New	
Tab: FAQs	Table		No server call	UI	table or list view of the same data
Tab: FAQs	List		No server call	UI	table or list view of the same data
Tab: FAQs	Edit	Edit FAQ: Question*; Answer*; Category [Bookings / Refunds / Payments / Safety / Trips / Offers]; Status [DRAFT / PUBLISHED]	No server call	UI	opens FAQ form
Tab: FAQs	Unpublish		POST /admin/faqs/:id/publish or /unpublish	New	
Tab: FAQs	View	FAQ1 (IdFAQ1QuestionHow do I cancel a booking?CategoryBookingsStatusDraftAThis is the)	GET /admin/content/:id	New	read one record for the detail window
Tab: FAQs > dialog	Delete		DELETE /admin/content/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/content/:id/restore

Where	Button	Opens	Endpoint	Status	Notes
Tab: FAQs	Publish		POST /admin/faqs/:id/publish or /unpublish	New	
Tab: Policy pages	Export CSV		GET /admin/content/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Policy pages	Table		No server call	UI	table or list view of the same data
Tab: Policy pages	List		No server call	UI	table or list view of the same data
Tab: Policy pages	Edit	Edit Terms and Conditions: Title; Content	No server call	UI	opens page editor
Tab: Policy pages > dialog	Save draft		PUT /admin/pages/:slug	New	draft or publish; creates a new version
Tab: Policy pages > dialog	Publish		PUT /admin/pages/:slug	New	draft or publish; creates a new version
Tab: Policy pages	Versions	Versions of Terms and Conditions (Version 412 Oct 2026Version 312 Sept 2026Version 213 Aug 2026Version 114 Jul 202)	GET /admin/pages/:slug/versions	New	
Tab: Policy pages	View	Terms and Conditions (IdPG1PageTerms and ConditionsStatusPublishedVersion4Updated12 Oct 2026, 05:39 pm)	GET /admin/content/:id	New	read one record for the detail window
Tab: Policy pages > dialog	Delete		DELETE /admin/content/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/content/:id/restore
Tab: Contact and hours	Save contact details		PUT /admin/contact-info	New	phone, WhatsApp, email, hours, emergency number

D.24 Messages sent (`messages`)

Item	Value
Menu group	Customers
Feature / phase	WhatsApp and email messages / Phase 0
API base	`/admin/messages`
Files	`messages_admin_manage_screen.dart` , `messages_admin_manage.ts` , `messages_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: View; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	Sent today | WhatsApp messages | Emails | Failed
Filters	Customer, phone, email, PNR | Channel [All / WhatsApp / Email] | Type [All / Booking confirmed / Booking cancelled]
Table columns	When ▼ | Channel | To | Type | Subject | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/messages/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Booking confirmed: ABRK5151A (ChannelEmailToAnanya Reddy - ananya.reddy@example.comTypeBooking confirmedSent12)	GET /admin/messages/:id	New	
Page > dialog	Send again		POST /admin/messages/:id/resend	New	queues again
Page	View	Booking confirmed: ABRK5151A (ChannelEmailToAnanya Reddy - ananya.reddy@example.comTypeBooking confirmedSent12)	GET /admin/messages/:id	New	
Page	Edit	Edit Ananya Reddy: When*; Channel [Email / Whatsapp]; To*; Type*; Subject*; Status*	No server call	UI	opens the edit form, filled from GET /admin/messages/:id
Page > dialog	Save changes		PATCH /admin/messages/:id	New	save changes; audit before and after
Page	Delete	Delete Ananya Reddy? (This removes it from every list. You can undo for a few seconds.)	DELETE /admin/messages/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/messages/:id/restore
Page > dialog	Delete		DELETE /admin/messages/:id	New	confirm of a delete or revoke; audit

D.25 Banners (`banners`)

Item	Value
Menu group	Marketing
Feature / phase	Banners / Phase 1
API base	`/admin/banners`
Files	`banners_admin_manage_screen.dart` , `banners_admin_manage.ts` , `banners_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Tabs	All banners | Analytics
Filters	Search | Status [All / Draft / Scheduled / Published / Archived]
Table columns	Banner | Status | Runs | Shown to | Opens | Order ▼ | Actions
Tab "Analytics"	columns: Banner, Impressions, Clicks, CTR, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/banners/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	New banner	New banner: Title*; Subtitle; Colour style [Navy to blue / Blue to sky / Green to navy]; Opens [An offer / Route search / An operator / A web link]; Offer code [None / ABRA100 / WEEKEND15 / UPIFEST / WOMEN10 / KAVERI50 ...]; Shown to [Everyone / New users / Signed-in users]; Starts; Ends; Order (higher shows first)	No server call	UI	opens banner form
Page > dialog	Save draft		POST /admin/banners or PUT /admin/banners/:id	Exists	
Page > dialog	Publish		POST /admin/banners or PUT /admin/banners/:id	Exists	
Page	Table		No server call	UI	table or list view of the same data

Where	Button	Opens	Endpoint	Status	Notes
Page	List		No server call	UI	table or list view of the same data
Page	(icon)		PATCH /admin/banners/order	New	move up or down
Page	Edit	Edit banner: Title*; Subtitle; Colour style [Navy to blue / Blue to sky / Green to navy]; Opens [An offer / Route search / An operator / A web link]; Offer code [None / ABRA100 / WEEKEND15 / UPIFEST / WOMEN10 / KAVERI50 ...]; Shown to [Everyone / New users / Signed-in users]; Starts; Ends; Order (higher shows first)	No server call	UI	opens banner form
Page	Unpublish		POST /admin/banners/:id/active	Exists	(unpublish)
Page	Duplicate		POST /admin/banners/:id/duplicate	New	new
Page	Archive		POST /admin/banners/:id/archive	New	new
Page	View	Festive special (copy) (IdBN-90001BannerFestive special (copy)SubFlat 15% off on selected routesCodeWEEK)	GET /admin/banners/:id	Exists	read one record for the detail window
Page > dialog	Delete		DELETE /admin/banners/:id	Exists	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/banners/:id/restore
Page	Publish		POST /admin/banners/:id/active	Exists	(publish)
Tab: Analytics	Export CSV		GET /admin/banners/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Analytics	Table		No server call	UI	table or list view of the same data
Tab: Analytics	List		No server call	UI	table or list view of the same data
Tab: Analytics	View	Festive special (IdBN1BannerFestive specialSubFlat 15% off on selected routesCodeWEEKEND15StatusP)	GET /admin/banners/:id	Exists	read one record for the detail window
Tab: Analytics > dialog	Edit		No server call	UI	opens banner form
Tab: Analytics > dialog	Delete		DELETE /admin/banners/:id	Exists	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/banners/:id/restore

D.26 Offers and coupons (`offers`)

Item	Value
Menu group	Marketing
Feature / phase	Offers and coupons / Phase 3
API base	`/admin/offers`
Files	`offers_admin_manage_screen.dart` , `offers_admin_manage.ts` , `offers_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Tabs	All offers | Test a coupon | Usage report
Filters	Search | Status [All / Active / Paused / Expired / Draft] | Funded by [All / Platform / Operator]
Table columns	Code | Discount | Applies to | Valid till | Used | Funded by | Status | Actions
Tab "All offers"	filters: Search, Status [All / Active / Paused / Expired / Draft], Funded by [All / Platform / Operator]. columns: Code, Discount, Applies to, Valid till, Used, Funded by, Status, Actions
Tab "Test a coupon"	filters: Code [ABRA100 / WEEKEND15 / UPIFEST / WOMEN10 / KAVERI50 / DIWALI20 / HYDBLR / SLEEP10], Booking amount (Rs), Customer [New customer / Existing customer], Day [Mon / Tue / Wed / Thu / Fri / Sat / Sun]
Tab "Usage report"	columns: Code, Times used, Discount given, Funded by, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/offers/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	New offer	New offer: Code*; Description (shown in the app)*; Type [Percentage / Flat amount]; Value (percent, or rupees); Maximum discount (Rs); Minimum booking (Rs); Valid from; Valid till; Total usage limit (0 = no limit); Uses per customer; Funded by [Abra Travels / The operator]; Who can use it [Everyone / New customers only / Women travellers]; Limit to (optional) [All / Route: Hyderabad to Bengaluru / Route: Bengaluru to Hyderabad / Route: Hyderabad to Chennai / Route: Chennai to Coimbatore / Route: Vijayawada to Visakhapatnam ...]; Terms (one per line)	No server call	UI	opens offer form
Page > dialog	Save offer		POST /admin/offers or PATCH /admin/offers/:id	New	audit
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	ABRA100 (DescriptionRs 100 off your first bookingDiscount₹100Minimum booking₹0Valid22 Sep)	GET /admin/offers/:id	New	detail with usage
Page	Edit	Edit ABRA100: Code*; Description (shown in the app)*; Type [Percentage / Flat amount]; Value (percent, or rupees); Maximum discount (Rs); Minimum booking (Rs); Valid from; Valid till; Total usage limit (0 = no limit); Uses per customer; Funded by [Abra Travels / The operator]; Who can use it [Everyone / New customers only / Women travellers]; Limit to (optional) [All / Route: Hyderabad to Bengaluru / Route: Bengaluru to Hyderabad / Route: Hyderabad to Chennai / Route: Chennai to Coimbatore / Route: Vijayawada to Visakhapatnam ...]; Terms (one per line)	No server call	UI	opens offer form
Page	Pause		PATCH /admin/offers/:id/active	New	pause or resume
Page	View	ABRA100 (DescriptionRs 100 off your first bookingDiscount₹100Minimum booking₹0Valid22 Sep)	GET /admin/offers/:id	New	detail with usage
Page	Delete	Delete ABRA100? (This removes it from every list. You can undo for a few seconds.)	DELETE /admin/offers/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/offers/:id/restore
Page > dialog	Delete		DELETE /admin/offers/:id	New	confirm of a delete or revoke; audit
Page	Resume		PATCH /admin/offers/:id/active	New	pause or resume
Tab: All offers	Export CSV		GET /admin/offers/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: All offers	New offer	New offer: Code*; Description (shown in the app)*; Type [Percentage / Flat amount]; Value (percent, or rupees); Maximum discount (Rs); Minimum booking (Rs); Valid from; Valid till; Total usage limit (0 = no limit); Uses per customer; Funded by [Abra Travels / The operator]; Who can use it [Everyone / New customers only / Women travellers]; Limit to (optional) [All / Route: Hyderabad to Bengaluru / Route: Bengaluru to Hyderabad / Route: Hyderabad to Chennai / Route: Chennai to Coimbatore / Route: Vijayawada to Visakhapatnam ...]; Terms (one per line)	No server call	UI	opens offer form

Where	Button	Opens	Endpoint	Status	Notes
Tab: All offers > dialog	Save offer		POST /admin/offers or PATCH /admin/offers/:id	New	audit
Tab: All offers	Table		No server call	UI	table or list view of the same data
Tab: All offers	List		No server call	UI	table or list view of the same data
Tab: All offers	Row (open the record)	ABRA100 (DescriptionRs 100 off your first bookingDiscount₹100Minimum booking₹0Valid22 Sep)	GET /admin/offers/:id	New	detail with usage
Tab: All offers	Edit	Edit ABRA100: Code*; Description (shown in the app)*; Type [Percentage / Flat amount]; Value (percent, or rupees); Maximum discount (Rs); Minimum booking (Rs); Valid from; Valid till; Total usage limit (0 = no limit); Uses per customer; Funded by [Abra Travels / The operator]; Who can use it [Everyone / New customers only / Women travellers]; Limit to (optional) [All / Route: Hyderabad to Bengaluru / Route: Bengaluru to Hyderabad / Route: Hyderabad to Chennai / Route: Chennai to Coimbatore / Route: Vijayawada to Visakhapatnam ...]; Terms (one per line)	No server call	UI	opens offer form
Tab: All offers	Pause		PATCH /admin/offers/:id/active	New	pause or resume
Tab: All offers	View	ABRA100 (DescriptionRs 100 off your first bookingDiscount₹100Minimum booking₹0Valid22 Sep)	GET /admin/offers/:id	New	detail with usage
Tab: All offers	Delete	Delete ABRA100? (This removes it from every list. You can undo for a few seconds.)	DELETE /admin/offers/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/offers/:id/restore
Tab: All offers > dialog	Delete		DELETE /admin/offers/:id	New	confirm of a delete or revoke; audit
Tab: All offers	Resume		PATCH /admin/offers/:id/active	New	pause or resume
Tab: Test a coupon	Check		POST /admin/offers/test	New	checks a coupon for a sample trip and user
Tab: Usage report	Export CSV		GET /admin/offers/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Usage report	Table		No server call	UI	table or list view of the same data
Tab: Usage report	List		No server call	UI	table or list view of the same data
Tab: Usage report	View	ABRA100 (IdOF1CodeABRA100DescRs 100 off your first bookingTypeFlatValue10000Maxdisc₹300Mi)	GET /admin/offers/:id	New	read one record for the detail window
Tab: Usage report > dialog	Edit		No server call	UI	opens offer form
Tab: Usage report > dialog	Delete		DELETE /admin/offers/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/offers/:id/restore

D.27 Announcements (`announce`)

Item	Value
Menu group	Marketing
Feature / phase	Announcements and popular routes / Phase 7
API base	`/admin/announcements`
Files	`announce_admin_manage_screen.dart` , `announce_admin_manage.ts` , `announce_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: None

On the screen	Contents
Filters	Search | Status [All / Sent / Scheduled / Draft]
Table columns	Message | Sent to | Channel | Status | Reached | When | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	New announcement	New announcement: Title*; Message*; Send to [All customers / Customers on a route / Customers of one trip / New users / Women travellers]; Channel [Push / Push, WhatsApp / WhatsApp]; When [Send now / Schedule]; Date and time (if scheduled)	No server call	UI	opens announcement form
Page > dialog	Save draft		POST /admin/announcements	New	draft, schedule or send now; send creates notifications and messages for the audience
Page > dialog	Send		POST /admin/announcements	New	draft, schedule or send now; send creates notifications and messages for the audience
Page	Export CSV		GET /admin/announcements/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	Heavy rain advisory on Hyderabad to Visakhapatnam (IdAN1MessageHeavy rain advisory on Hyderabad to VisakhapatnamBodyTrips may run l)	GET /admin/announcements/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/announcements/:id
Page > dialog	Delete		DELETE /admin/announcements/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/announcements/:id/restore

D.28 Popular routes (`popular`)

Item	Value
Menu group	Marketing
Feature / phase	Announcements and popular routes / Phase 7
API base	`/admin/popular-routes`
Files	`announce_admin_manage_screen.dart` , `announce_admin_manage.ts` , `announce_admin_manage.service.ts`

Item	Value
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Table columns	Order ▲ | Route | Searches (30 d) | Bookings (30 d) | Shown | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Add route	Add popular route: Route [Mumbai to Pune / Hyderabad to Tirupati / Bengaluru to Kochi / Tirupati to Chennai]	No server call	UI	opens add form
Page > dialog	Add		POST /admin/popular-routes	New	
Page	Export CSV		GET /admin/popular-routes/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	(icon)		PATCH /admin/popular-routes/order	New	
Page	(icon)		PATCH /admin/popular-routes/:id/active	New	
Page	View	Hyderabad to Bengaluru (IdPO1RouteHyderabad to BengaluruOrder1ShownNoSearches (30 d)2247Bookings (30 d)4)	GET /admin/popular-routes/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/popular-routes/:id
Page > dialog	Delete		DELETE /admin/popular-routes/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/popular-routes/:id/restore

D.29 SOS board (`sos`)

Item	Value
Menu group	Safety
Feature / phase	SOS and women safety / Phase 5
API base	`/admin/sos`
Files	`sos_admin_manage_screen.dart` , `sos_admin_manage.ts` , `sos_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: None

On the screen	Contents
Sections of the page	Left: a card for each SOS alert (customer name, status, "Raised by customer or driver", route, time). Right: the selected alert with map position, customer, driver, operator, bus, contacts alerted, the log of what happened, and the action buttons
Detail page #/sos/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	Row (open the record)	Opens page #/sos/:id	GET /admin/sos/:id	New	board card and detail; new alerts arrive by Socket.IO
Detail page #/sos/:id	Call		No server call	UI	dials the number
Detail page #/sos/:id	Customer		No server call	UI	dials the number
Detail page #/sos/:id	Driver		No server call	UI	dials the number
Detail page #/sos/:id	Operator		No server call	UI	dials the number
Detail page #/sos/:id	Police		No server call	UI	dials the number
Detail page #/sos/:id	Acknowledge		POST /admin/sos/:id/acknowledge	New	tells the customer or driver and the operator
Detail page #/sos/:id	Add note	Add note: Note*	No server call	UI	opens note form
Detail page #/sos/:id > dialog	Save		POST /admin/sos/:id/notes	New	
Detail page #/sos/:id	Escalate		POST /admin/sos/:id/escalate	New	police and safety lead; audit
Detail page #/sos/:id	Resolve	Resolve SOS: How was it resolved?*	No server call	UI	opens resolve form (note and proof required)
Detail page #/sos/:id > dialog	Resolve		POST /admin/sos/:id/resolve	New	note, proof; notifies the customer or driver
Detail page #/sos/:id	False alarm	Mark as false alarm: What did the customer say?*	No server call	UI	opens false alarm form (proof)
Detail page #/sos/:id > dialog	Mark false alarm		POST /admin/sos/:id/false-alarm	New	what the customer said, proof

D.30 Incidents and sharing (`incidents`)

Item	Value
Menu group	Safety
Feature / phase	Incidents / Phase 5
API base	`/admin/incidents`
Files	`incidents_admin_manage_screen.dart` , `incidents_admin_manage.ts` , `incidents_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: None

On the screen	Contents
Tabs	Incidents | Trip sharing audit
Filters	Search | Status [All / Open / In progress / Closed] | Severity [All / High / Medium / Low]
Table columns	Incident | Reported by | About | Severity | Status | Trip | Reported | Actions
Tab "Incidents"	filters: Search, Status [All / Open / In progress / Closed], Severity [All / High / Medium / Low]. columns: Incident, Reported by, About, Severity, Status, Trip, Reported, Actions
Tab "Trip sharing audit"	filters: Search, Status [All / Active / Expired]. columns: Customer, Trip, Link, Times opened, Shared, Expires, Status, Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/incidents/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Log incident	Log incident: What happened*; Reported by [Aarav Sharma / Priya Nair / Rohan Mehta / Ananya Reddy / Vikram Singh / Divya Iyer ...]; About [Driver / Conductor / Co-passenger / Operator]; Severity [HIGH / MEDIUM / LOW]; Trip [TRP-3001 / TRP-3002 / TRP-3003 / TRP-3004 / TRP-3005 / TRP-3006 ...]	No server call	UI	opens incident form
Page > dialog	Log		POST /admin/incidents	New	with proof

Where	Button	Opens	Endpoint	Status	Notes
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	INC-3001 - Rash driving reported: Change status [OPEN / IN_PROGRESS / CLOSED]; Add note	GET /admin/incidents/:id	New	
Page > dialog	Save		PATCH /admin/incidents/:id	New	status and note; audit
Page	View	INC-3001 - Rash driving reported: Change status [OPEN / IN_PROGRESS / CLOSED]; Add note	GET /admin/incidents/:id	New	
Page	Edit	Edit Rash driving reported: Reported by*; About*; Severity [Medium / High / Low]; Status [Closed / Open / In progress]; Trip*; Reported*	No server call	UI	opens the edit form, filled from GET /admin/incidents/:id
Page > dialog	Save changes		PATCH /admin/incidents/:id	New	save changes; audit before and after
Tab: Incidents	Export CSV		GET /admin/incidents/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Incidents	Log incident	Log incident: What happened*; Reported by [Aarav Sharma / Priya Nair / Rohan Mehta / Ananya Reddy / Vikram Singh / Divya Iyer ...]; About [Driver / Conductor / Co-passenger / Operator]; Severity [HIGH / MEDIUM / LOW]; Trip [TRP-3001 / TRP-3002 / TRP-3003 / TRP-3004 / TRP-3005 / TRP-3006 ...]	No server call	UI	opens incident form
Tab: Incidents > dialog	Log		POST /admin/incidents	New	with proof
Tab: Incidents	Table		No server call	UI	table or list view of the same data
Tab: Incidents	List		No server call	UI	table or list view of the same data
Tab: Incidents	Row (open the record)	INC-3001 - Rash driving reported: Change status [OPEN / IN_PROGRESS / CLOSED]; Add note	GET /admin/incidents/:id	New	
Tab: Incidents > dialog	Save		PATCH /admin/incidents/:id	New	status and note; audit
Tab: Incidents	View	INC-3001 - Rash driving reported: Change status [OPEN / IN_PROGRESS / CLOSED]; Add note	GET /admin/incidents/:id	New	
Tab: Incidents	Edit	Edit Rash driving reported: Reported by*; About*; Severity [Medium / High / Low]; Status [Closed / Open / In progress]; Trip*; Reported*	No server call	UI	opens the edit form, filled from GET /admin/incidents/:id
Tab: Incidents > dialog	Save changes		PATCH /admin/incidents/:id	New	save changes; audit before and after
Tab: Trip sharing audit	Export CSV		GET /admin/incidents/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Trip sharing audit	Table		No server call	UI	table or list view of the same data
Tab: Trip sharing audit	List		No server call	UI	table or list view of the same data
Tab: Trip sharing audit	Revoke link		DELETE /admin/trip-shares/:id	New	revokes a sharing link; audit
Tab: Trip sharing audit	View	SH1 (IdSH1CustomerAarav SharmaTripTRP-3004Linkabra.in/s/349787Times opened4Shared12 O)	GET /admin/incidents/:id	New	read one record for the detail window
Tab: Trip sharing audit > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/incidents/:id
Tab: Trip sharing audit > dialog	Delete		DELETE /admin/incidents/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /admin/incidents/:id/restore

D.31 Safety settings (`safety`)

Item	Value
Menu group	Safety
Feature / phase	SOS and women safety / Phase 5
API base	`/admin/safety`
Files	`sos_admin_manage_screen.dart` , `sos_admin_manage.ts` , `sos_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Sections of the page	Safety settings | Escalation timing (acknowledge minutes, escalate minutes, SOS hold seconds) | Who is alerted (operator, SMS, WhatsApp switches) | Safety team (people who receive alerts) | Message sent to emergency contacts
Filters	Acknowledge within (minutes) | Escalate to safety lead after (minutes) | Police control room number

Where	Button	Opens	Endpoint	Status	Notes
Page	Save		PUT /admin/safety/settings	New	acknowledge minutes, escalate minutes, police number, SOS hold seconds; audit
Page	(icon)		PUT /admin/safety/settings	New	SMS switch
Page	(icon)		PUT /admin/safety/settings	New	WhatsApp switch
Page	(icon)		PUT /admin/safety/settings	New	notify operator switch
Page	Add person	Add to safety team: Name and role*	No server call	UI	opens add form
Page > dialog	Add		POST /admin/safety/team	New	name and role
Page	Remove		DELETE /admin/safety/team/:id	New	

D.32 Reports (`reports`)

Item	Value
Menu group	Insights
Feature / phase	Dashboards and reports / Phase 7
API base	`/admin/reports`
Files	`dashboard_admin_manage_screen.dart` , `dashboard_admin_manage.ts` , `dashboard_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: View

On the screen	Contents
Sections of the page	Reports (Sales, Operators, Routes) | Gross sales by day chart | rows that open their detail inline on the same page | Export these rows | Print

On the screen	Contents
Tabs	Sales | Operators | Routes | Cancellations | Refunds | Occupancy | Commission
Filters	Search
Table columns	Day | Bookings | Gross sales | Commission
Tab "Sales"	filters: Search. columns: PNR, Passenger, Route, Operator, Amount, Commission, Status. columns: Day, Bookings, Gross sales, Commission
Tab "Operators"	filters: Search. columns: Operator, Bookings, Sales, Cancelled, Rating
Tab "Routes"	filters: Search. columns: Route, Bookings, Sales, Occupancy
Tab "Cancellations"	filters: Search. columns: PNR, Route, Operator, Amount, Booked
Tab "Refunds"	filters: Search. columns: Refund, PNR, Amount, Paid by, Status, Actions
Tab "Occupancy"	filters: Search. columns: Trip, Route, Operator, Occupancy
Tab "Commission"	filters: Search. columns: Operator, Bookings, Commission earned

Where	Button	Opens	Endpoint	Status	Notes
Page	7 days		GET /admin/reports/:type?range=	New	
Page	30 days		GET /admin/reports/:type?range=	New	
Page	90 days		GET /admin/reports/:type?range=	New	
Page	Print		No server call	UI	browser print of the report
Page	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Page	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Tab: Sales	7 days		GET /admin/reports/:type?range=	New	
Tab: Sales	30 days		GET /admin/reports/:type?range=	New	
Tab: Sales	90 days		GET /admin/reports/:type?range=	New	
Tab: Sales	Print		No server call	UI	browser print of the report
Tab: Sales	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Sales	Export these rows		GET /admin/reports/:type/export	New	rows of the open detail
Tab: Sales	Close details		No server call	UI	closes the detail panel
Tab: Sales	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Sales	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Sales	Table		No server call	UI	table or list view of the same data
Tab: Sales	List		No server call	UI	table or list view of the same data
Tab: Operators	7 days		GET /admin/reports/:type?range=	New	
Tab: Operators	30 days		GET /admin/reports/:type?range=	New	
Tab: Operators	90 days		GET /admin/reports/:type?range=	New	
Tab: Operators	Print		No server call	UI	browser print of the report
Tab: Operators	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Operators	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Operators	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Operators	Table		No server call	UI	table or list view of the same data
Tab: Operators	List		No server call	UI	table or list view of the same data
Tab: Routes	7 days		GET /admin/reports/:type?range=	New	
Tab: Routes	30 days		GET /admin/reports/:type?range=	New	
Tab: Routes	90 days		GET /admin/reports/:type?range=	New	
Tab: Routes	Print		No server call	UI	browser print of the report
Tab: Routes	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Routes	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Routes	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Routes	Table		No server call	UI	table or list view of the same data
Tab: Routes	List		No server call	UI	table or list view of the same data
Tab: Cancellations	7 days		GET /admin/reports/:type?range=	New	
Tab: Cancellations	30 days		GET /admin/reports/:type?range=	New	
Tab: Cancellations	90 days		GET /admin/reports/:type?range=	New	
Tab: Cancellations	Print		No server call	UI	browser print of the report
Tab: Cancellations	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Cancellations	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Cancellations	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Cancellations	Table		No server call	UI	table or list view of the same data
Tab: Cancellations	List		No server call	UI	table or list view of the same data
Tab: Refunds	7 days		GET /admin/reports/:type?range=	New	
Tab: Refunds	30 days		GET /admin/reports/:type?range=	New	

Where	Button	Opens	Endpoint	Status	Notes
Tab: Refunds	90 days		GET /admin/reports/:type?range=	New	
Tab: Refunds	Print		No server call	UI	browser print of the report
Tab: Refunds	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Refunds	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Refunds	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Refunds	Table		No server call	UI	table or list view of the same data
Tab: Refunds	List		No server call	UI	table or list view of the same data
Tab: Refunds	View	ABRX7013B (RefundRF-4001BookingidBK-7011PNRABRX7013BCustomerAnjali DasOperatorVindhya Trave)	GET /admin/reports/:id	New	read one record for the detail window
Tab: Refunds > dialog	Edit		No server call	UI	opens the edit form, filled from GET /admin/reports/:id
Tab: Occupancy	7 days		GET /admin/reports/:type?range=	New	
Tab: Occupancy	30 days		GET /admin/reports/:type?range=	New	
Tab: Occupancy	90 days		GET /admin/reports/:type?range=	New	
Tab: Occupancy	Print		No server call	UI	browser print of the report
Tab: Occupancy	TRP-300193%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-300589%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-302289%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-300874%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-304270%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-302068%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-300264%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	TRP-300664%		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Occupancy	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Occupancy	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Occupancy	Table		No server call	UI	table or list view of the same data
Tab: Occupancy	List		No server call	UI	table or list view of the same data
Tab: Commission	7 days		GET /admin/reports/:type?range=	New	
Tab: Commission	30 days		GET /admin/reports/:type?range=	New	
Tab: Commission	90 days		GET /admin/reports/:type?range=	New	
Tab: Commission	Print		No server call	UI	browser print of the report
Tab: Commission	Row (open the record)		GET /admin/reports/:type/rows?key=	New	detail panel inline on the Reports page
Tab: Commission	Export CSV		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Commission	Excel (CSV)		GET /admin/reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Commission	Table		No server call	UI	table or list view of the same data
Tab: Commission	List		No server call	UI	table or list view of the same data

D.33 AI assistant (`ai`)

Item	Value
Menu group	Insights
Feature / phase	AI assistant (Groq) / Phase 0
API base	`/admin/settings/assistant`
Files	`assistant_admin_manage_screen.dart` , `assistant_admin_manage.ts` , `assistant_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit; Support: View; Accounts: View

On the screen	Contents
Sections of the page	AI assistant settings | Provider and model | Safety rule (what the assistant may and may not do) | What the assistant can do | Actions done by the assistant (log)
Number cards (each opens the records behind it)	Questions today | Actions done by AI | Waiting for confirmation | -Helpful answers
Filters	Provider | Model [llama-3.3-70b-versatile / openai/gpt-oss-120b / llama-3.1-8b-instant] | Messages per admin per day | Saved on the server only | Search
Table columns	Tool | What it does | Access needed | Confirmation

Where	Button	Opens	Endpoint	Status	Notes
Page	Test connection		POST /settings/integrations/assistant/test-connection	New	built; checks the Groq key and the model name
Page	Save		PUT /settings/integrations/assistant	New	built; provider, Groq key (secret), model, "may propose actions"; audit
Page	(icon)		No server call	UI	show or hide the API key field
Page	Export CSV		GET /admin/settings/assistant/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data

D.34 Settings and integrations (`settings`)

Item	Value
Menu group	System
Feature / phase	Integrations, feature flags and maintenance / Phase 0
API base	`/admin/settings`
Files	`settings_admin_manage_screen.dart` , `settings_admin_manage.ts` , `settings_admin_manage.service.ts`

Item	Value
Starting access	Super Admin: Edit+Delete; Admin: View; Support: None; Accounts: None

On the screen	Contents
Sections of the page	Settings and integrations | WhatsApp | Email (SMTP) | SMS | Razorpay | Maps | Push (Firebase) | Company | Booking rules | Security | Message templates
Tabs	Integrations | Company | Booking rules | Security | Message templates
Tab "Company"	filters: Company name, Email, Phone, GSTIN, Address
Tab "Booking rules"	filters: Seat hold time (seconds), Most seats per booking, Stop sales before departure (minutes), Book up to (days ahead)
Tab "Security"	filters: Sign out after (minutes of no activity), Ask for a new password every (days)
Tab "Message templates"	columns: Message, Channels, Languages

Where	Button	Opens	Endpoint	Status	Notes
Page	Set up	Set up whatsapp: Phone number id; Access token	No server call	UI	opens the integration form (values and write-only secrets)
Page > dialog	Save		PUT /settings/integrations/:key	Exists	built; key = razorpay, maps, sms, push or assistant (WhatsApp and email keep their own PUT routes, which); secrets stored encrypted, never returned; reason required; audit settings.integration.changed
Page	Test connection		POST /settings/integrations/:key/test-connection	Exists	built; asks the provider a harmless question; nothing is sent to a customer; WhatsApp and email have their own test routes ()
Tab: Integrations	Set up	Set up whatsapp: Phone number id; Access token	No server call	UI	opens the integration form (values and write-only secrets)
Tab: Integrations > dialog	Save		PUT /settings/integrations/:key	Exists	built; key = razorpay, maps, sms, push or assistant (WhatsApp and email keep their own PUT routes, which); secrets stored encrypted, never returned; reason required; audit settings.integration.changed
Tab: Integrations	Test connection		POST /settings/integrations/:key/test-connection	Exists	built; asks the provider a harmless question; nothing is sent to a customer; WhatsApp and email have their own test routes ()
Tab: Company	Save		PUT /admin/settings/company	New	audit
Tab: Booking rules	Save	Reason for change: Why are you changing this?*	No server call	UI	opens the reason form
Tab: Booking rules > dialog	Confirm		PUT /admin/settings/booking	New	hold minutes, seat limit and similar; a reason is required; audit
Tab: Security	(icon)		PUT /admin/settings/security/mfa	New	require MFA for roles
Tab: Security	Save		PUT /admin/settings/security	New	password and session rules; audit
Tab: Message templates	Table		No server call	UI	table or list view of the same data
Tab: Message templates	List		No server call	UI	table or list view of the same data
Tab: Message templates	Preview	Message preview (Booking confirmedHi Priya, your seat 12A on Hyderabad to Bengaluru is confirmed.)	GET /admin/message-templates/:id/preview	New	

D.35 Staff and roles (`staff`)

Item	Value
Menu group	System
Feature / phase	Staff, roles and access matrix / Phase 7
API base	`/admin/staff`
Files	`staff_admin_manage_screen.dart` , `staff_admin_manage.ts` , `staff_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: None; Support: None; Accounts: None

On the screen	Contents
Tabs	Staff | Permission matrix | Sessions
Filters	Search | Role [All / Super Admin / Admin / Support / Accounts] | Status [All / Active / Invited / Deactivated]
Table columns	Person | Role | Two-step | Status | Overrides | Last seen | Actions
Tab "Staff"	filters: Search, Role [All / Super Admin / Admin / Support / Accounts], Status [All / Active / Invited / Deactivated]. columns: Person, Role, Two-step, Status, Overrides, Last seen, Actions
Tab "Permission matrix"	filters: Set every area for Admin [Set all to... / None / View / Edit / Edit + Delete], Set every area for Support [Set all to... / None / View / Edit / Edit + Delete], Set every area for Accounts [Set all to... / None / View / Edit / Edit + Delete], Admin, Dashboard [None / View / Edit / Edit + Delete], Support, Dashboard [None / View / Edit / Edit + Delete], Accounts, Dashboard [None / View / Edit / Edit + Delete], Admin, Invitations [None / View / Edit / Edit + Delete], Support, Invitations [None / View / Edit / Edit + Delete], Accounts, Invitations [None / View / Edit / Edit + Delete], Admin, KYC reviews [None / View / Edit / Edit + Delete], Support, KYC reviews [None / View / Edit / Edit + Delete], Accounts, KYC reviews [None / View / Edit / Edit + Delete], Admin, Operators [None / View / Edit / Edit + Delete], Support, Operators [None / View / Edit / Edit + Delete], Accounts, Operators [None / View / Edit / Edit + Delete], Admin, Driver reports [None / View / Edit / Edit + Delete], Support, Driver reports [None / View / Edit / Edit + Delete], Accounts, Driver reports [None / View / Edit / Edit + Delete], Admin, Driver rules (defaults) [None / View / Edit / Edit + Delete], Support, Driver rules (defaults) [None / View / Edit / Edit + Delete], Accounts, Driver rules (defaults) [None / View / Edit / Edit + Delete], Admin, Commission [None / View / Edit / Edit + Delete], Support, Commission [None / View / Edit / Edit + Delete], Accounts, Commission [None / View / Edit / Edit + Delete]. columns: Area, Super Admin Always full access, AdminSet all to...NoneViewEditEdit + Delete, SupportSet all to...NoneViewEditEdit + Delete, AccountsSet all to...NoneViewEditEdit + Delete
Tab "Sessions"	columns: Person, Device, IP address, Signed in

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /admin/staff/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add staff	Add staff: Full name*; Email*; Role [Super Admin / Admin / Support / Accounts]	No server call	UI	opens staff form
Page > dialog	Send invitation		POST /admin/staff (invitation) or PATCH /admin/staff/:id	New	
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Edit	Edit staff: Full name*; Email*; Role [Super Admin / Admin / Support / Accounts]	No server call	UI	opens staff form
Page > dialog	Save		POST /admin/staff (invitation) or PATCH /admin/staff/:id	New	
Page	Deactivate		POST /admin/staff/:id/deactivate	New	audit
Page	View	Admin Super (IdSF1PersonAdmin SuperTitleSuper AdminRoleSuper adminStatusActiveEmailadmin.supe)	GET /admin/staff/:id	New	read one record for the detail window
Page	Activate		POST /admin/staff/:id/activate	New	audit
Tab: Staff	Export CSV		GET /admin/staff/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Staff	Add staff	Add staff: Full name*; Email*; Role [Super Admin / Admin / Support / Accounts]	No server call	UI	opens staff form
Tab: Staff > dialog	Send invitation		POST /admin/staff (invitation) or PATCH /admin/staff/:id	New	

Where	Button	Opens	Endpoint	Status	Notes
Tab: Staff	Table		No server call	UI	table or list view of the same data
Tab: Staff	List		No server call	UI	table or list view of the same data
Tab: Staff	Edit	Edit staff: Full name*; Email*; Role [Super Admin / Admin / Support / Accounts]	No server call	UI	opens staff form
Tab: Staff > dialog	Save		POST /admin/staff (invitation) or PATCH /admin/staff/:id	New	
Tab: Staff	Deactivate		POST /admin/staff/:id/deactivate	New	audit
Tab: Staff	View	Admin Super (IdSF1PersonAdmin SuperTitleSuper AdminRoleSuper adminStatusActiveEmailadmin.supe)	GET /admin/staff/:id	New	read one record for the detail window
Tab: Permission matrix	Add role	Add role: Role name*; Start with the access of [No access (set it after) / Admin / Support / Accounts]	No server call	UI	opens add role form
Tab: Permission matrix > dialog	Add role		POST /access-matrix/roles	New	new role, optionally copying another role's access; Super Admin only; audit role.created
Tab: Sessions	Table		No server call	UI	table or list view of the same data
Tab: Sessions	List		No server call	UI	table or list view of the same data
Tab: Sessions	Sign out		DELETE /me/sessions/:sessionId	Exists	

D.36 Feature flags and maintenance (`flags`)

Item	Value
Menu group	System
Feature / phase	Integrations, feature flags and maintenance / Phase 0
API base	`/admin/feature-flags`
Files	`settings_admin_manage_screen.dart` , `settings_admin_manage.ts` , `settings_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: None; Accounts: None

On the screen	Contents
Sections of the page	Feature flags (switch each feature on or off) | Maintenance mode (global, customer app, operator app, partner; message and window)
Filters	Starts | Ends

Where	Button	Opens	Endpoint	Status	Notes
Page	(icon)	Turn off Live tracking (Customers lose this feature straight away.)	PUT /admin/feature-flags/:key	New	built; body on and a reason; audit feature-flag.changed; reaches every server within seconds; the public map GET /features follows
Page > dialog	Turn off		DELETE /admin/feature-flags/:id	New	confirm of a delete or revoke; audit
Page	(icon)	Global maintenance (Customer, operator and partner apps stop working. Admin staff still get in.)	PUT /admin/maintenance	New	built; body scopes (global, customer, operator, partner, admin) and a reason; audit maintenance.changed; the guard enforces it on every request
Page > dialog	Turn on		DELETE /admin/feature-flags/:id	New	confirm of a delete or revoke; audit
Page	Save message and schedule		PUT /admin/maintenance	New	built; body message, from, to (end must be after start) and a reason; same endpoint as the switches

D.37 Audit log (`audit`)

Item	Value
Menu group	System
Feature / phase	Audit log / Phase 0
API base	`/audit-logs`
Files	`audit_admin_manage_screen.dart` , `audit_admin_manage.ts` , `audit_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: View; Support: None; Accounts: None

On the screen	Contents
Filters	Search | Person [All / Admin super / Abishek / Mynampati asha / Divya rao / Kiran kumar / Ravi teja / Sunita paul / Naveen babu] | Type [All / Sensitive only] | Action [All / Created coupon / Changed commission / Added staff user / Changed setting / Approved kyc submission / Suspended operator / Blocked customer / Approved settlement / Rejected document]
Table columns	When ▼ | Who | Action | Target | IP address | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /audit-logs/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Audit entry (When12 Oct 2026, 04:40 pmWhoMynampati AshaRoleAdminActionCreated couponTargetST-)	GET /audit-logs/:id	New	detail with before and after
Page	View	Audit entry (When12 Oct 2026, 04:40 pmWhoMynampati AshaRoleAdminActionCreated couponTargetST-)	GET /audit-logs/:id	New	detail with before and after

D.38 Notifications (`notifications`)

Item	Value
Menu group	System
Feature / phase	Notifications and event router / Phase 0
API base	`/notifications`
Files	`notifications_admin_manage_screen.dart` , `notifications_admin_manage.ts` , `notifications_admin_manage.service.ts`
Starting access	Super Admin: Edit+Delete; Admin: Edit+Delete; Support: Edit; Accounts: Edit

On the screen	Contents
Filters	Search | Status [All / Unread / Read]
Table columns	Notification | When ▼ | Actions
Detail page #/kyc/:id	detail view

On the screen	Contents
Detail page #/settlements	cards: Still to pay, Paid this cycle, Waiting for approval, On hold. columns: Settlement, Operator, Gross, Commission, Refunds, Net payable, Mode, Status, Actions
Detail page #/tickets/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	Mark all read		POST /notifications/read-all	Exists	
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	Opens page #/kyc/:id	POST /notifications/:id/read	Exists	then opens the target
Page	Edit	Edit N0: Notification*; When*	No server call	UI	opens the edit form, filled from GET /notifications/:id
Page > dialog	Save changes		PATCH /notifications/:id	New	save changes; audit before and after
Page	Delete	Delete N0? (This removes it from every list. You can undo for a few seconds.)	DELETE /notifications/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /notifications/:id/restore
Page > dialog	Delete		DELETE /notifications/:id	New	confirm of a delete or revoke; audit
Page	Row (open the record)	Opens page #/settlements	POST /notifications/:id/read	Exists	then opens the target
Detail page #/kyc/:id	History	History (Submitted12 Oct 2026, 05:40 pm)	GET /onboarding/submissions/:id/history	New	new
Detail page #/kyc/:id	Reject submission		POST /onboarding/submissions/:id/reject	Exists	notifies the operator with the rejected documents
Detail page #/kyc/:id	Approve		POST /onboarding/submissions/:id/approve	Exists	notifies the operator
Detail page #/kyc/:id	View	GST certificate (GST_scan.pdfDocument preview placeholder. The real portal shows the uploaded PDF)	GET /files/:id/content	Exists	
Detail page #/kyc/:id	Accept		POST /onboarding/submissions/:id/documents/:documentTypeId/review	Exists	accept
Detail page #/kyc/:id	Reject	Reject GST certificate: Reason*	No server call	UI	opens the reason form
Detail page #/kyc/:id > dialog	Reject document		POST /onboarding/submissions/:id/documents/:documentTypeId/review	Exists	reject with reason
Detail page #/settlements	Generate settlements	Generate settlements: Cycle [Last 7 days / Last 14 days / Custom range]; Operator [All operators / Kaveri Express / Godavari Lines / Deccan Voyager / Sahyadri Coaches / Nilgiri Travels ...]	No server call	UI	opens generate form
Detail page #/settlements > dialog	Generate		POST /admin/settlements/generate	New	cycle and operator; gross minus commission minus refunds plus adjustments
Detail page #/settlements	Payout file		GET /admin/settlements/payout-file?cycle=	New	bank file for manual mode
Detail page #/settlements	GST statement		GET /admin/settlements/gst-statement?cycle=	New	file
Detail page #/settlements	Export CSV		GET /notifications/export?format=csv|xlsx	New	same filters, search and permissions as the list
Detail page #/settlements	Table		No server call	UI	table or list view of the same data
Detail page #/settlements	List		No server call	UI	table or list view of the same data
Detail page #/settlements	View	Opens page #/settlements/:id	GET /admin/settlements/:id	New	with booking lines
Detail page #/settlements	Edit	Edit ST-6001: Operator*; Gross (Rs)*; Commission (Rs)*; Refunds (Rs)*; Net payable (Rs)*; Mode [Auto / Manual]; Status [On hold / Paid / Draft / Approved]	No server call	UI	opens the edit form, filled from GET /notifications/:id
Detail page #/settlements > dialog	Save changes		PATCH /notifications/:id	New	save changes; audit before and after
Detail page #/tickets/:id	Assign	Assign ticket: Assign to [Nobody / Divya (Support) / Kiran (Support)]	No server call	UI	opens assign form
Detail page #/tickets/:id > dialog	Assign		POST /admin/tickets/:id/assign	New	notifies the assignee
Detail page #/tickets/:id	Mark resolved		POST /admin/tickets/:id/resolve	New	audit
Detail page #/tickets/:id	Send reply		POST /admin/tickets/:id/replies	New	reply with proof files; notifies the customer

D.x Buttons that only appear in some states

These buttons are in the demo code but were not on screen in the default state (for example Approve only shows on a waiting refund). They are part of the build.

Action	Where	Endpoint	Status	Notes
ai_fb	AI assistant	POST /assistant/messages/:turnId/feedback	New	built; thumbs up or down on one answer
ai_go	AI assistant	No server call	UI	opens the page the answer points to
ai_no	AI assistant	POST /assistant/actions/:id/decline	New	built; audit assistant.action.declined
ai_ok	AI assistant	POST /assistant/actions/:id/confirm	New	built; checks the permission again, makes the change once, audit assistant.action.confirmed
ai_rep_csv	AI assistant	GET /admin/reports/trip-revenue/export	New	the report the assistant showed
ban_img	banners	POST /admin/banners/:id/image	Exists	JPG, PNG or WebP, up to 1 MB
bus_bulk_m	buses	POST /admin/buses/bulk-status	New	selected active buses to maintenance; audit
bus_bulk_x	buses	GET /catalogue/buses/export?ids=	New	export of the selected rows
cu_unblock	customers	POST /admin/customers/:id/unblock	New	audit
dr_appr	drules / dreports	No server call	UI	operator only; admin reads driver reports
dr_reset	drules / dreports	No server call	UI	operator only; admin edits the defaults instead
dr_ret	drules / dreports	No server call	UI	operator only

Action	Where	Endpoint	Status	Notes
dr_ret_ok	drules / dreports	No server call	UI	operator only
file_view	proof files	GET /files/:id/content	Exists	viewer and download for proofs
gs_go	global search	No server call	UI	opens the chosen result
gs_run	global search	GET /admin/search?q=	New	PNR, phone, email, operator, bus, trip across collections; only what the role may see
gsearch	global search	No server call	UI	global search box
imp_dry	import dialog on places, buses, fares, schedules	POST /shell/import?dryRun=true	New	row-by-row result
imp_file	import dialog on places, buses, fares, schedules	No server call	UI	choose file
imp_go	import dialog on places, buses, fares, schedules	POST /shell/import?dryRun=false	New	saves the valid rows
imp_sample	import dialog on places, buses, fares, schedules	GET /shell/import-template	New	sample file
imp_tpl	import dialog on places, buses, fares, schedules	GET /shell/import-template	New	
kpi	number cards	GET /shell?<card filter>	New	each number card opens the records behind it
kpi_exp	number cards	GET /shell/export?<card filter>	New	export of the records behind a card
mfa	account menu	No server call	UI	opens the MFA dialog (GET /me/mfa exists)
mfa_ok	account menu	POST /me/mfa/totp/activate	Exists	
op_react	operators	POST /catalogue/operators/:id/status	Exists	body status ACTIVE; notifies operator
perm_all	staff	PUT /access-matrix/roles/:roleKey	New	built; one level for every page; Super Admin only; audit
perm_set	staff	PUT /access-matrix/roles/:roleKey/pages/:pageId	New	built; level None, View, Edit or Edit+Delete; Super Admin only; audit role.access.changed; holders get new tokens
pf_pick	proof files	No server call	UI	choose proof files (up to 5, 600 KB each) uploaded with POST /files | exists
rec_add	every table	POST /shell	New	create from the generic form; audit
rf_bulk	refunds	POST /admin/refunds/bulk-approve	New	selected initiated refunds
rf_no	refunds	No server call	UI	opens reject form (proof required)
rf_no_ok	refunds	POST /admin/refunds/:id/reject	New	reason and proof; notifies the customer
rf_ok	refunds	POST /admin/refunds/:id/approve	New	starts the gateway refund
rf_retry	refunds	POST /admin/refunds/:id/retry	New	after a gateway failure
role	staff	No server call	UI	demo only: switching the role to preview access. Real build: the signed-in role comes from the token
role_del	staff	DELETE /access-matrix/roles/:roleKey	New	custom roles only; refused while any user holds it; audit role.deleted
role_ren	staff	No server call	UI	opens rename form
role_ren_ok	staff	PATCH /access-matrix/roles/:roleKey	New	custom roles only; audit role.renamed
rq_del	drules / dreports	DELETE /admin/driver-rules/defaults/reports/:id/questions/:qid	New	
rq_mv	drules / dreports	PATCH /admin/driver-rules/defaults/reports/:id/questions/order	New	
rq_req	drules / dreports	PATCH /admin/driver-rules/defaults/reports/:id/questions/:qid	New	required or not
st_appr	settlements	POST /admin/settlements/:id/approve	New	audit
st_bulk	settlements	POST /admin/settlements/bulk-approve	New	skips ones with open queries
st_hold	settlements	No server call	UI	opens hold form (reason and proof)
st_hold_ok	settlements	POST /admin/settlements/:id/hold	New	notifies the operator
st_pay	settlements	No server call	UI	opens pay form (reference)
st_pay_ok	settlements	POST /admin/settlements/:id/pay	New	manual mode marks paid with reference; automatic mode triggers the Razorpay Route transfer; notifies the operator
st_reply	settlements	No server call	UI	opens reply form
st_reply_ok	settlements	POST /admin/settlements/:id/query/reply	New	answers the operator's query; notifies the operator
tk_canned	tickets	GET /admin/tickets/canned-replies	New	list used by the reply box
tk_pick	tickets	No server call	UI	choose proof files
tk_reopen	tickets	POST /admin/tickets/:id/reopen	New	audit
trk_op	tracking	No server call	UI	filter
trk_replay	tracking	GET /admin/tracking/trips/:id/replay	New	stored positions
trk_st	tracking	No server call	UI	filter

Appendix E. Client (operator) portal: every page, button and endpoint

How to read each page: first its facts (feature, base path, files), then what is on the screen, then one row per button. "Where" says whether the button is on the page, on a tab, in a dialog or on a detail page. "Opens" lists the dialog and its fields (a star marks a required field). "Status" is **Exists** when the backend already has the route, **New** when it must be built, **UI** when no server call is needed. Row actions (View, Edit, Delete) exist on every table as in the demo; where changing a field would be unsafe (money, KYC status) the server refuses and says why.

> **Rule:** Build the real screen to look and behave exactly like the demo page of the same name. This appendix lists what the demo page contains; the demo file itself (`prototypes/client_demo_model.html`) is the visual reference.

E.1 Dashboard (`dashboard`)

Item	Value
Menu group	Overview
Feature / phase	Dashboards and reports / Phase 7
API base	`/client/dashboard`
Files	`dashboard_client_manage_screen.dart` , `dashboard_client_manage.ts` , `dashboard_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: View

On the screen	Contents
Sections of the page	Needs your attention | Bookings, last 14 days (chart) | Booking status (chart) | Your earnings per day, after commission (chart) | Occupancy by route (chart)
Number cards (each opens the records behind it)	Bookings today | Your earnings, 7 days | Average occupancy | Upcoming trips | Active buses | Buses live now | Drivers on duty | ustomer rating

E.2 KYC verification (`kyc`)

Item	Value
Menu group	Company
Feature / phase	KYC review and company onboarding / Phase 1
API base	`/onboarding/submissions`
Files	`kyc_client_manage_screen.dart` , `kyc_client_manage.ts` , `kyc_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: View

On the screen	Contents
Sections of the page	Step-by-step KYC wizard: questionnaire, then one upload box per required document with status (waiting, accepted, rejected with reason, replace), Send for review; bus KYC has the same steps

E.3 Company profile (`company`)

Item	Value
Menu group	Company
Feature / phase	KYC review and company onboarding / Phase 1
API base	`/client/company`
Files	`kyc_client_manage_screen.dart` , `kyc_client_manage.ts` , `kyc_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: View

On the screen	Contents
Sections of the page	Company profile (name, owner, GST, PAN, address, contact) | Bank account for payout (with verification status set by Abra)
Filters	Company name * | Owner name * | Email * | Support phone | City | GSTIN | Account holder | Account number | IFSC

Where	Button	Opens	Endpoint	Status	Notes
Page	Save		PUT /client/company	New	company profile; audit
Page	Save bank details		PUT /client/company/bank	New	bank details; admin verifies

E.4 Buses (`buses`)

Item	Value
Menu group	Fleet
Feature / phase	Buses and seat layouts / Phase 2
API base	`/catalogue/buses`
Files	`buses_client_manage_screen.dart` , `buses_client_manage.ts` , `buses_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	Buses | Active | Waiting for approval | Insurance expiring in 30 days
Filters	Search | Status [All / Active / Pending approval / Maintenance / Retired]
Table columns	Registration | Type | Seats | Layout | Insurance | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Add bus	Add bus: Registration number*; Bus name*; Bus type [AC Sleeper / Non-AC Sleeper / AC Seater / Non-AC Seater / Volvo Multi-Axle AC / Semi-Sleeper AC]; Seats*; Seat layout [2+2 Seater / 2+1 Sleeper / 1+2 Sleeper / 2+1 Semi-sleeper]; Insurance valid till*	No server call	UI	opens bus form
Page > dialog	Send for approval		POST /catalogue/buses (sends the bus for KYC approval) or PUT /catalogue/buses/:id	Exists	
Page	Import	Import buses (Upload a CSV file with these columns: registration, name, type, seats. Excel fil)	POST /catalogue/buses/import?dryRun=true then false	New	upload, row-by-row errors, then save; GET /catalogue/buses/import-template for the file
Page	Export CSV		GET /catalogue/buses/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data

Where	Button	Opens	Endpoint	Status	Notes
Page	Row (open the record)	KA01 AB 3469 · Kaveri Prime 39 (Active Semi-sleeper ac 45 seats · 2+2 Seater Insurance valid till19 Dec 2026Amen)	GET /catalogue/buses/:id	Exists	
Page > dialog	Seat layout		No server call	UI	opens the seat layout of the bus
Page > dialog	Edit		No server call	UI	opens bus form
Page > dialog	Mark maintenance		POST /catalogue/buses/:id/status	Exists	MAINTENANCE
Page	View	KA01 AB 3469 · Kaveri Prime 39 (Active Semi-sleeper ac 45 seats · 2+2 Seater Insurance valid till19 Dec 2026Amen)	GET /catalogue/buses/:id	Exists	

E.5 Seat layouts (`layouts`)

Item	Value
Menu group	Fleet
Feature / phase	Buses and seat layouts / Phase 2
API base	`/client/seat-layouts`
Files	`buses_client_manage_screen.dart` , `buses_client_manage.ts` , `buses_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: None

On the screen	Contents
Sections of the page	Seat layout designer: tools (Seat, Sleeper, Aisle, Remove), grid with Add row and Remove row, Reset, Save layout, Summary (seat counts, deck, ladies seats); choose the bus
Filters	Bus [KA01 AB 3469 · 2+2 Seater / KA01 AB 4945 · 2+2 Seater]

Where	Button	Opens	Endpoint	Status	Notes
Page	Seat		No server call	UI	chooses the drawing tool (seat or sleeper)
Page	Sleeper		No server call	UI	chooses the drawing tool (seat or sleeper)
Page	Ladies seat		No server call	UI	chooses the drawing tool (seat or sleeper)
Page	Aisle		No server call	UI	chooses the drawing tool (seat or sleeper)
Page	Empty		No server call	UI	chooses the drawing tool (seat or sleeper)
Page	Seat		No server call	UI	places or removes a seat in the drawing
Page	Add row		No server call	UI	adds or removes a row in the drawing
Page	Remove row		No server call	UI	adds or removes a row in the drawing
Page	Reset		No server call	UI	clears the drawing
Page	Save layout		PUT /client/seat-layouts/:busId	New	saves a new version; trips already published keep their copy

E.6 Drivers and crew (`crew`)

Item	Value
Menu group	Fleet
Feature / phase	Crew: drivers and conductors / Phase 2
API base	`/client/crew`
Files	`crew_client_manage_screen.dart` , `crew_client_manage.ts` , `crew_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	Drivers | Active | On leave | icences expiring in 30 days
Filters	Search | Role [All / Driver / Conductor / Manager] | Status [All / Active / On leave / Inactive]
Table columns	Name | Role | Licence | Trips in 7 days | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Add person	Add person: Full name*; Role [Driver / Conductor / Manager]; Phone*; Status [Active / On leave / Inactive]; Driving licence number; Licence valid till	No server call	UI	opens person form
Page > dialog	Add		POST /client/crew or PATCH /client/crew/:id	New	licence number and expiry; audit
Page	Export CSV		GET /client/crew/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	Ramesh K (IdST101OperatoridOP1NameRamesh KRoleDriverPhone+91 9100697329LicenceTS12 2022 00)	GET /client/crew/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form
Page > dialog	Delete		DELETE /client/crew/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /client/crew/:id/restore

E.7 Driver rules (`drules`)

Item	Value
Menu group	Fleet
Feature / phase	Driver rules and trip checklists / Phase 9
API base	`/client/driver-rules`
Files	`driver_rules_client_manage_screen.dart` , `driver_rules_client_manage.ts` , `driver_rules_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: None

On the screen	Contents
Tabs	Before the trip | After the trip | Reports

On the screen	Contents
Table columns	Item | Rule | Photo | Applies to | On
Tab "Before the trip"	columns: Item, Rule, Photo, Applies to, On
Tab "After the trip"	columns: Item, Rule, Photo, Applies to, On
Tab "Reports"	columns: Report, How often, Due, Questions, On

Where	Button	Opens	Endpoint	Status	Notes
Page	Reset to Abra defaults	Reset to Abra defaults (Your checklist items and report questions go back to the Abra defaults. Reports)	POST /client/driver-rules/reset	New	back to the Abra defaults
Page > dialog	Reset		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Page	Export CSV		GET /client/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add an item	Add an item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]	No server call	UI	opens item form
Page > dialog	Add		POST /client/driver-rules/items or PATCH /client/driver-rules/items/:id	New	this operator's copy
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	(icon)		PATCH /client/driver-rules/items/:id/active	New	locked items cannot be switched off
Page	(icon)		PATCH /client/driver-rules/order	New	
Page	Edit	Edit item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]	No server call	UI	opens item form
Page > dialog	Save		POST /client/driver-rules/items or PATCH /client/driver-rules/items/:id	New	this operator's copy
Page	Remove	Remove this item (Drivers will no longer be asked: I am fit to drive and well rested. Old checklis)	DELETE /client/driver-rules/items/:id	New	locked items cannot be removed
Page > dialog	Remove		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Before the trip	Reset to Abra defaults	Reset to Abra defaults (Your checklist items and report questions go back to the Abra defaults. Reports)	POST /client/driver-rules/reset	New	back to the Abra defaults
Tab: Before the trip > dialog	Reset		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Before the trip	Export CSV		GET /client/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Before the trip	Add an item	Add an item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]	No server call	UI	opens item form
Tab: Before the trip > dialog	Add		POST /client/driver-rules/items or PATCH /client/driver-rules/items/:id	New	this operator's copy
Tab: Before the trip	Table		No server call	UI	table or list view of the same data
Tab: Before the trip	List		No server call	UI	table or list view of the same data
Tab: Before the trip	(icon)		PATCH /client/driver-rules/items/:id/active	New	locked items cannot be switched off
Tab: Before the trip	(icon)		PATCH /client/driver-rules/order	New	
Tab: Before the trip	Edit	Edit item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]	No server call	UI	opens item form
Tab: Before the trip > dialog	Save		POST /client/driver-rules/items or PATCH /client/driver-rules/items/:id	New	this operator's copy
Tab: Before the trip	Remove	Remove this item (Drivers will no longer be asked: I am fit to drive and well rested. Old checklis)	DELETE /client/driver-rules/items/:id	New	locked items cannot be removed
Tab: Before the trip > dialog	Remove		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: After the trip	Reset to Abra defaults	Reset to Abra defaults (Your checklist items and report questions go back to the Abra defaults. Reports)	POST /client/driver-rules/reset	New	back to the Abra defaults
Tab: After the trip > dialog	Reset		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: After the trip	Export CSV		GET /client/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: After the trip	Add an item	Add an item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]	No server call	UI	opens item form
Tab: After the trip > dialog	Add		POST /client/driver-rules/items or PATCH /client/driver-rules/items/:id	New	this operator's copy
Tab: After the trip	Table		No server call	UI	table or list view of the same data
Tab: After the trip	List		No server call	UI	table or list view of the same data
Tab: After the trip	(icon)		PATCH /client/driver-rules/items/:id/active	New	locked items cannot be switched off
Tab: After the trip	(icon)		PATCH /client/driver-rules/order	New	

Where	Button	Opens	Endpoint	Status	Notes
Tab: After the trip	Edit	Edit item: Item (what the driver reads)*; Answer type [Yes or no / Choose one / A number]; Choices (for "Choose one", separate with commas); What counts as a problem; If it is a problem [Blocks the trip / Warns only]; Photo needed [No / Yes, when there is a problem or a number]; Applies to [Every trip / Long routes (8 hours or more) / Sleeper buses / Seater buses]	No server call	UI	opens item form
Tab: After the trip > dialog	Save		POST /client/driver-rules/items or PATCH /client/driver-rules/items/:id	New	this operator's copy
Tab: After the trip	Remove	Remove this item (Drivers will no longer be asked: End odometer reading (km). Old checklists keep)	DELETE /client/driver-rules/items/:id	New	locked items cannot be removed
Tab: After the trip > dialog	Remove		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Reports	Reset to Abra defaults	Reset to Abra defaults (Your checklist items and report questions go back to the Abra defaults. Reports)	POST /client/driver-rules/reset	New	back to the Abra defaults
Tab: Reports > dialog	Reset		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit
Tab: Reports	Export CSV		GET /client/driver-rules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Reports	Add a report	Add a report: Report name*; How often [Every day / Every week / When needed]; Due time (daily and weekly); Due day (weekly) [Mon / Tue / Wed / Thu / Fri / Sat ...]	No server call	UI	opens the report type form
Tab: Reports > dialog	Add		POST /client/driver-rules/reports or PATCH /client/driver-rules/reports/:id	New	this operator's report types
Tab: Reports	Table		No server call	UI	table or list view of the same data
Tab: Reports	List		No server call	UI	table or list view of the same data
Tab: Reports	(icon)		PATCH /client/driver-rules/reports/:id/active	New	
Tab: Reports	Questions	Questions: Question; Answer type [Text / Number / Yes or no / Choose one / Photo / File]; Choices (for "Choose one"); Required [Optional / Required]	No server call	UI	opens question list
Tab: Reports > dialog	Add this question		POST /client/driver-rules/reports/:id/questions	New	
Tab: Reports	Edit	Edit report: Report name*; How often [Every day / Every week / When needed]; Due time (daily and weekly); Due day (weekly) [Mon / Tue / Wed / Thu / Fri / Sat ...]	No server call	UI	opens the report type form
Tab: Reports > dialog	Save		POST /client/driver-rules/reports or PATCH /client/driver-rules/reports/:id	New	this operator's report types
Tab: Reports	Remove	Remove this report (Drivers will no longer see Daily report. Reports already sent are kept.)	DELETE /client/driver-rules/reports/:id	New	
Tab: Reports > dialog	Remove		DELETE /client/driver-rules/:id	New	confirm of a delete or revoke; audit

E.8 Driver reports (`dreports`)

Item	Value
Menu group	Fleet
Feature / phase	Driver reports / Phase 9
API base	`/client/driver-reports`
Files	`driver_reports_client_manage_screen.dart` , `driver_reports_client_manage.ts` , `driver_reports_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	Reports due or late6 overdue | Waiting for review | Sent back to drivers | Trips started past a failed check
Tabs	Reports | Missing | Trip checklists
Filters	Driver or report | Status [All / Waiting for review / Approved / Sent back] | Report [All / Daily report / Weekly report] | Driver [All / Ramesh k / Sunil y]
Table columns	Report | Driver | Sent ▼ | Status | Files
Tab "Reports"	cards: Reports due or late6 overdue, Waiting for review, Sent back to drivers, Trips started past a failed check. filters: Driver or report, Status [All / Waiting for review / Approved / Sent back], Report [All / Daily report / Weekly report], Driver [All / Ramesh k / Sunil y]. columns: Report, Driver, Sent ▼, Status, Files
Tab "Missing"	cards: Reports due or late6 overdue, Waiting for review, Sent back to drivers, Trips started past a failed check. filters: Status [All / Overdue / Due soon]. columns: Driver, Report, Due, Status
Tab "Trip checklists"	cards: Reports due or late6 overdue, Waiting for review, Sent back to drivers, Trips started past a failed check. filters: Search, Result [All / Passed / Warnings / Started anyway], When [All / Before the trip / After the trip]. columns: Trip, Driver, When, Time ▼, Odometer, Result

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /client/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Daily report - Ramesh K (DriverRamesh KOperatorKaveri ExpressReport11 Oct 2026Sent11 Oct 2026, 07:40 pmSt)	GET /client/driver-reports/:id	New	system numbers (frozen) and answers
Page > dialog	Send back		No server call	UI	opens the note form
Page > dialog	Approve		POST /client/driver-reports/:id/approve	New	notifies the driver
Tab: Reports	Export CSV		GET /client/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Reports	Table		No server call	UI	table or list view of the same data
Tab: Reports	List		No server call	UI	table or list view of the same data
Tab: Reports	Row (open the record)	Daily report - Ramesh K (DriverRamesh KOperatorKaveri ExpressReport11 Oct 2026Sent11 Oct 2026, 07:40 pmSt)	GET /client/driver-reports/:id	New	system numbers (frozen) and answers
Tab: Reports > dialog	Send back		No server call	UI	opens the note form
Tab: Reports > dialog	Approve		POST /client/driver-reports/:id/approve	New	notifies the driver
Tab: Missing	Export CSV		GET /client/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Missing	Table		No server call	UI	table or list view of the same data
Tab: Missing	List		No server call	UI	table or list view of the same data

Where	Button	Opens	Endpoint	Status	Notes
Tab: Missing	Remind the driver		POST /client/driver-reports/remind	New	tells the driver
Tab: Trip checklists	Export CSV		GET /client/driver-reports/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Trip checklists	Table		No server call	UI	table or list view of the same data
Tab: Trip checklists	List		No server call	UI	table or list view of the same data
Tab: Trip checklists	Row (open the record)	After the trip: TRP-3018 (DriverRamesh KTripTRP-3018 Vijayawada to VisakhapatnamWhen11 Oct 2026, 07:40 amO)	GET /client/trip-checks/:id	New	checklist result

E.9 Routes and stops (`routes`)

Item	Value
Menu group	Network
Feature / phase	Routes, stops and boarding points / Phase 2
API base	`/catalogue/routes`
Files	`routes_client_manage_screen.dart` , `routes_client_manage.ts` , `routes_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: None

On the screen	Contents
Filters	Search
Table columns	Route | Distance | Duration | Stops | Your trips | Status | Actions
Detail page #/routes/:id	detail view

Where	Button	Opens	Endpoint	Status	Notes
Page	New route	Add a report: Report name*; How often [Every day / Every week / When needed]; Due time (daily and weekly); Due day (weekly) [Mon / Tue / Wed / Thu / Fri / Sat ...]	No server call	UI	opens route form
Page > dialog	Add		POST /catalogue/routes or PUT /catalogue/routes/:id	Exists	stops with times, boarding and dropping points
Page	Export CSV		GET /catalogue/routes/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/routes/:id	GET /catalogue/routes/:id	Exists	
Page	View	Opens page #/routes/:id	GET /catalogue/routes/:id	Exists	
Page	Edit	Edit report: Report name*; How often [Every day / Every week / When needed]; Due time (daily and weekly); Due day (weekly) [Mon / Tue / Wed / Thu / Fri / Sat ...]	No server call	UI	opens route form
Page > dialog	Save		POST /catalogue/routes or PUT /catalogue/routes/:id	Exists	stops with times, boarding and dropping points
Detail page #/routes/:id	Copy as my route		POST /client/routes/copy-from/:id	New	copies a public route as this operator's own

E.10 Places (`places`)

Item	Value
Menu group	Network
Feature / phase	Cities and places / Phase 1
API base	`/client/place-requests`
Files	`places_client_manage_screen.dart` , `places_client_manage.ts` , `places_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: None

On the screen	Contents
Filters	Search | City [All / Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore / Kochi / Mumbai / Pune]
Table columns 1	Place | City | Requested | Status | Actions
Table columns 2	Place | City | Used for | Landmark

Where	Button	Opens	Endpoint	Status	Notes
Page	Request a place	Request a place: City [Hyderabad / Bengaluru / Chennai / Vijayawada / Visakhapatnam / Coimbatore ...]; Place name*; Address*; Landmark	No server call	UI	opens place request form
Page > dialog	Send request		POST /client/place-requests	New	admin approves or rejects; the operator is notified
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	Hosur Bus Stand Gate 2 (IdPR-100OperatorKaveri ExpressCityHosurPlaceHosur Bus Stand Gate 2AddressHosur M)	GET /client/place-requests/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /client/place-requests/:id
Page	Export CSV		GET /client/place-requests/export?format=csv|xlsx	New	same filters, search and permissions as the list

E.11 Fares (`fares`)

Item	Value
Menu group	Sales setup
Feature / phase	Fares and schedules / Phase 2
API base	`/client/fares`
Files	`fares_client_manage_screen.dart` , `fares_client_manage.ts` , `fares_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: View

On the screen	Contents
Filters	Search | Status [All / Active / Inactive]
Table columns	Route | Seat type | Weekday | Weekend | Festival | Customer pays (weekday) | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	New fare	New fare: Route [Hyderabad to Bengaluru / Bengaluru to Hyderabad / Vijayawada to Visakhapatnam]; Seat type [AC Sleeper / Non-AC Sleeper / AC Seater / Non-AC Seater / Volvo Multi-Axle AC / Semi-Sleeper AC]; Weekday fare (Rs)*; Weekend fare (Rs)*; Festival fare (Rs)*; Effective from*; Status [ACTIVE / INACTIVE]	No server call	UI	opens fare form
Page > dialog	Save fare		POST /client/fares or PATCH /client/fares/:id	New	weekday, weekend, festival fare per seat type; effective from
Page	Export CSV		GET /client/fares/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	Vijayawada to Visakhapatnam (IdFR17OperatoridOP1RouteidRT5RouteVijayawada to VisakhapatnamSeat typeNon-AC Sle)	GET /client/fares/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens fare form
Page > dialog	Delete		DELETE /client/fares/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /client/fares/:id/restore

E.12 Schedules (`schedules`)

Item	Value
Menu group	Sales setup
Feature / phase	Fares and schedules / Phase 2
API base	`/client/schedules`
Files	`fares_client_manage_screen.dart` , `fares_client_manage.ts` , `fares_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: None

On the screen	Contents
Sections of the page	List of recurring schedules (route, bus, time, days, from, till, status) and the form to create them; Generate trips
Filters	Search

Where	Button	Opens	Endpoint	Status	Notes
Page	New schedule	New schedule: Route [Hyderabad to Bengaluru / Bengaluru to Hyderabad / Vijayawada to Visakhapatnam]; Bus [KA01 AB 3469 · Semi-Sleeper AC / KA01 AB 4945 · Non-AC Sleeper]; Departure time*; Status [ACTIVE / INACTIVE]; From*; Till*	No server call	UI	opens schedule form
Page > dialog	Save		POST /client/schedules or PATCH /client/schedules/:id	New	recurring trip
Page	Export CSV		GET /client/schedules/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data

E.13 Trips (`trips`)

Item	Value
Menu group	Sales setup
Feature / phase	Trips and seat inventory / Phase 2
API base	`/catalogue/trips`
Files	`trips_client_manage_screen.dart` , `trips_client_manage.ts` , `trips_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: Edit; Accounts: None

On the screen	Contents
Number cards (each opens the records behind it)	On sale | Drafts | Running now | Without a driver
Filters	Search | Status [All / Draft / Scheduled / Boarding / In transit / Delayed / Arrived / Cancelled] | When [All / Upcoming / Today / Past]
Table columns	Trip | Departs ▲ | Bus | Driver | Booked | Fare | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	New trip	New trip: Route [Hyderabad to Bengaluru / Bengaluru to Hyderabad / Vijayawada to Visakhapatnam]; Bus [KA01 AB 3469 · Semi-Sleeper AC · 45 seats / KA01 AB 4945 · Non-AC Sleeper · 38 seats]; Date*; Departure time*; Driver [Assign later / Ramesh K / Sunil Y / Anwar S]; Window / corner seat (Rs per seat)*; Normal seat (Rs per seat)*	No server call	UI	opens trip form
Page > dialog	Create trip		POST /catalogue/trips	Exists	draft with fares by seat type
Page	Export CSV		GET /catalogue/trips/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	TRP-4001 · Hyderabad to Bengaluru (Arrived 8 Oct 2026, 08:30 pm · KA01 AB 3469 · Semi-Sleeper AC · ₹1,100 per seat)	GET /catalogue/trips/:id	Exists	plus seat map GET /client/trips/:id/seats
Page > dialog	Download manifest		GET /client/trips/:id/manifest.pdf	New	passenger list with seats and boarding points
Page	View	TRP-4001 · Hyderabad to Bengaluru (Arrived 8 Oct 2026, 08:30 pm · KA01 AB 3469 · Semi-Sleeper AC · ₹1,100 per seat)	GET /catalogue/trips/:id	Exists	plus seat map GET /client/trips/:id/seats
Page	Edit	Edit Hyderabad to Bengaluru: Departs*; Bus*; Driver*; Booked*; Fare (Rs)*; Status [Arrived / Scheduled / Delayed / In transit / Draft / Cancelled]	No server call	UI	opens the edit form, filled from GET /catalogue/trips/:id
Page > dialog	Save changes		PATCH /catalogue/trips/:id	New	save changes; audit before and after

E.14 Bookings (`bookings`)

Item	Value
Menu group	Sales
Feature / phase	Booking, passengers and bookings lists / Phase 3
API base	`/client/bookings`
Files	`booking_client_manage_screen.dart` , `booking_client_manage.ts` , `booking_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: Edit; Accounts: View

On the screen	Contents
Number cards (each opens the records behind it)	Bookings | onfirmed | ancelled | Your share
Filters	Search | Status [All / Confirmed / Completed / Cancelled] | Route [All / Vijayawada to visakhapatnam / Hyderabad to bengaluru / Bengaluru to hyderabad]
Table columns	PNR | Passenger | Route | Travel ▼ | Seats | Fare | You get | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /client/bookings/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	ABRK5152A (PassengerRahul VermaPhone+91 9777868637TripTRP-4022 · Bengaluru to HyderabadTrav)	GET /client/bookings/:id	New	detail
Page > dialog	Call passenger		No server call	UI	dials the passenger
Page > dialog	Cancel booking		No server call	UI	opens cancel form (proof)
Page	View	ABRK5152A (PassengerRahul VermaPhone+91 9777868637TripTRP-4022 · Bengaluru to HyderabadTrav)	GET /client/bookings/:id	New	detail
Page	Edit	Edit ABRK5152A: PNR*; Passenger*; Route*; Travel*; Seats*; Fare (Rs)*; Status [Confirmed / Completed / Cancelled]	No server call	UI	opens the edit form, filled from GET /client/bookings/:id
Page > dialog	Save changes		PATCH /client/bookings/:id	New	save changes; audit before and after

E.15 Offers (`offers`)

Item	Value
Menu group	Sales
Feature / phase	Offers and coupons / Phase 3
API base	`/client/offer-requests`
Files	`offers_client_manage_screen.dart` , `offers_client_manage.ts` , `offers_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: None

On the screen	Contents
Filters	Search
Table columns	Code | Discount | Valid till | Used | Cost to you | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Request an offer	Request an offer: Code*; Type [Flat amount / Percentage]; Value (Rs, or percent)*; Maximum discount (Rs); Valid till*; Total uses (0 = no limit); Description (customers see it)*	No server call	UI	opens offer request form
Page > dialog	Send to Abra		POST /client/offer-requests	New	arrives in admin as an offer request to approve
Page	Export CSV		GET /client/offer-requests/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	KAVERI50 (IdOF5CodeKAVERI50DescRs 50 off Kaveri Express busesTypeFlatDiscount5000Maxdisc₹5)	GET /client/offer-requests/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /client/offer-requests/:id
Page > dialog	Delete		DELETE /client/offer-requests/:id	New	soft delete, refused if the record is in use; audit; "Undo" calls POST /client/offer-requests/:id/restore

E.16 Earnings and settlements (`earnings`)

Item	Value
Menu group	Money
Feature / phase	Settlements and earnings / Phase 6
API base	`/client/settlements`
Files	`settlements_client_manage_screen.dart` , `settlements_client_manage.ts` , `settlements_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: Edit

On the screen	Contents
Number cards (each opens the records behind it)	Gross sales, 30 days | Abra commission | Refunds you bore | Your net, 30 days
Filters	Search
Table columns	Settlement | Gross | Commission | Refunds | Net payable | Status | Actions

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /client/settlements/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data

Where	Button	Opens	Endpoint	Status	Notes
Page	Row (open the record)	ST-6001 · 1 Oct to 7 Oct 2026 (Gross₹34,453.35Abra commission- ₹3,445.34Refunds- ₹2,742.97Adjustments₹0Net paya)	GET /client/settlements/:id	New	detail with lines
Page > dialog	Download statement		GET /client/settlements/:id/statement.pdf	New	
Page > dialog	Raise a query		No server call	UI	opens query form (proof allowed)
Page	View	ST-6001 · 1 Oct to 7 Oct 2026 (Gross₹34,453.35Abra commission- ₹3,445.34Refunds- ₹2,742.97Adjustments₹0Net paya)	GET /client/settlements/:id	New	detail with lines
Page	Edit	Edit ST-6001: Gross (Rs)*; Commission (Rs)*; Refunds (Rs)*; Net payable (Rs)*; Status [On hold / Paid / Draft / Approved]	No server call	UI	opens the edit form, filled from GET /client/settlements/:id
Page > dialog	Save changes		PATCH /client/settlements/:id	New	save changes; audit before and after

E.17 Commission (`commission`)

Item	Value
Menu group	Money
Feature / phase	Commission rules / Phase 2
API base	`/client/commission`
Files	`commission_client_manage_screen.dart` , `commission_client_manage.ts` , `commission_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: None; Accounts: View

On the screen	Contents
Sections of the page	Rules that apply to you (read-only, set by Abra) | Price calculator: fare in, customer price and your payable out
Filters	Your fare per seat (Rs) | Route [Hyderabad to Bengaluru / Bengaluru to Hyderabad / Vijayawada to Visakhapatnam] | Seat type [Semi-Sleeper AC / Non-AC Sleeper]

Where	Button	Opens	Endpoint	Status	Notes
Page	Calculate		POST /client/commission/calculate	New	shows how a fare becomes the customer price and the operator payable

E.18 Live buses and safety (`live`)

Item	Value
Menu group	Operations
Feature / phase	Live tracking / Phase 5
API base	`/client/live`
Files	`tracking_client_manage_screen.dart` , `tracking_client_manage.ts` , `tracking_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: View; Accounts: None

On the screen	Contents
Sections of the page	Live buses and safety | Running now (list with map, position, ETA, passengers, checked in) | SOS history on this operator's buses (customer, driver, status)

Where	Button	Opens	Endpoint	Status	Notes
Page	Call		No server call	UI	dials the passenger
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data

E.19 Notifications (`notifications`)

Item	Value
Menu group	System
Feature / phase	Notifications and event router / Phase 0
API base	`/notifications`
Files	`notifications_client_manage_screen.dart` , `notifications_client_manage.ts` , `notifications_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: Edit+Delete; Booking staff: Edit; Accounts: Edit

On the screen	Contents
Filters	Search
Table columns	Notification | When ▼
Detail page #/buses	cards: Buses, Active, Waiting for approval, Insurance expiring in 30 days. columns: Registration, Type, Seats, Layout, Insurance, Status, Actions
Detail page #/settlements	cards: Bookings today, Your earnings, 7 days, Average occupancy, Upcoming trips, Active buses, Buses live now, Drivers on duty, ustomer rating

Where	Button	Opens	Endpoint	Status	Notes
Page	Mark all read		POST /notifications/read-all	Exists	
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	Row (open the record)	Opens page #/buses	POST /notifications/:id/read	Exists	then opens the target
Detail page #/buses	Add bus	Add bus: Registration number*; Bus name*; Bus type [AC Sleeper / Non-AC Sleeper / AC Seater / Non-AC Seater / Volvo Multi-Axle AC / Semi-Sleeper AC]; Seats*; Seat layout [2+2 Seater / 2+1 Sleeper / 1+2 Sleeper / 2+1 Semi-sleeper]; Insurance valid till*	No server call	UI	opens bus form
Detail page #/buses > dialog	Send for approval		POST /catalogue/buses (sends the bus for KYC approval) or PUT /catalogue/buses/:id	Exists	
Detail page #/buses	Import	Import buses (Upload a CSV file with these columns: registration, name, type, seats. Excel fil)	POST /notifications/import?dryRun=true then false	New	upload, row-by-row errors, then save; GET /notifications/import-template for the file
Detail page #/buses	Export CSV		GET /notifications/export?format=csv|xlsx	New	same filters, search and permissions as the list

Where	Button	Opens	Endpoint	Status	Notes
Detail page #/buses	Table		No server call	UI	table or list view of the same data
Detail page #/buses	List		No server call	UI	table or list view of the same data
Detail page #/buses	View	KA01 AB 3469 · Kaveri Prime 39 (Active Semi-sleeper ac 45 seats · 2+2 Seater Insurance valid till19 Dec 2026Amen)	GET /catalogue/buses/:id	Exists	
Detail page #/buses > dialog	Seat layout		No server call	UI	opens the seat layout of the bus
Detail page #/buses > dialog	Edit		No server call	UI	opens bus form
Detail page #/buses > dialog	Mark maintenance		POST /catalogue/buses/:id/status	Exists	MAINTENANCE

E.20 Staff and roles (`staff`)

Item	Value
Menu group	System
Feature / phase	Staff, roles and access matrix / Phase 7
API base	`/client/users`
Files	`staff_client_manage_screen.dart` , `staff_client_manage.ts` , `staff_client_manage.service.ts`
Starting access	Owner: Edit+Delete; Manager: None; Booking staff: None; Accounts: None

On the screen	Contents
Tabs	Portal users | Permission matrix
Filters	Search
Table columns	Person | Role | Status | Last seen | Actions
Tab "Portal users"	filters: Search. columns: Person, Role, Status, Last seen, Actions
Tab "Permission matrix"	filters: Set every area for Manager [Set all to... / None / View / Edit / Edit + Delete], Set every area for Booking staff [Set all to... / None / View / Edit / Edit + Delete], Set every area for Accounts [Set all to... / None / View / Edit / Edit + Delete], Manager, Dashboard [None / View / Edit / Edit + Delete], Booking staff, Dashboard [None / View / Edit / Edit + Delete], Accounts, Dashboard [None / View / Edit / Edit + Delete], Manager, KYC verification [None / View / Edit / Edit + Delete], Booking staff, KYC verification [None / View / Edit / Edit + Delete], Accounts, KYC verification [None / View / Edit / Edit + Delete], Manager, Company profile [None / View / Edit / Edit + Delete], Booking staff, Company profile [None / View / Edit / Edit + Delete], Accounts, Company profile [None / View / Edit / Edit + Delete], Manager, Buses [None / View / Edit / Edit + Delete], Booking staff, Buses [None / View / Edit / Edit + Delete], Accounts, Buses [None / View / Edit / Edit + Delete], Manager, Seat layouts [None / View / Edit / Edit + Delete], Booking staff, Seat layouts [None / View / Edit / Edit + Delete], Accounts, Seat layouts [None / View / Edit / Edit + Delete], Manager, Drivers and crew [None / View / Edit / Edit + Delete], Booking staff, Drivers and crew [None / View / Edit / Edit + Delete], Accounts, Drivers and crew [None / View / Edit / Edit + Delete], Manager, Driver reports [None / View / Edit / Edit + Delete], Booking staff, Driver reports [None / View / Edit / Edit + Delete], Accounts, Driver reports [None / View / Edit / Edit + Delete]. columns: Area, Owner Always full access, ManagerSet all to...NoneViewEditEdit + Delete, Booking staffSet all to...NoneViewEditEdit + Delete, AccountsSet all to...NoneViewEditEdit + Delete

Where	Button	Opens	Endpoint	Status	Notes
Page	Export CSV		GET /client/users/export?format=csv|xlsx	New	same filters, search and permissions as the list
Page	Add user	Add user: Full name*; Email*; Role [Manager / Booking staff / Accounts]	No server call	UI	opens user form
Page > dialog	Send invitation		POST /client/users (invitation) or PATCH /client/users/:id	New	
Page	Table		No server call	UI	table or list view of the same data
Page	List		No server call	UI	table or list view of the same data
Page	View	Ramesh Naidu (IdOU1OperatoridOP1PersonRamesh NaiduEmailramesh.naidu@example.comRoleSuper admin)	GET /client/users/:id	New	read one record for the detail window
Page > dialog	Edit		No server call	UI	opens the edit form, filled from GET /client/users/:id
Tab: Portal users	Export CSV		GET /client/users/export?format=csv|xlsx	New	same filters, search and permissions as the list
Tab: Portal users	Add user	Add user: Full name*; Email*; Role [Manager / Booking staff / Accounts]	No server call	UI	opens user form
Tab: Portal users > dialog	Send invitation		POST /client/users (invitation) or PATCH /client/users/:id	New	
Tab: Portal users	Table		No server call	UI	table or list view of the same data
Tab: Portal users	List		No server call	UI	table or list view of the same data
Tab: Portal users	View	Ramesh Naidu (IdOU1OperatoridOP1PersonRamesh NaiduEmailramesh.naidu@example.comRoleSuper admin)	GET /client/users/:id	New	read one record for the detail window
Tab: Portal users > dialog	Edit		No server call	UI	opens the edit form, filled from GET /client/users/:id
Tab: Permission matrix	Add role	Add role: Role name*; Start with the access of [No access (set it after) / Manager / Booking staff / Accounts]	No server call	UI	opens role form
Tab: Permission matrix > dialog	Add role		POST /access-matrix/roles	New	audience OPERATOR; phase 8 lets the owner manage own roles (operatorId on the role)

E.x Buttons that only appear in some states

These buttons are in the demo code but were not on screen in the default state (for example Approve only shows on a waiting refund). They are part of the build.

Action	Where	Endpoint	Status	Notes
au_acc	sign-in and KYC wizard	POST /onboarding/invitations/accept	Exists	
au_back	sign-in and KYC wizard	No server call	UI	back
au_in	sign-in and KYC wizard	POST /auth/login	Exists	(password) or POST /auth/social/google
au_inv	sign-in and KYC wizard	POST /onboarding/invitations/inspect	Exists	shows the invited company
bd_fix	sign-in and KYC wizard	PUT /onboarding/submissions/:id/documents/:documentTypeId	Exists	
bd_resend	sign-in and KYC wizard	POST /onboarding/submissions/:id/submit	Exists	
bdoc	sign-in and KYC wizard	No server call	UI	choose the bus document
cai_go	AI assistant	No server call	UI	opens the page the answer points to
cai_no	AI assistant	POST /assistant/actions/:id/decline	New	built

Action	Where	Endpoint	Status	Notes
cai_ok	AI assistant	POST /assistant/actions/:id/confirm	New	built
cb_cancel_ok	bookings	POST /client/bookings/:id/cancel	New	operator-side cancellation; refund by the operator's slab; audit
cs_q_ok	earnings	POST /client/settlements/:id/query	New	notifies admin
d_reset	drules / dreports	No server call	UI	confirm reset
dop	drules / dreports	No server call	UI	opens an option list
dr_push	drules / dreports	No server call	UI	admin only
dr_ret_ok	drules / dreports	POST /client/driver-reports/:id/return	New	note required; notifies the driver
file_view	proof files	GET /files/:id/content	Exists	
gs_go	global search	No server call	UI	opens the result
gs_run	global search	GET /client/search?q=	New	only this company's data
gsearch	global search	No server call	UI	search box
imp_dry	import dialog on places, buses, fares, schedules	POST /shell/import?dryRun=true	New	row-by-row result
imp_file	import dialog on places, buses, fares, schedules	No server call	UI	choose file
imp_go	import dialog on places, buses, fares, schedules	POST /shell/import?dryRun=false	New	
imp_sample	import dialog on places, buses, fares, schedules	GET /shell/import-template	New	sample file
imp_tpl	import dialog on places, buses, fares, schedules	GET /shell/import-template	New	
kd_fix	sign-in and KYC wizard	PUT /onboarding/submissions/:id/documents/:documentTypeId	Exists	replaces a rejected file
kd_resend	sign-in and KYC wizard	POST /onboarding/submissions/:id/submit	Exists	sends corrections
kd_submit	sign-in and KYC wizard	POST /onboarding/submissions/:id/submit	Exists	
kd_up	sign-in and KYC wizard	PUT /onboarding/submissions/:id/documents/:documentTypeId	Exists	
kpi	number cards	GET /shell?<card filter>	New	every number card opens the records behind it
kpi_exp	number cards	GET /shell/export?<card filter>	New	
kq	sign-in and KYC wizard	PUT /onboarding/submissions/:id/answers	New	saves a questionnaire answer
kq_in	sign-in and KYC wizard	PUT /onboarding/submissions/:id/answers	New	
kstep	sign-in and KYC wizard	No server call	UI	wizard step
lay_bus	layouts	GET /client/seat-layouts?busId=	New	layout of the chosen bus
mfa	account menu	No server call	UI	MFA dialog
mfa_ok	account menu	POST /me/mfa/totp/activate	Exists	
perm_all	staff	PUT /access-matrix/roles/:roleKey	New	one level for every page
perm_set	staff	PUT /access-matrix/roles/:roleKey/pages/:pageId	New	the owner only (phase 8)
pf_pick	proof files	No server call	UI	choose files; POST /files | exists
rec_add	every table	POST /shell	New	create from the generic form; audit
role	staff	No server call	UI	demo only: preview as another role
role_del	staff	DELETE /access-matrix/roles/:roleKey	New	
role_ren	staff	No server call	UI	rename
role_ren_ok	staff	PATCH /access-matrix/roles/:roleKey	New	
rq_del	drules / dreports	DELETE /client/driver-rules/reports/:id/questions/:qid	New	
rq_mv	drules / dreports	PATCH /client/driver-rules/reports/:id/questions/order	New	
rq_req	drules / dreports	PATCH /client/driver-rules/reports/:id/questions/:qid	New	
rt_add	drules / dreports	No server call	UI	adds a stop row
rt_mv	drules / dreports	No server call	UI	moves a stop
rt_off	drules / dreports	No server call	UI	sets the minutes from start
rt_pl	drules / dreports	GET /catalogue/locations?cityId=	Exists	places for a stop
rt_rm	drules / dreports	No server call	UI	removes a stop row
rt_ty	drules / dreports	No server call	UI	sets the stop type (boarding, dropping, both)
sc_day	schedules	No server call	UI	picks days of the week
sc_gen	schedules	POST /client/schedules/:id/generate	New	creates the trips for the date range
sch_edit	schedules	No server call	UI	opens schedule form
tr_bulk_pub	trips	POST /client/trips/bulk-publish	New	selected drafts
tr_cancel_ok	trips	POST /catalogue/trips/:id/cancel	Exists	refunds all bookings; notifies customers, driver, admin
tr_delay_ok	trips	POST /client/trips/:id/delay	New	minutes, reason, proof; notifies customers and admin
tr_drv_ok	trips	PUT /client/trips/:id/driver	New	notifies the driver
tr_pub	trips	POST /catalogue/trips/:id/publish	Exists	only if company and bus KYC are approved
tr_refares	trips	POST /client/trips/:id/refares	New	reapplies the fare rule to unsold trips (does not change published trips unless republished)
up_ex	sign-in and KYC wizard	No server call	UI	choose file
up_go	sign-in and KYC wizard	POST /files	Exists	

Appendix F. Customer app: every screen and endpoint

The customer app is mostly built; this lists every screen so the remaining ones and the real connections are finished against the demo. Endpoints under `/customer` need a signed-in customer; the others are public.

> **Rule:** Build the real screens to look and behave exactly like the demo (`prototypes/customer_demo_model.html`): same layout, wording, order, colours, dialogs and states.

F.1 Welcome, onboarding and language

Item	Value
Feature / phase	Customers and profile / Phase 4
Files	`customers_customer_screen.dart` , `customers_customer.ts` , `customers_customer.service.ts`
On the screen	Three intro cards (Get started, Skip, Back, Continue); language chooser with the 8 languages (names in English next to the native script).

Button / element	What it does	Endpoint	Status	Data
Get started / Skip / Continue	Stores the chosen language and marks onboarding as done on the device	none (device only)	UI	Language is saved to the profile after sign-in with PATCH /customer/profile
Language chooser	Sets app language	GET /translations/:lang (bundle, cached)	New	translations

F.2 Home

Item	Value
Feature / phase	Search and trip details / Phase 3
Files	`discovery_customer_screen.dart` , `discovery_customer.ts` , `discovery_customer.service.ts`
On the screen	Search card (From, To, Date, Swap, Today/Tomorrow chips), recent searches, banner carousel, offers strip, popular routes, running-trip banner with SOS and Track buttons, bell with unread count.

Button / element	What it does	Endpoint	Status	Data
City picker (From / To)	Searches cities as the user types	GET /cities?q=	Exists	cities (exists)
Swap	Swaps From and To	none	UI	
Recent searches	Last 4 searches kept on the device	none (device only)	UI	
Banner tap	Opens the linked offer or route	GET /banners?placement=home (exists) and POST /banners/:id/click	Exists	banners
Popular routes	Fills the search card	GET /popular-routes	New	popular
Search	Opens results	GET /trips/search?from&to&date (exists)	Exists	trips
Running-trip banner: Track / SOS	Opens tracking or the safety screen of the trip in transit	GET /customer/bookings?status=RUNNING	New	bookings, trips
Bell	Opens notifications	GET /notifications/unread-count (exists)	Exists	inbox

F.3 Search results

Item	Value
Feature / phase	Search and trip details / Phase 3
Files	`discovery_customer_screen.dart` , `discovery_customer.ts` , `discovery_customer.service.ts`
On the screen	Day chips, sort sheet, filter sheet (time of day, bus type, AC, operator, amenities, maximum fare), list of trips with operator, times, seats left, price from.

Button / element	What it does	Endpoint	Status	Data
Day chips	Changes the date	GET /trips/search (exists)	Exists	
Sort / Filters / Apply	Re-runs the search with filters	GET /trips/search/filters (exists), GET /trips/search	Exists	
Trip card	Opens trip details	GET /trips/:tripId (exists)	Exists	

F.4 Trip details

Item	Value
Feature / phase	Search and trip details / Phase 3
Files	`discovery_customer_screen.dart` , `discovery_customer.ts` , `discovery_customer.service.ts`
On the screen	Operator, bus type, amenities, route timeline with stops, boarding and dropping points, per-seat-type prices, cancellation policy, rating, Select seats.

Button / element	What it does	Endpoint	Status	Data
Select seats	Opens the seat map	GET /trips/:tripId/seats	New	inventory
Bus no longer available	Shown when the trip was cancelled, sold out or unpublished	GET /trips/:tripId returns 410	New	

F.5 Select seats

Item	Value
Feature / phase	Seat selection and hold / Phase 3
Files	`seats_customer_screen.dart` , `seats_customer.ts` , `seats_customer.service.ts`
On the screen	Seat map (seater and sleeper, ladies seats, blocked, held, sold), price on each seat by seat type, running total, limit of 6.

Button / element	What it does	Endpoint	Status	Data
Tap a seat	Selects or unselects locally; the limit of 6 is shown	none until Hold	UI	
Continue (hold)	Holds the seats for 10 minutes atomically; asks to sign in if needed	POST /trips/:tripId/seat-locks	New	Redis lock + inventory; returns holdId and expiry
Back / timer end	Releases the hold	DELETE /trips/:tripId/seat-locks/:holdId	New	
Live seat changes	Other people's holds and sales appear without reload	Socket.IO room trip:{id}	New	

F.6 Sign in sheet (opens at Continue)

Item	Value
Feature / phase	Sign-in and access control / Phase 0

Item	Value
Files	`auth_access_customer_screen.dart` , `auth_access_customer.ts` , `auth_access_customer.service.ts`
On the screen	Mobile number, Continue, code boxes, Verify, Change number, Resend, Continue with Google, Continue with Apple.

Button / element	What it does	Endpoint	Status	Data
Continue	Sends the WhatsApp code	POST /auth/otp/send (exists)	Exists	sessions
Verify	Checks the code and signs in	POST /auth/otp/verify (exists)	Exists	
Resend / Change number	Resend after a countdown; change goes back	POST /auth/otp/send (exists)	Exists	
Google / Apple	Signs in with the provider token	POST /auth/social/google, POST /auth/social/apple	Exists	Google exists for admin and client; Apple is new

F.7 Boarding and dropping

Item	Value
Feature / phase	Booking, passengers and bookings lists / Phase 3
Files	`booking_customer_screen.dart` , `booking_customer.ts` , `booking_customer.service.ts`
On the screen	Two tabs (Boarding, Dropping), points with time and address, map link, Continue.

Button / element	What it does	Endpoint	Status	Data
Boarding / Dropping tabs	Switches the list	GET /trips/:tripId/points (new, also returned inside trip details)	New	places
Open in maps	Opens the map app for the point	none (deep link)	UI	
Continue	Stores the chosen points in the draft booking	PATCH /customer/booking-drafts/:holdId	New	bookings (draft)

F.8 Passenger details

Item	Value
Feature / phase	Booking, passengers and bookings lists / Phase 3
Files	`booking_customer_screen.dart` , `booking_customer.ts` , `booking_customer.service.ts`
On the screen	One block per seat (Full name, Age, Gender; ladies-only seat rule), Use my profile, Contact (Mobile *, Email *), GST details (GSTIN, Company name).

Button / element	What it does	Endpoint	Status	Data
Use my profile	Fills passenger 1 and the contact	GET /customer/profile	New	customers
Continue	Validates and saves passengers and contact; 10-digit mobile, valid email, GSTIN pattern, age 1 to 99, female-only seats	PATCH /customer/booking-drafts/:holdId	New	

F.9 Review booking

Item	Value
Feature / phase	Booking, passengers and bookings lists / Phase 3
Files	`booking_customer_screen.dart` , `booking_customer.ts` , `booking_customer.service.ts`
On the screen	Passengers, coupon box (Apply, Remove), fare breakup (fare, convenience fee, GST, discount, total), cancellation policy, agree checkbox, Proceed to pay.

Button / element	What it does	Endpoint	Status	Data
Apply coupon	Validates the code for this trip and user	POST /customer/coupons/validate	New	offers
Remove coupon	Removes it	DELETE /customer/booking-drafts/:holdId/coupon	New	
Fare breakup	Price computed on the server: fare + commission + fee + GST - discount	GET /customer/booking-drafts/:holdId/price	New	commissions, payoutCfg
Proceed to pay	Creates the booking in PENDING and the Razorpay order (idempotent)	POST /customer/bookings (Idempotency-Key header)	New	bookings, payments

F.10 Payment

Item	Value
Feature / phase	Payments / Phase 3
Files	`payments_customer_screen.dart` , `payments_customer.ts` , `payments_customer.service.ts`
On the screen	Payment methods (UPI apps, card, netbanking, wallet), processing screen, failure screen (Retry payment, Change payment method, Back to review).

Button / element	What it does	Endpoint	Status	Data
Pay now	Opens Razorpay checkout with the server order	POST /customer/bookings/:id/pay (returns Razorpay order)	New	payments
Payment result	App sends the Razorpay result; the server verifies the signature and also trusts its own webhook	POST /customer/bookings/:id/verify-payment and POST /webhooks/razorpay	New	payments, bookings
Retry / Change method	New attempt on the same booking while the hold lasts	POST /customer/bookings/:id/pay	New	

F.11 Booking confirmed

Item	Value
Feature / phase	Ticket, PNR and QR / Phase 3
Files	`tickets_pdf_customer_screen.dart` , `tickets_pdf_customer.ts` , `tickets_pdf_customer.service.ts`
On the screen	PNR (copy), ticket, share trip, add to calendar, View the messages, My trips.

Button / element	What it does	Endpoint	Status	Data
Copy PNR	Copies	none	UI	
View ticket	Opens e-ticket	GET /customer/bookings/:id	New	bookings
Add to calendar	Downloads an .ics file	GET /customer/bookings/:id/calendar.ics	New	
View the messages	Opens the messages page	GET /customer/messages	New	messages

F.12 E-ticket

Item	Value
Feature / phase	Ticket, PNR and QR / Phase 3

Item	Value
Files	`tickets_pdf_customer_screen.dart` , `tickets_pdf_customer.ts` , `tickets_pdf_customer.service.ts`
On the screen	QR to show the conductor, passengers and seats, boarding and dropping, operator contact, PDF, share.

Button / element	What it does	Endpoint	Status	Data
Call operator	Dials the operator contact	none (tel link)	UI	
Download PDF	Ticket PDF	GET /customer/bookings/:id/ticket.pdf	New	
Share trip	Opens the sharing screen	see Trip sharing	New	

F.13 My trips (tab)

Item	Value
Feature / phase	My Trips / Phase 4
Files	`my_trips_customer_screen.dart` , `my_trips_customer.ts` , `my_trips_customer.service.ts`
On the screen	Tabs Upcoming, Completed, Cancelled; booking cards.

Button / element	What it does	Endpoint	Status	Data
Tabs	Filter by status	GET /customer/bookings?status=	New	bookings
Card tap	Opens booking details	GET /customer/bookings/:id	New	

F.14 Booking details

Item	Value
Feature / phase	My Trips / Phase 4
Files	`my_trips_customer_screen.dart` , `my_trips_customer.ts` , `my_trips_customer.service.ts`
On the screen	Status, PNR, route, seats, amount, boarding point, buttons Track, Ticket, Share, Cancel, Refund status, Support.

Button / element	What it does	Endpoint	Status	Data
Track	Opens live tracking when the trip is running	GET /customer/bookings/:id/tracking	New	live
Cancel booking	Opens cancel screen	GET /customer/bookings/:id/cancel-quote	New	cancelSlabs
Refund status	Opens refund screen	GET /customer/refunds/:bookingId	New	refunds

F.15 Cancel booking

Item	Value
Feature / phase	Cancellation and refunds / Phase 4
Files	`refunds_customer_screen.dart` , `refunds_customer.ts` , `refunds_customer.service.ts`
On the screen	Select seats to cancel, refund estimate from the slab, reason, Cancel booking.

Button / element	What it does	Endpoint	Status	Data
Refund estimate	Slab by hours before departure	GET /customer/bookings/:id/cancel-quote?seats=	New	
Cancel	Cancels the chosen seats, frees them, starts the refund, sends messages	POST /customer/bookings/:id/cancel	New	bookings, refunds, inventory, messages

F.16 Refund status

Item	Value
Feature / phase	Cancellation and refunds / Phase 4
Files	`refunds_customer_screen.dart` , `refunds_customer.ts` , `refunds_customer.service.ts`
On the screen	Amount, status steps (initiated, processing, completed), reference.

Button / element	What it does	Endpoint	Status	Data
Open	Shows the refund	GET /customer/refunds/:bookingId	New	refunds

F.17 Live tracking

Item	Value
Feature / phase	Live tracking / Phase 5
Files	`tracking_customer_screen.dart` , `tracking_customer.ts` , `tracking_customer.service.ts`
On the screen	Map with the bus, ETA, driver and bus details, Call, SOS and safety button, share.

Button / element	What it does	Endpoint	Status	Data
Live position	Receives GPS every few seconds	Socket.IO room tracking:{tripId} and GET /customer/bookings/:id/tracking	New	live
SOS and safety	Opens the safety screen	see Women safety	New	
Call driver or operator	Dial	none (tel link)	UI	

F.18 Women safety and SOS

Item	Value
Feature / phase	SOS and women safety / Phase 5
Files	`sos_customer_screen.dart` , `sos_customer.ts` , `sos_customer.service.ts`
On the screen	Press-and-hold SOS ring, I am safe (cancel), emergency contacts (add, remove), helplines, share trip.

Button / element	What it does	Endpoint	Status	Data
Hold SOS	Sends the alert with live location; nothing is sent until the ring is full	POST /customer/sos	New	sos
I am safe	Cancels a fresh alert	POST /customer/sos/:id/cancel	New	sos
Add / remove contact	Emergency contacts of the customer	POST /customer/emergency-contacts, DELETE /customer/emergency-contacts/:id	New	customers

Button / element	What it does	Endpoint	Status	Data
Status updates	Acknowledged, resolved, false alarm arrive as notifications	Socket.IO + notification	New	

F.19 Trip sharing

Item	Value
Feature / phase	Trip sharing / Phase 5
Files	`sharing_customer_screen.dart` , `sharing_customer.ts` , `sharing_customer.service.ts`
On the screen	Share link (WhatsApp, SMS, Copy), people with access, Stop sharing.

Button / element	What it does	Endpoint	Status	Data
Start sharing	Creates the link, expires after the trip	POST /customer/trip-shares	New	sharing
WhatsApp / SMS / Copy	Sends the link	POST /customer/trip-shares/:id/send	New	
Stop sharing	Revokes	DELETE /customer/trip-shares/:id	New	
Family view (public link)	Read-only live page	GET /public/share/:token	New	

F.20 Notifications

Item	Value
Feature / phase	Notifications and event router / Phase 0
Files	`notifications_customer_screen.dart` , `notifications_customer.ts` , `notifications_customer.service.ts`
On the screen	List of notifications with read and unread, Mark all read.

Button / element	What it does	Endpoint	Status	Data
List / open	Marks one read	GET /notifications, POST /notifications/:id/read (exist)	Exists	inbox
Mark all read		POST /notifications/read-all (exists)	Exists	

F.21 Offers and promotions

Item	Value
Feature / phase	Offers and coupons / Phase 3
Files	`offers_customer_screen.dart` , `offers_customer.ts` , `offers_customer.service.ts`
On the screen	Offer cards with code, Copy, Terms apply.

Button / element	What it does	Endpoint	Status	Data
List	Active offers	GET /offers	New	offers
Terms	Shows terms	GET /offers/:id	New	

F.22 Messages sent to me

Item	Value
Feature / phase	WhatsApp and email messages / Phase 0
Files	`messages_customer_screen.dart` , `messages_customer.ts` , `messages_customer.service.ts`
On the screen	Copies of the WhatsApp messages and emails Abra sent, open one to read.

Button / element	What it does	Endpoint	Status	Data
List / open		GET /customer/messages, GET /customer/messages/:id	New	messages

F.23 Support (tab) and ticket

Item	Value
Feature / phase	Support tickets / Phase 4
Files	`support_customer_screen.dart` , `support_customer.ts` , `support_customer.service.ts`
On the screen	Help centre (FAQs), Contact support (call), My tickets, New ticket with category, subject, message and proof file; ticket thread with replies.

Button / element	What it does	Endpoint	Status	Data
FAQs	Help centre	GET /faqs	New	faqs
New ticket	Creates a ticket with proof files	POST /customer/tickets	New	tickets
My tickets	List and thread	GET /customer/tickets, GET /customer/tickets/:id	New	
Reply	Adds a message with proof	POST /customer/tickets/:id/replies	New	
Contact support	Dial the support number	GET /contact-info	New	contact

F.24 Profile (tab), edit, notification settings

Item	Value
Feature / phase	Customers and profile / Phase 4
Files	`customers_customer_screen.dart` , `customers_customer.ts` , `customers_customer.service.ts`
On the screen	Signed-in details, Edit profile (Full name, Email *, mobile read-only), Notification settings, Language, Help centre, Messages, Delete account, Sign out.

Button / element	What it does	Endpoint	Status	Data
Save profile	Name and email	PATCH /customer/profile	New	customers
Notification settings	Toggles per channel and topic	PUT /customer/notification-settings	New	customers
Language	Saves the language	PATCH /customer/profile	New	
Sign out	Revokes this session	POST /auth/logout (exists)	Exists	sessions
Delete account	Deletes personal data after confirmation; past financial records stay	DELETE /customer/account	New	customers

F.25 Information pages

Item	Value
Feature / phase	Information pages and FAQs / Phase 4
Files	`content_customer_screen.dart` , `content_customer.ts` , `content_customer.service.ts`
On the screen	About, Terms, Privacy, Refund policy, Contact.

Button / element	What it does	Endpoint	Status	Data
Open page		GET /pages/:slug	New	pages

F.26 AI assistant

Item	Value
Feature / phase	AI assistant (Groq) / Phase 0
Files	`assistant_customer_screen.dart` , `assistant_customer.ts` , `assistant_customer.service.ts`
On the screen	Chat sheet with suggestions (my trips, track, refund, safety), voice input, answers in the user's language.

Button / element	What it does	Endpoint	Status	Data
Send / voice	Asks the assistant	POST /assistant/chat (streams)	New	Groq via backend; sees only this customer's data

Appendix G. Driver app (new app, Android): every screen and endpoint

The driver app does not exist yet. Phase 5 builds sign-in, trips, check-in, start and end, delay, GPS and SOS; Phase 9 adds reports and checklists.

> **Rule:** Build the real screens to look and behave exactly like the demo (`prototypes/driver_demo_model.html`): same layout, wording, order, colours, dialogs and states.

G.1 Sign in

Item	Value
Feature / phase	Driver app: trips and check-in / Phase 5
Files	`driver_core_driver_screen.dart` , `driver_core_driver.ts` , `driver_core_driver.service.ts`
On the screen	Phone and code (demo picks a driver).

Button / element	What it does	Endpoint	Status	Data
Sign in	Driver sign-in with the WhatsApp code	POST /auth/otp/send, POST /auth/otp/verify (exist)	Exists	opStaff, sessions

G.2 Home (tab)

Item	Value
Feature / phase	Driver app: trips and check-in / Phase 5
Files	`driver_core_driver_screen.dart` , `driver_core_driver.ts` , `driver_core_driver.service.ts`
On the screen	Greeting, next trip card, running trip, reports-to-send banner, SOS button always visible.

Button / element	What it does	Endpoint	Status	Data
Trip cards	Trips assigned to this driver	GET /driver/trips?status=	New	trips
Reports to send banner	Count of due and overdue reports	GET /driver/reports/due	New	driverReports
SOS	Raises a driver SOS	POST /driver/sos	New	sos

G.3 Your trip (tab)

Item	Value
Feature / phase	Driver app: trips and check-in / Phase 5
Files	`driver_core_driver_screen.dart` , `driver_core_driver.ts` , `driver_core_driver.service.ts`
On the screen	Trip details, passenger list with check-in, buttons Board, Start, Resume, End, Delay.

Button / element	What it does	Endpoint	Status	Data
Passenger list	Booked passengers and seats	GET /driver/trips/:id/manifest	New	bookings
Check in	Marks a passenger checked in (QR or manual)	POST /driver/trips/:id/check-in	New	trips.checkedIn
Board	Status BOARDING; notifies customers	POST /driver/trips/:id/board	New	
Start (opens pre-trip checklist)	Runs the checklist first, then starts	POST /driver/trips/:id/start (with checklist)	New	tripChecks, trips
Delay	Sets minutes and reason with proof; notifies customers, operator, admin	POST /driver/trips/:id/delay	New	
End (opens end checklist)	Runs the end checklist, then ends	POST /driver/trips/:id/end (with checklist)	New	
GPS	Sends position while the trip runs	POST /driver/trips/:id/location (batched) or Socket.IO	New	live

G.4 Reports (tab)

Item	Value
Feature / phase	Driver reports / Phase 9
Files	`driver_reports_driver_screen.dart` , `driver_reports_driver.ts` , `driver_reports_driver.service.ts`
On the screen	To send (Daily, Weekly, each with state Due, Overdue, Sent back), Report something now (Incident, Bus problem, Expense claim), Sent by you.

Button / element	What it does	Endpoint	Status	Data
To send list	Reports due for this driver	GET /driver/reports/due	New	driverReports
Open a report	Shows system numbers (trips, distance, passengers, no-shows, delays, on-time, SOS, checklists) and the questions the operator set	GET /driver/reports/preview?type&period	New	computed on the server from trips, bookings, tripChecks, sos
Save draft / Send report	Stores answers and files; on send, freezes the system numbers	POST /driver/reports, PATCH /driver/reports/:id, POST /driver/reports/:id/send	New	driverReports
Report something now	Incident, bus problem or expense claim	POST /driver/reports (type event)	New	
Sent by you	List and view	GET /driver/reports	New	

G.5 Before you start

Item	Value
Feature / phase	Driver rules and trip checklists / Phase 9
Files	`driver_rules_driver_screen.dart` , `driver_rules_driver.ts` , `driver_rules_driver.service.ts`
On the screen	The checklist the operator set: automatic checks (licence, bus papers, location, rest hours, passengers checked in), yes/no, choice and number items with photo where required; summary (all clear, warnings, blocked); override with reason and proof for non-locked failures.

Button / element	What it does	Endpoint	Status	Data
Automatic checks	Server evaluates licence, papers, location, rest, manifest	GET /driver/trips/:id/checklist	New	driverRules, opStaff, buses
Answer items	Stored on the device until Start	none	UI	
Start trip	Server re-checks; blocks if a blocking item failed; with override reason and proof it starts and notifies operator and admin	POST /driver/trips/:id/start	New	tripChecks

G.6 End of trip

Item	Value
Feature / phase	Driver rules and trip checklists / Phase 9
Files	`driver_rules_driver_screen.dart` , `driver_rules_driver.ts` , `driver_rules_driver.service.ts`
On the screen	End odometer with photo, new damage, lost items, fuel filled, cash collected, bus parked and locked.

Button / element	What it does	Endpoint	Status	Data
End trip	Saves the end check and closes the trip	POST /driver/trips/:id/end	New	tripChecks, trips

G.7 Notifications

Item	Value
Feature / phase	Notifications and event router / Phase 0
Files	`notifications_driver_screen.dart` , `notifications_driver.ts` , `notifications_driver.service.ts`
On the screen	List, Mark all read.

Button / element	What it does	Endpoint	Status	Data
List / read all		GET /notifications, POST /notifications/read-all (exist)	Exists	inbox

G.8 Profile

Item	Value
Feature / phase	Driver app: trips and check-in / Phase 5
Files	`driver_core_driver_screen.dart` , `driver_core_driver.ts` , `driver_core_driver.service.ts`
On the screen	Name, company, licence expiry, Sign out.

Button / element	What it does	Endpoint	Status	Data
Sign out		POST /auth/logout (exists)	Exists	

G.9 AI assistant

Item	Value
Feature / phase	AI assistant (Groq) / Phase 0
Files	`assistant_driver_screen.dart` , `assistant_driver.ts` , `assistant_driver.service.ts`
On the screen	Chat and voice: start trip, end trip, report delay, my trips.

Button / element	What it does	Endpoint	Status	Data
Send / voice	Start and end now open the checklists	POST /assistant/chat	New	

Appendix H. Public website (Next.js): every screen and endpoint

Server-rendered pages for search engines; booking hands over to the customer app.

> **Rule:** Build the real screens to look and behave exactly like the demo (`prototypes/website_demo_model.html`): same layout, wording, order, colours, dialogs and states.

H.1 Home

Item	Value
Feature / phase	Public website / Phase 10
Files	`website_public_page.tsx` , `website_public.ts` , `website_public.service.ts`
On the screen	Header (logo, language, menu), search box with Swap and Search, banners, offers, popular routes, footer.

Button / element	What it does	Endpoint	Status	Data
Search	Opens results and hands over to the customer app to book	GET /trips/search (exists), GET /cities (exists)	Exists	trips, cities
Banners / offers / popular routes	Public lists	GET /banners, GET /offers, GET /popular-routes	New	

H.2 Search results

Item	Value
Feature / phase	Public website / Phase 10
Files	`website_public_page.tsx` , `website_public.ts` , `website_public.service.ts`
On the screen	Trips for the route with price from, Book in app.

Button / element	What it does	Endpoint	Status	Data
Book in app	Deep link to the customer app	none	UI	

H.3 Popular routes

Item	Value
Feature / phase	Public website / Phase 10
Files	`website_public_page.tsx` , `website_public.ts` , `website_public.service.ts`
On the screen	Route pages for search engines.

Button / element	What it does	Endpoint	Status	Data
Route page	Server-rendered page with structured data	GET /public/routes/:slug	New	routes

H.4 Offers

Item	Value
Feature / phase	Public website / Phase 10
Files	`website_public_page.tsx` , `website_public.ts` , `website_public.service.ts`
On the screen	Offer cards with Copy code.

Button / element	What it does	Endpoint	Status	Data
List		GET /offers	New	offers

H.5 Track my bus

Item	Value
Feature / phase	Live tracking / Phase 5
Files	`tracking_public_page.tsx` , `tracking_public.ts` , `tracking_public.service.ts`
On the screen	PNR and phone, Track.

Button / element	What it does	Endpoint	Status	Data
Track	Shows the bus position for a booking	POST /public/track (PNR + phone)	New	bookings, live

H.6 Help and FAQs

Item	Value
Feature / phase	Information pages and FAQs / Phase 4
Files	`content_public_page.tsx` , `content_public.ts` , `content_public.service.ts`
On the screen	FAQs, Contact us.

Button / element	What it does	Endpoint	Status	Data
FAQs		GET /faqs	New	faqs

H.7 Contact us

Item	Value
Feature / phase	Information pages and FAQs / Phase 4
Files	`content_public_page.tsx` , `content_public.ts` , `content_public.service.ts`
On the screen	Contact form.

Button / element	What it does	Endpoint	Status	Data
Send	Creates an enquiry for admin	POST /public/enquiries (type contact)	New	contact

H.8 Partner with us

Item	Value
Feature / phase	Public website / Phase 10
Files	`website_public_page.tsx` , `website_public.ts` , `website_public.service.ts`
On the screen	Form (company, city, phone, email).

Button / element	What it does	Endpoint	Status	Data
Send	Creates a partner enquiry for admin	POST /public/enquiries (type partner)	New	contact

H.9 About, Terms, Privacy

Item	Value
Feature / phase	Information pages and FAQs / Phase 4
Files	`content_public_page.tsx` , `content_public.ts` , `content_public.service.ts`
On the screen	Static pages from the CMS.

Button / element	What it does	Endpoint	Status	Data
Open		GET /pages/:slug	New	pages

Appendix I. The backend today

Read from the controllers in `backend/src`: 26 controllers, 113 routes. Everything above marked "Exists" is in this list. Everything marked "New" is not.

Module	Routes
admin-banners	GET /admin/banners GET /admin/banners/:id POST /admin/banners PUT /admin/banners/:id POST /admin/banners/:id/active POST /admin/banners/:id/image DELETE /admin/banners/:id/image DELETE /admin/banners/:id GET /admin/banners GET /admin/banners/:id/image
audit-logs	GET /audit-logs
authentication	POST /auth/otp/send POST /auth/otp/verify POST /auth/register POST /auth/login POST /auth/mfa/verify POST /auth/refresh POST /auth/logout POST /auth/password/reset GET /me POST /me/password POST /auth/logout-all GET /me/mfa POST /me/mfa/totp/enroll POST /me/mfa/totp/activate POST /me/mfa/totp/disable
bus-types	GET /catalogue/bus-types POST /catalogue/bus-types PUT /catalogue/bus-types/:id POST /catalogue/bus-types/:id/active
buses	GET /catalogue/buses GET /catalogue/buses/:id POST /catalogue/buses PUT /catalogue/buses/:id POST /catalogue/buses/:id/status
cities	GET /catalogue/cities GET /catalogue/cities/:id POST /catalogue/cities PUT /catalogue/cities/:id POST /catalogue/cities/:id/active GET /cities
document-types	GET /catalogue/document-types POST /catalogue/document-types PUT /catalogue/document-types/:id POST /catalogue/document-types/:id/active
files	POST /files GET /files/:id GET /files/:id/content
health	GET /health/live GET /health/ready
locations	GET /catalogue/locations GET /catalogue/locations/:id POST /catalogue/locations PUT /catalogue/locations/:id POST /catalogue/locations/:id/active
notifications	GET /notifications GET /notifications/unread-count POST /notifications/read-all POST /notifications/devices POST /notifications/devices/remove POST /notifications/:id/read
onboarding	POST /onboarding/invitations GET /onboarding/invitations POST /onboarding/invitations/:id/resend POST /onboarding/invitations/:id/revoke POST /onboarding/invitations/inspect POST /onboarding/invitations/accept GET /onboarding/submissions GET /onboarding/submissions/:id POST /onboarding/submissions PUT /onboarding/submissions/:id/documents/:documentTypeId DELETE /onboarding/submissions/:id/documents/:documentTypeId POST /onboarding/submissions/:id/submit POST /onboarding/submissions/:id/documents/:documentTypeId/review POST /onboarding/submissions/:id/approve POST /onboarding/submissions/:id/reject
operators	GET /catalogue/operators GET /catalogue/operators/:id POST /catalogue/operators PUT /catalogue/operators/:id POST /catalogue/operators/:id/status
rbac	GET /permissions GET /roles GET /me/permissions GET /users/:userId/permissions PUT /users/:userId/roles PUT /users/:userId/permissions/:permissionKey DELETE /users/:userId/permissions/:permissionKey PUT /users/:userId/data-scope

Module	Routes
routes	GET /catalogue/routes GET /catalogue/routes/:id POST /catalogue/routes PUT /catalogue/routes/:id POST /catalogue/routes/:id/status
sessions	GET /me/sessions DELETE /me/sessions/:sessionId
system-settings	GET /settings/integrations/email PUT /settings/integrations/email POST /settings/integrations/email/test-connection POST /settings/integrations/email/test-email GET /settings/integrations/whatsapp POST /settings/integrations/whatsapp/test-connection PUT /settings/integrations/whatsapp
trips	GET /trips/search GET /trips/search/filters GET /trips/:tripId GET /catalogue/trips GET /catalogue/trips/:id POST /catalogue/trips PUT /catalogue/trips/:id POST /catalogue/trips/:id/publish POST /catalogue/trips/:id/cancel
users	GET /users

I.2 Modules with no controller yet

`analytics`, `api-clients`, `bookings`, `boarding-points`, `chatbot`, `check-in`, `commissions`, `content`, `coupons`, `customers`, `drivers`, `fares`, `feature-flags`, `inventory`, `live-tracking`, `maintenance`, `notification-templates`, `offers`, `operator-staff`, `partner-webhooks`, `partners`, `passengers`, `payments`, `popular-routes`, `reconciliation`, `refunds`, `reports`, `schedules`, `seat-layouts`, `settlements`, `support`, `tickets`, `translations`, `trip-operations`, `whatsapp`, `women-safety`. These folders exist as empty placeholders; they get their three-file packages in the phases above.

I.3 Coverage of the demo buttons

App	Buttons mapped to an existing endpoint	Buttons that need a new endpoint	UI-only buttons	Not mapped
Admin portal	35	221	123	0
Client portal	12	57	53	0

Customer app: 26 screens, 78 mapped elements. Driver app: 9 screens, 23. Website: 9 pages, 10.

Appendix J. Feature file index (the three files of every feature)

Status was last updated 13 October 2026 (Part 9 has the detail). One row per feature and audience. Paths are relative to the repository root. Flutter files live in the app's `lib/features/<feature>/` folder; backend files in `backend/src/modules/<first module>/`. A "(backend only)" feature has no screen. Update the Status column as work finishes.

Phase	Feature	Screen file (all widgets inside)	Backend file (schema, DTOs, controller, module)	Service file
Phase 0	Sign-in and access control	apps/customer_app/lib/features/auth_access/auth_access_customer_screen.dart	backend/src/modules/authentication/auth_access_customer.ts	backend/src/modules/authentication/auth_access_customer.se
Phase 0	Sign-in and access control	apps/driver_app/lib/features/auth_access/auth_access_driver_screen.dart	backend/src/modules/authentication/auth_access_driver.ts	backend/src/modules/authentication/auth_access_driver.servic
Phase 0	Audit log	apps/admin_app/lib/features/audit/audit_admin_manage_screen.dart	backend/src/modules/audit-logs/audit_admin_manage.ts	backend/src/modules/audit-logs/audit_admin_manage.service
Phase 0	Notifications and event router	apps/admin_app/lib/features/notifications/notifications_admin_manage_screen.dart	backend/src/modules/notifications/notifications_admin_manage.ts	backend/src/modules/notifications/notifications_admin_manag
Phase 0	Notifications and event router	apps/client_portal/lib/features/notifications/notifications_client_manage_screen.dart	backend/src/modules/notifications/notifications_client_manage.ts	backend/src/modules/notifications/notifications_client_manag
Phase 0	Notifications and event router	apps/customer_app/lib/features/notifications/notifications_customer_screen.dart	backend/src/modules/notifications/notifications_customer.ts	backend/src/modules/notifications/notifications_customer.serv
Phase 0	Notifications and event router	apps/driver_app/lib/features/notifications/notifications_driver_screen.dart	backend/src/modules/notifications/notifications_driver.ts	backend/src/modules/notifications/notifications_driver.service
Phase 0	File storage and proof attachments	(backend only)	backend/src/modules/files/files_core.ts	backend/src/modules/files/files_core.service.ts
Phase 0	WhatsApp and email messages	apps/admin_app/lib/features/messages/messages_admin_manage_screen.dart	backend/src/modules/whatsapp/messages_admin_manage.ts	backend/src/modules/whatsapp/messages_admin_manage.ser
Phase 0	WhatsApp and email messages	apps/customer_app/lib/features/messages/messages_customer_screen.dart	backend/src/modules/whatsapp/messages_customer.ts	backend/src/modules/whatsapp/messages_customer.service.ts
Phase 0	Integrations, feature flags and maintenance	apps/admin_app/lib/features/settings/settings_admin_manage_screen.dart	backend/src/modules/system-settings/settings_admin_manage.ts	backend/src/modules/system-settings/settings_admin_manag
Phase 0	AI assistant (Groq)	apps/admin_app/lib/features/assistant/assistant_admin_manage_screen.dart	backend/src/modules/chatbot/assistant_admin_manage.ts	backend/src/modules/chatbot/assistant_admin_manage.servic
Phase 0	AI assistant (Groq)	apps/customer_app/lib/features/assistant/assistant_customer_screen.dart	backend/src/modules/chatbot/assistant_customer.ts	backend/src/modules/chatbot/assistant_customer.service.ts
Phase 0	AI assistant (Groq)	apps/driver_app/lib/features/assistant/assistant_driver_screen.dart	backend/src/modules/chatbot/assistant_driver.ts	backend/src/modules/chatbot/assistant_driver.service.ts
Phase 0	Seed data and live updates	(backend only)	backend/src/modules/database/seed_core.ts	backend/src/modules/database/seed_core.service.ts
Phase 1	Cities and places	apps/admin_app/lib/features/places/places_admin_manage_screen.dart	backend/src/modules/cities/places_admin_manage.ts	backend/src/modules/cities/places_admin_manage.service.ts
Phase 1	Cities and places	apps/client_portal/lib/features/places/places_client_manage_screen.dart	backend/src/modules/cities/places_client_manage.ts	backend/src/modules/cities/places_client_manage.service.ts
Phase 1	Cities and places	apps/customer_app/lib/features/places/places_customer_screen.dart	backend/src/modules/cities/places_customer.ts	backend/src/modules/cities/places_customer.service.ts
Phase 1	Bus types and amenities	apps/admin_app/lib/features/bus_types/bus_types_admin_manage_screen.dart	backend/src/modules/bus-types/bus_types_admin_manage.ts	backend/src/modules/bus-types/bus_types_admin_manage.se
Phase 1	Required documents and KYC questionnaire	apps/admin_app/lib/features/kyc_config/kyc_config_admin_manage_screen.dart	backend/src/modules/document-types/kyc_config_admin_manage.ts	backend/src/modules/document-types/kyc_config_admin_manage.service.ts
Phase 1	Operator invitations	apps/admin_app/lib/features/invitations/invitations_admin_manage_screen.dart	backend/src/modules/onboarding/invitations_admin_manage.ts	backend/src/modules/onboarding/invitations_admin_manage.

Phase	Feature	Screen file (all widgets inside)	Backend file (schema, DTOs, controller, module)	Service file
Phase 1	KYC review and company onboarding	apps/admin_app/lib/features/kyc/kyc_admin_manage_screen.dart	backend/src/modules/onboarding/kyc_admin_manage.ts	backend/src/modules/onboarding/kyc_admin_manage.service
Phase 1	KYC review and company onboarding	apps/client_portal/lib/features/kyc/kyc_client_manage_screen.dart	backend/src/modules/onboarding/kyc_client_manage.ts	backend/src/modules/onboarding/kyc_client_manage.service.t
Phase 1	Operators	apps/admin_app/lib/features/operators/operators_admin_manage_screen.dart	backend/src/modules/operators/operators_admin_manage.ts	backend/src/modules/operators/operators_admin_manage.ser
Phase 1	Banners	apps/admin_app/lib/features/banners/banners_admin_manage_screen.dart	backend/src/modules/admin-banners/banners_admin_manage.ts	backend/src/modules/admin-banners/banners_admin_manage
Phase 1	Banners	apps/customer_app/lib/features/banners/banners_customer_screen.dart	backend/src/modules/admin-banners/banners_customer.ts	backend/src/modules/admin-banners/banners_customer.servi
Phase 1	Banners	apps/public_website/app/banners/banners_public_page.tsx	backend/src/modules/admin-banners/banners_public.ts	backend/src/modules/admin-banners/banners_public.service.t
Phase 2	Buses and seat layouts	apps/admin_app/lib/features/buses/buses_admin_manage_screen.dart	backend/src/modules/buses/buses_admin_manage.ts	backend/src/modules/buses/buses_admin_manage.service.ts
Phase 2	Buses and seat layouts	apps/client_portal/lib/features/buses/buses_client_manage_screen.dart	backend/src/modules/buses/buses_client_manage.ts	backend/src/modules/buses/buses_client_manage.service.ts
Phase 2	Crew: drivers and conductors	apps/client_portal/lib/features/crew/crew_client_manage_screen.dart	backend/src/modules/operator-staff/crew_client_manage.ts	backend/src/modules/operator-staff/crew_client_manage.servi
Phase 2	Routes, stops and boarding points	apps/client_portal/lib/features/routes/routes_client_manage_screen.dart	backend/src/modules/routes/routes_client_manage.ts	backend/src/modules/routes/routes_client_manage.service.ts
Phase 2	Fares and schedules	apps/client_portal/lib/features/fares/fares_client_manage_screen.dart	backend/src/modules/fares/fares_client_manage.ts	backend/src/modules/fares/fares_client_manage.service.ts
Phase 2	Trips and seat inventory	apps/admin_app/lib/features/trips/trips_admin_manage_screen.dart	backend/src/modules/trips/trips_admin_manage.ts	backend/src/modules/trips/trips_admin_manage.service.ts
Phase 2	Trips and seat inventory	apps/client_portal/lib/features/trips/trips_client_manage_screen.dart	backend/src/modules/trips/trips_client_manage.ts	backend/src/modules/trips/trips_client_manage.service.ts
Phase 2	Commission rules	apps/admin_app/lib/features/commission/commission_admin_manage_screen.dart	backend/src/modules/commissions/commission_admin_manage.ts	backend/src/modules/commissions/commission_admin_mana
Phase 2	Commission rules	apps/client_portal/lib/features/commission/commission_client_manage_screen.dart	backend/src/modules/commissions/commission_client_manage.ts	backend/src/modules/commissions/commission_client_manag
Phase 2	Payout and fee settings	apps/admin_app/lib/features/payout_cfg/payout_cfg_admin_manage_screen.dart	backend/src/modules/settlements/payout_cfg_admin_manage.ts	backend/src/modules/settlements/payout_cfg_admin_manage
Phase 3	Search and trip details	apps/customer_app/lib/features/discovery/discovery_customer_screen.dart	backend/src/modules/trips/discovery_customer.ts	backend/src/modules/trips/discovery_customer.service.ts
Phase 3	Search and trip details	apps/public_website/app/discovery/discovery_public_page.tsx	backend/src/modules/trips/discovery_public.ts	backend/src/modules/trips/discovery_public.service.ts
Phase 3	Seat selection and hold	apps/customer_app/lib/features/seats/seats_customer_screen.dart	backend/src/modules/inventory/seats_customer.ts	backend/src/modules/inventory/seats_customer.service.ts
Phase 3	Booking, passengers and bookings lists	apps/admin_app/lib/features/booking/booking_admin_manage_screen.dart	backend/src/modules/bookings/booking_admin_manage.ts	backend/src/modules/bookings/booking_admin_manage.servi
Phase 3	Booking, passengers and bookings lists	apps/client_portal/lib/features/booking/booking_client_manage_screen.dart	backend/src/modules/bookings/booking_client_manage.ts	backend/src/modules/bookings/booking_client_manage.servic
Phase 3	Booking, passengers and bookings lists	apps/customer_app/lib/features/booking/booking_customer_screen.dart	backend/src/modules/bookings/booking_customer.ts	backend/src/modules/bookings/booking_customer.service.ts
Phase 3	Offers and coupons	apps/admin_app/lib/features/offers/offers_admin_manage_screen.dart	backend/src/modules/coupons/offers_admin_manage.ts	backend/src/modules/coupons/offers_admin_manage.service.t
Phase 3	Offers and coupons	apps/client_portal/lib/features/offers/offers_client_manage_screen.dart	backend/src/modules/coupons/offers_client_manage.ts	backend/src/modules/coupons/offers_client_manage.service.ts
Phase 3	Offers and coupons	apps/customer_app/lib/features/offers/offers_customer_screen.dart	backend/src/modules/coupons/offers_customer.ts	backend/src/modules/coupons/offers_customer.service.ts
Phase 3	Offers and coupons	apps/public_website/app/offers/offers_public_page.tsx	backend/src/modules/coupons/offers_public.ts	backend/src/modules/coupons/offers_public.service.ts
Phase 3	Payments	apps/admin_app/lib/features/payments/payments_admin_manage_screen.dart	backend/src/modules/payments/payments_admin_manage.ts	backend/src/modules/payments/payments_admin_manage.ser
Phase 3	Payments	apps/customer_app/lib/features/payments/payments_customer_screen.dart	backend/src/modules/payments/payments_customer.ts	backend/src/modules/payments/payments_customer.service.ts
Phase 3	Ticket, PNR and QR	apps/customer_app/lib/features/tickets_pdf/tickets_pdf_customer_screen.dart	backend/src/modules/tickets/tickets_pdf_customer.ts	backend/src/modules/tickets/tickets_pdf_customer.service.ts
Phase 4	My Trips	apps/customer_app/lib/features/my_trips/my_trips_customer_screen.dart	backend/src/modules/bookings/my_trips_customer.ts	backend/src/modules/bookings/my_trips_customer.service.ts
Phase 4	Cancellation and refunds	apps/admin_app/lib/features/refunds/refunds_admin_manage_screen.dart	backend/src/modules/refunds/refunds_admin_manage.ts	backend/src/modules/refunds/refunds_admin_manage.service
Phase 4	Cancellation and refunds	apps/customer_app/lib/features/refunds/refunds_customer_screen.dart	backend/src/modules/refunds/refunds_customer.ts	backend/src/modules/refunds/refunds_customer.service.ts
Phase 4	Support tickets	apps/admin_app/lib/features/support/support_admin_manage_screen.dart	backend/src/modules/support/support_admin_manage.ts	backend/src/modules/support/support_admin_manage.service
Phase 4	Support tickets	apps/customer_app/lib/features/support/support_customer_screen.dart	backend/src/modules/support/support_customer.ts	backend/src/modules/support/support_customer.service.ts
Phase 4	Support tickets	apps/public_website/app/support/support_public_page.tsx	backend/src/modules/support/support_public.ts	backend/src/modules/support/support_public.service.ts

Phase	Feature	Screen file (all widgets inside)	Backend file (schema, DTOs, controller, module)	Service file
Phase 4	Customers and profile	apps/admin_app/lib/features/customers/customers_admin_manage_screen.dart	backend/src/modules/customers/customers_admin_manage.ts	backend/src/modules/customers/customers_admin_manage.se
Phase 4	Customers and profile	apps/customer_app/lib/features/customers/customers_customer_screen.dart	backend/src/modules/customers/customers_customer.ts	backend/src/modules/customers/customers_customer.service.
Phase 4	Information pages and FAQs	apps/admin_app/lib/features/content/content_admin_manage_screen.dart	backend/src/modules/content/content_admin_manage.ts	backend/src/modules/content/content_admin_manage.service
Phase 4	Information pages and FAQs	apps/customer_app/lib/features/content/content_customer_screen.dart	backend/src/modules/content/content_customer.ts	backend/src/modules/content/content_customer.service.ts
Phase 4	Information pages and FAQs	apps/public_website/app/content/content_public_page.tsx	backend/src/modules/content/content_public.ts	backend/src/modules/content/content_public.service.ts
Phase 5	Driver app: trips and check-in	apps/driver_app/lib/features/driver_core/driver_core_driver_screen.dart	backend/src/modules/drivers/driver_core_driver.ts	backend/src/modules/drivers/driver_core_driver.service.ts
Phase 5	Live tracking	apps/admin_app/lib/features/tracking/tracking_admin_manage_screen.dart	backend/src/modules/live-tracking/tracking_admin_manage.ts	backend/src/modules/live-tracking/tracking_admin_manage.se
Phase 5	Live tracking	apps/client_portal/lib/features/tracking/tracking_client_manage_screen.dart	backend/src/modules/live-tracking/tracking_client_manage.ts	backend/src/modules/live-tracking/tracking_client_manage.ser
Phase 5	Live tracking	apps/customer_app/lib/features/tracking/tracking_customer_screen.dart	backend/src/modules/live-tracking/tracking_customer.ts	backend/src/modules/live-tracking/tracking_customer.service.t
Phase 5	Live tracking	apps/driver_app/lib/features/tracking/tracking_driver_screen.dart	backend/src/modules/live-tracking/tracking_driver.ts	backend/src/modules/live-tracking/tracking_driver.service.ts
Phase 5	Live tracking	apps/public_website/app/tracking/tracking_public_page.tsx	backend/src/modules/live-tracking/tracking_public.ts	backend/src/modules/live-tracking/tracking_public.service.ts
Phase 5	SOS and women safety	apps/admin_app/lib/features/sos/sos_admin_manage_screen.dart	backend/src/modules/women-safety/sos_admin_manage.ts	backend/src/modules/women-safety/sos_admin_manage.servi
Phase 5	SOS and women safety	apps/customer_app/lib/features/sos/sos_customer_screen.dart	backend/src/modules/women-safety/sos_customer.ts	backend/src/modules/women-safety/sos_customer.service.ts
Phase 5	SOS and women safety	apps/driver_app/lib/features/sos/sos_driver_screen.dart	backend/src/modules/women-safety/sos_driver.ts	backend/src/modules/women-safety/sos_driver.service.ts
Phase 5	Trip sharing	apps/customer_app/lib/features/sharing/sharing_customer_screen.dart	backend/src/modules/women-safety/sharing_customer.ts	backend/src/modules/women-safety/sharing_customer.service
Phase 5	Incidents	apps/admin_app/lib/features/incidents/incidents_admin_manage_screen.dart	backend/src/modules/women-safety/incidents_admin_manage.ts	backend/src/modules/women-safety/incidents_admin_manage
Phase 6	Settlements and earnings	apps/admin_app/lib/features/settlements/settlements_admin_manage_screen.dart	backend/src/modules/settlements/settlements_admin_manage.ts	backend/src/modules/settlements/settlements_admin_manage
Phase 6	Settlements and earnings	apps/client_portal/lib/features/settlements/settlements_client_manage_screen.dart	backend/src/modules/settlements/settlements_client_manage.ts	backend/src/modules/settlements/settlements_client_manage
Phase 7	Dashboards and reports	apps/admin_app/lib/features/dashboard/dashboard_admin_manage_screen.dart	backend/src/modules/reports/dashboard_admin_manage.ts	backend/src/modules/reports/dashboard_admin_manage.serv
Phase 7	Dashboards and reports	apps/client_portal/lib/features/dashboard/dashboard_client_manage_screen.dart	backend/src/modules/reports/dashboard_client_manage.ts	backend/src/modules/reports/dashboard_client_manage.servic
Phase 7	Announcements and popular routes	apps/admin_app/lib/features/announce/announce_admin_manage_screen.dart	backend/src/modules/content/announce_admin_manage.ts	backend/src/modules/content/announce_admin_manage.servi
Phase 7	Announcements and popular routes	apps/public_website/app/announce/announce_public_page.tsx	backend/src/modules/content/announce_public.ts	backend/src/modules/content/announce_public.service.ts
Phase 7	Staff, roles and access matrix	apps/admin_app/lib/features/staff/staff_admin_manage_screen.dart	backend/src/modules/rbac/staff_admin_manage.ts	backend/src/modules/rbac/staff_admin_manage.service.ts
Phase 7	Staff, roles and access matrix	apps/client_portal/lib/features/staff/staff_client_manage_screen.dart	backend/src/modules/rbac/staff_client_manage.ts	backend/src/modules/rbac/staff_client_manage.service.ts
Phase 9	Driver rules and trip checklists	apps/admin_app/lib/features/driver_rules/driver_rules_admin_manage_screen.dart	backend/src/modules/drivers/driver_rules_admin_manage.ts	backend/src/modules/drivers/driver_rules_admin_manage.serv
Phase 9	Driver rules and trip checklists	apps/client_portal/lib/features/driver_rules/driver_rules_client_manage_screen.dart	backend/src/modules/drivers/driver_rules_client_manage.ts	backend/src/modules/drivers/driver_rules_client_manage.servi
Phase 9	Driver rules and trip checklists	apps/driver_app/lib/features/driver_rules/driver_rules_driver_screen.dart	backend/src/modules/drivers/driver_rules_driver.ts	backend/src/modules/drivers/driver_rules_driver.service.ts
Phase 9	Driver reports	apps/admin_app/lib/features/driver_reports/driver_reports_admin_manage_screen.dart	backend/src/modules/drivers/driver_reports_admin_manage.ts	backend/src/modules/drivers/driver_reports_admin_manage.se
Phase 9	Driver reports	apps/client_portal/lib/features/driver_reports/driver_reports_client_manage_screen.dart	backend/src/modules/drivers/driver_reports_client_manage.ts	backend/src/modules/drivers/driver_reports_client_manage.se
Phase 9	Driver reports	apps/driver_app/lib/features/driver_reports/driver_reports_driver_screen.dart	backend/src/modules/drivers/driver_reports_driver.ts	backend/src/modules/drivers/driver_reports_driver.service.ts
Phase 10	Public website	apps/public_website/app/website/website_public_page.tsx	backend/src/modules/public/website_public.ts	backend/src/modules/public/website_public.service.ts
Phase 10	Partner app and partner API	(backend only)	backend/src/modules/partners/partner_core.ts	backend/src/modules/partners/partner_core.service.ts

Appendix K. Glossary and languages

Term	Meaning
Operator / client / bus owner	The bus company. The client portal is the operator app.
KYC	Know your customer: company and bus documents checked by Abra.
PNR	The booking reference printed on the ticket.
Hold	Seats reserved for 10 minutes while a customer pays.
Manifest	The passenger list of a trip.
Settlement	What Abra pays an operator for a cycle: gross sales minus commission minus refunds plus adjustments.
Razorpay Route	Razorpay's automatic split of one payment between Abra and the operator.
Proof	A picture, PDF or text file attached to a sensitive action.
Locked item	A driver checklist item Abra requires; an operator cannot remove or switch it off.
Snapshot	The system numbers of a driver report, frozen when the driver sends it.
Access matrix	None, View, Edit, Edit+Delete per role per page.

Term	Meaning
Seed	The script that loads the sample data into MongoDB.

Language	Code	Shown in the picker as
English	en	English
Telugu	te	Telugu (తెలుగు)
Hindi	hi	Hindi (हिन्दी)
Kannada	kn	Kannada (ಕನ್ನಡ)
Tamil	ta	Tamil (தமிழ்)
Malayalam	ml	Malayalam (മലയാളം)
Marathi	mr	Marathi (मराठी)
Bengali	bn	Bengali (বাংলা)